

1510.06699: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00001	1	0.755	$\{ \}^{\{1,2, \mathrm{a}\}}$	1,2,a	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00002	1	0.755	$\{ \}^{\{1, \mathrm{~b}\}}$	1, b	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00003	1	0.969	$\{ \}^{\{1\}}$	1)	¹⁾ <i>Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK</i>
1510.06699_F00004	1	0.979	$\{ \}^{\{2\}}$	2)	²⁾ <i>Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK</i>
1510.06699_F00005	1	0.947	$\{ \}^{\{\underline{1}\}}$	<u>1</u>	Following an initial attempt by Utiyama ¹ , the idea
1510.06699_F00006	1	0.998	$\mathrm{\varphi}$	φ	(or fields) φ with energy-momentum and, in general,
1510.06699_F00007	1	0.798	$\{ \}^{\{\text{a}\}}$	a)	^{a)} Electronic mail: a.n.lasenby@mrao.cam.ac.uk
1510.06699_F00008	1	0.997	$\{ \}^{\{\text{b}\}}$	b)	^{b)} Electronic mail: mph@mrao.cam.ac.uk
1510.06699_F00009	1	1.000	$S_{\mathrm{M}}=$	$S_{\mathrm{M}}=$	field dynamics are described by a matter action $S_{\mathrm{M}}=$
1510.06699_F00010	1	1.000	$\int \mathrm{L}_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) \mathrm{d}^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$ that is invariant under global Poincaré
1510.06699_F00011	1	1.000	h_{a}^{μ}	h_a^{μ}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00012	1	1.000	$A^{\mathrm{a} \mathrm{~b}}_{\mu}$	A_{μ}^{ab}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00013	1	0.990	$\mathrm{D}_{\mathrm{a}} \varphi$	$\mathcal{D}_a \varphi$	a covariant derivative $\mathcal{D}_a \varphi$, and the matter action in the
1510.06699_F00014	1	1.000	$S_{\mathrm{M}}=\int h^{-1} \mathrm{L}_{\mathrm{M}}(\varphi, \mathrm{D}_{\mathrm{a}} \varphi) \mathrm{d}^4 x$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$	to obtain $S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$ (where the factor
1510.06699_F00015	1	1.000	$h \equiv \operatorname{det}(\mathrm{h}_{\mathrm{a}}^{\mu})$	$h \equiv \det(h_a^{\mu})$	containing $h \equiv \det(h_a^{\mu})$ is required to make the inte-
1510.06699_F00016	2	1.000	$[\mathrm{D}_{\mathrm{a}}, \mathrm{D}_{\mathrm{b}}] \varphi$	$[\mathcal{D}_a, \mathcal{D}_b] \varphi$	ering $[\mathcal{D}_a, \mathcal{D}_b] \varphi$. This procedure yields two field strength
1510.06699_F00017	2	1.000	$\mathrm{R}^{\mathrm{a} \mathrm{~b}}_{\mathrm{cd}}(h, A, \partial A)$	$\mathcal{R}_{cd}^{ab}(h, A, \partial A)$	tensors: $\mathcal{R}_{cd}^{ab}(h, A, \partial A)$ corresponding to the gauge field
1510.06699_F00018	2	0.999	$\mathrm{T}^{\mathrm{a}}_{\mathrm{bc}}(h, \partial h, A)$	$\mathcal{T}_{bc}^a(h, \partial h, A)$	Poincaré group; and $\mathcal{T}_{bc}^a(h, \partial h, A)$ corresponding to
1510.06699_F00019	2	1.000	$S_{\mathrm{G}}=\int h^{-1} \mathrm{L}_{\mathrm{G}}(\mathrm{R}^{\mathrm{a} \mathrm{~b}}_{\mathrm{cd}}, \mathrm{T}^{\mathrm{a}}_{\mathrm{bc}}) \mathrm{d}^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a\right) d^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a\right) d^4 x$, where the Lagrangian
1510.06699_F00020	2	1.000	L_{G}	L_{G}	L_{G} is a scalar depending on the two field strengths.
1510.06699_F00021	2	1.000	$\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{\mathrm{a}}^{\mu}$	$\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$	energy-momentum $\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$ and spin-angular-
1510.06699_F00022	2	1.000	$\sigma_{\mathrm{ab}}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{\mathrm{ab}}$	$\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$	momentum $\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$ of the matter field act
1510.06699_F00023	2	1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} \mathrm{L}_{\mathrm{M}}(\varphi, \mathrm{D}_{\mathrm{a}} \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$	as sources, where $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$. In contrast to
1510.06699_F00024	2	1.000	$\mathcal{R} \equiv \mathcal{R}^{\mathrm{ab}}_{\mathrm{ab}}$	$\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$	derivatives of the gauge fields, namely $\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$. In-
1510.06699_F00025	2	1.000	$L_{\mathrm{G}} \propto \mathcal{R}$	$L_{\mathrm{G}} \propto \mathcal{R}$	deed, Kibble chose simply to set $L_{\mathrm{G}} \propto \mathcal{R}$, which defines
1510.06699_F00026	2	0.733	$\{ \}^{\{\underline{5}\}}$	<u>5</u>	transformations has been criticised by Hehl et al. ⁵ , who
1510.06699_F00027	2	1.000	$\delta \varphi \equiv \varphi^{\prime}(x^{\prime})-\varphi(x)$	$\delta \varphi \equiv \varphi^{\prime}(x')-\varphi(x)$	ble considered the ‘total’ variation $\delta \varphi \equiv \varphi^{\prime}(x')-\varphi(x)$ of
1510.06699_F00028	2	1.000	$\delta_0 \varphi \equiv \varphi^{\prime}(x)-\varphi(x)$	$\delta_0 \varphi \equiv \varphi^{\prime}(x)-\varphi(x)$	ation $\delta_0 \varphi \equiv \varphi^{\prime}(x)-\varphi(x)$. The latter approach is closer
1510.06699_F00029	2	1.000	$\delta_0 \varphi$	$\delta_0 \varphi$	matter field transformation $\delta_0 \varphi$, the partial derivative in
1510.06699_F00030	2	0.988	$\{ \}^{\{6,9\}}$	6,9	theory in Minkowski spacetime ^{6,9} , in the same way as the
1510.06699_F00031	2	0.991	$\{ \}^{\{\underline{10}\}}$	<u>10</u>	see Blagojevic ¹⁰ , in which many of the issues above are
1510.06699_F00032	2	1.000	$\mathrm{R}^{\mathrm{a} \mathrm{~b}}_{\mathrm{cd}}$	$\mathcal{R}_{cd}^{ab}$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00033	2	1.000	$\mathrm{T}^{\mathrm{a}}_{\mathrm{bc}}$	$\mathcal{T}_{bc}^a$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00034	2	1.000	$\{ \}^{\{11,12\}}$	11,12	adding the linear Hilbert-Einstein term ^{11,12} . Hence, one

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1510.06699_F00035	2	■ 1.000	$\backslash\kappa=8 \backslash\pi G / c^{\wedge\{4\}}$	$\kappa = 8\pi G/c^4$	where $\kappa = 8\pi G/c^4$ is Einstein’s gravitational constant,
1510.06699_F00036	2	■ 1.000	$\backslash\Lambda$	Λ	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00037	2	■ 1.000	a	a	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00038	3	■ 1.000	$\backslash\alpha_{\{i\}}$	α_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00039	3	■ 1.000	$\backslash\beta_{\{i\}}$	β_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00040	3	■ 0.362	$\{ \}^{\wedge\{\backslash\mathrm{underline}\{12,14\}-\backslash\mathrm{underline}\{16\}\}}$	$\underline{12,14-16}$	investigated previously ^{12,14–16} and may, in principle, be
1510.06699_F00041	3	■ 1.000	$\backslash\alpha_{\{1\}}, \backslash\alpha_{\{3\}}$	α_1, α_3	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00042	3	■ 1.000	$\backslash\alpha_{\{6\}}$	α_6	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00043	3	■ 0.999	$c=\hbar=1$	$c = \hbar = 1$	in the form (1) so that, in natural units ($c = \hbar = 1$,
1510.06699_F00044	3	■ 0.944	$\{ \}^{\wedge\{-4\}}$	-4	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00045	3	■ 0.944	$S_{\{\mathrm{G}\}}$	S_G	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00046	3	■ 1.000	$\backslash\kappa$	κ	mensionless; note that in natural units κ has dimensions
1510.06699_F00047	3	■ 0.986	$\{ \}^{\wedge\{-2\}}$	-2	of (length) ² and Λ has dimensions of (length) ^{−2} .
1510.06699_F00048	3	■ 0.585	$\{ \}^{\wedge\{19,20\}}$	19,20	suffer from Ostrogradsky’s instability ^{19,20} , wherein the
1510.06699_F00049	3	■ 1.000	$b^{\{a\}\{ \}_{\{\mu\}}}$	b^a_{μ}	b^a_{μ} of the inverse h -field) and A^{ab}_{μ} , the ‘curvature’ and
1510.06699_F00050	3	■ 1.000	h	h	b^a_{μ} of the inverse h -field) and A^{ab}_{μ} , the ‘curvature’ and
1510.06699_F00051	3	■ 1.000	$b^{\{a\}\{ \}_{\{0\}}}$	b^a_0	derivatives of b^a_0 and A^{ab}_0 ; thus 10 of the field equa-
1510.06699_F00052	3	■ 1.000	$A^{\{a \ b\}\{ \}_{\{0\}}}$	A^{ab}_0	derivatives of b^a_0 and A^{ab}_0 ; thus 10 of the field equa-
1510.06699_F00053	3	■ 1.000	$b^{\{a\}\{ \}_{\{\alpha\}}}$	b^a_{α}	remaining 30 variables b^a_{α} and A^{ab}_{α} ($\alpha = 1, 2, 3$), one
1510.06699_F00054	3	■ 1.000	$A^{\{a \ b\}\{ \}_{\{\alpha\}}(\backslash\alpha=1,2,3)$	$A^{ab}_{\alpha}(\alpha = 1, 2, 3)$	remaining 30 variables b^a_{α} and A^{ab}_{α} ($\alpha = 1, 2, 3$), one
1510.06699_F00055	3	■ 1.000	$\backslash\mathrm{cal}\{R\}+\backslash\mathrm{cal}\{T\}^{\wedge\{2\}}$	$\mathcal{R} + \mathcal{T}^2$	so-called $\mathcal{R} + \mathcal{T}^2$ theory, for which $L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})$.
1510.06699_F00056	3	■ 1.000	$L_{\{\mathrm{G}\}}=\kappa^{\wedge\{-1\}}\backslash\mathrm{left}(a \ \backslash\mathrm{cal}\{R\}+L_{\{\backslash\mathrm{cal}\{T\}^{\wedge\{2\}}\}}\backslash\mathrm{right})$	$L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})$	so-called $\mathcal{R} + \mathcal{T}^2$ theory, for which $\boxed{L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})}$.
1510.06699_F00057	3	■ 1.000	A	A	namics imposes an algebraic relation between the A -field
1510.06699_F00058	3	■ 1.000	$L_{\{\mathrm{cal}\{R\}^{\wedge\{2\}}\}}$	$L_{\mathcal{R}^2}$	in $L_{\mathcal{R}^2}$, in which case the ‘torsion’ becomes a propagating
1510.06699_F00059	3	■ 1.000	$\backslash\mathrm{cal}\{R\}+\backslash\mathrm{cal}\{R\}^{\wedge\{2\}}$	$\mathcal{R} + \mathcal{R}^2$	field. Consequently, a number of $\mathcal{R} + \mathcal{R}^2$ -type theories
1510.06699_F00060	3	■ 0.970	$\backslash\mathrm{cal}\{R\}^{\wedge\{2\}}+\backslash\mathrm{cal}\{T\}^{\wedge\{2\}}$	$\mathcal{R}^2 + \mathcal{T}^2$	have been investigated, as have $\mathcal{R}^2 + \mathcal{T}^2$ theories ²¹ . It
1510.06699_F00061	3	■ 0.986	$\backslash\mathrm{cal}\{R\}+\backslash\mathrm{cal}\{R\}^{\wedge\{2\}}+\backslash\mathrm{cal}\{T\}^{\wedge\{2\}}$	$\mathcal{R} + \mathcal{R}^2 + \mathcal{T}^2$	$\mathcal{R} + \mathcal{R}^2 + \mathcal{T}^2$ theories is given by Obukhov et al. ¹² . The
1510.06699_F00062	3	■ 1.000	$\backslash\mathrm{cal}\{R\}$	$\mathcal{R}$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00063	3	■ 1.000	$\backslash\mathrm{cal}\{T\}^{\wedge\{2\}}$	$\mathcal{T}^2$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00064	3	■ 1.000	$\backslash\mathrm{cal}\{R\}^{\wedge\{2\}}$	$\mathcal{R}^2$	quantisation. On the other hand, the $\mathcal{R}^2$ terms contain
1510.06699_F00065	3	■ 1.000	$\{ \}^{\wedge\{\backslash\mathrm{underline}\{25\}\}}$	$\underline{25}$	can eliminate such higher derivatives from the theory ²⁵ ,
1510.06699_F00066	3	■ 0.881	$\{ \}^{\wedge\{\backslash\mathrm{underline}\{26\}\}}$	$\underline{26}$	well-defined initial value problem ²⁶ . Nonetheless, the
1510.06699_F00067	3	■ 0.995	$a, \backslash\alpha_{\{i\}}$	a, α_i	that the parameters a , α_i and β_i in the Lagrangian L_G
1510.06699_F00068	3	■ 0.924	$\{ \}^{\wedge\{\backslash\mathrm{underline}\{27\}\}}$	$\underline{27}$	obey certain conditions ²⁷ .
1510.06699_F00069	4	■ 1.000	$h_{\{a\}^{\wedge\{\mu\}}}$	h_a^{μ}	fields h_a^{μ} and A^{ab}_{μ} about the trivial vacuum solution to
1510.06699_F00070	4	■ 1.000	$L_{\{\mathrm{G}\}}^{\wedge\{(2)\}}$	$L_G^{(2)}$	fields to obtain $L_G^{(2)}$. After transforming to momentum
1510.06699_F00071	4	■ 1.000	k	k	structed using k -space projection operators ²⁸ , which de-
1510.06699_F00072	4	■ 1.000	$J^{\{P\}}$	J^P	duce to states of definite spin-parity J^P in a ‘rest-frame’
1510.06699_F00073	4	■ 1.000	$k^{\{a\}}$	k^a	where k^a has only a time component, and are thus ir-
1510.06699_F00074	4	■ 0.998	$J^{\{P\}}=2^{\wedge\{+\}}$	$J^P = 2^+$	ton (with spin-parity $J^P = 2^+$) and the decomposition
1510.06699_F00075	4	■ 1.000	$J^{\{P\}}=2^{\wedge\{ \backslash\mathrm{pm}\}}, \ 1^{\wedge\{ \backslash\mathrm{pm}\}}, \ 0^{\wedge\{ \backslash\mathrm{pm}\}}$	$J^P = 2^{\pm}, 1^{\pm}, 0^{\pm}$	(or <i>rotons</i>) with spin-parities of $J^P = 2^{\pm}, 1^{\pm}, 0^{\pm}$, with
1510.06699_F00076	4	■ 1.000	$2(5+3+1)=18$	$2(5 + 3 + 1) = 18$	$\boxed{2(5 + 3 + 1) = 18}$ degrees of freedom. The propagator is
1510.06699_F00077	4	■ 1.000	S	S	field energy, which destroys the unitarity of the S matrix)
1510.06699_F00078	4	■ 1.000	$k^{\{2\}}$	k^2	k^2 , whereas the condition for no ghosts is that the ma-
1510.06699_F00079	4	■ 1.000	$-R /\backslash\mathrm{left}(k^{\{2\}}-m^{\{2\}}\backslash\mathrm{right})$	$-R/(k^2 - m^2)$	tor must be of the form $-R/(k^2 - m^2)$ with $R > 0$ and
1510.06699_F00080	4	■ 1.000	$R>0$	$R > 0$	tor must be of the form $-R/(k^2 - m^2)$ with $R > 0$ and
1510.06699_F00081	4	■ 1.000	$m^{\{2\}} \backslash\mathrm{geq} \ 0$	$m^2 \geq 0$	$m^2 \geq 0$ (assuming a ‘mostly minus’ metric signature).

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1510.06699_F00082	4	■ 1.000	{ }^{18,33}	18,33	ativity). In this case, the results are not so complete ^{18,33} .
1510.06699_F00083	4	■ 1.000	J^{P}=1^{-}	$J^P = 1^-$	that propagates (only) a massless $J^P = 1^-$ tordion and
1510.06699_F00084	4	■ 1.000	\sim 1 / k^4	$\sim 1/k^4$	as $\sim 1/k^4$, whereas those for the tordions should be
1510.06699_F00085	4	■ 1.000	\sim 1 / k^2	$\sim 1/k^2$	$\sim 1/k^2$, since tordion vertices are better behaved than
1510.06699_F00086	4	■ 1.000	{ }^{29,31,35}	29,31,35	those of gravitons ^{29,31,35} . Pure $1/k^4$ poles can occur in
1510.06699_F00087	4	■ 1.000	1 / k^4	$1/k^4$	those of gravitons ^{29,31,35} . Pure $1/k^4$ poles can occur in
1510.06699_F00088	4	■ 1.000	\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}\right)^{-1}=	$(k^2 - m_1^2)^{-1} (k^2 - m_2^2)^{-1} =$	pole structures of the form $(k^2 - m_1^2)^{-1}(k^2 - m_2^2)^{-1} =$
1510.06699_F00089	4	■ 1.000	\left(m_2^2-m_1^2\right)^{-1}\left[\left(k^2-m_2^2\right)^{-1}-\left(k^2-m_1^2\right)^{-1}\right]\right)^{-1}=	$(m_2^2 - m_1^2)^{-1} \left[(k^2 - m_2^2)^{-1} - (k^2 - m_1^2)^{-1} \right]$	$(m_2^2 - m_1^2)^{-1} [(k^2 - m_2^2)^{-1} - (k^2 - m_1^2)^{-1}]$. These can
1510.06699_F00090	4	■ 1.000	{ }^{6,39}	6,39	have been investigated ^{6,39} . More innovative approaches
1510.06699_F00091	4	■ 0.945	{ }^{40,41}	40,41	to quantisation of PGT have been pursued ^{40,41} , but are
1510.06699_F00092	4	■ 0.991	{ }^{\underline{31,43}}	<u>31,43</u>	tary and renormalizable theory ^{31,43} , without having to
1510.06699_F00093	4	■ 0.566	{ }^{44-46}	44–46	resort to non-local theories ^{44–46} . In particular, perhaps
1510.06699_F00094	4	■ 1.000	m^2 \neq 0	$m^2 \neq 0$	either continuous (if $m^2 \neq 0$) or all the masses vanish.
1510.06699_F00095	4	■ 0.838	\mathrm{C}(1,3)^{\underline{48-50}}	$C(1,3)^{48-50}$	of the full conformal group $C(1,3)^{48-50}$. This may be for-
1510.06699_F00096	5	■ 1.000	S_{\mathrm{M}}=\int L_{\mathrm{M}}(\varphi,\partial_{\mu}\varphi)\,d^4x	$S_M = \int L_M(\varphi, \partial_\mu \varphi) d^4x$	ter action $S_M = \int L_M(\varphi, \partial_\mu \varphi) d^4x$ that is invariant under
1510.06699_F00097	5	■ 1.000	B_{\mu}	B_μ	a new gravitational gauge field B_μ corresponding to the
1510.06699_F00098	5	■ 0.870	{ }^{51-53}	51–53	dilation part of the Weyl group ^{51–53} . Similar to PGT, the
1510.06699_F00099	5	■ 1.000	h_a{}^{\mu}, A^{ab}{}_{\mu}	$h_a{}^\mu, A^{ab}{}_\mu$	gravitational gauge fields $h_a{}^\mu$, $A^{ab}{}_\mu$ and B_μ are used to
1510.06699_F00100	5	■ 0.991	\mathcal{D}_a{}^*\varphi	$\mathcal{D}_a^* \varphi$	assemble a covariant derivative $\mathcal{D}_a^* \varphi$, and the matter ac-
1510.06699_F00101	5	■ 1.000	S_{\mathrm{M}}=\int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a{}^*\varphi)\,d^4x	$S_M = \int h^{-1} L_M(\varphi, \mathcal{D}_a^* \varphi) d^4x$	coupling procedure, so that $S_M = \int h^{-1} L_M(\varphi, \mathcal{D}_a^* \varphi) d^4x$.
1510.06699_F00102	5	■ 1.000	L_{\mathrm{M}}	L_M	cise form of L_M . In particular, since h^{-1} has a Weyl
1510.06699_F00103	5	■ 1.000	h^{-1}	h^{-1}	cise form of L_M . In particular, since h^{-1} has a Weyl
1510.06699_F00104	5	■ 0.788	w\left(h^{-1}\right)=4	$w(h^{-1}) = 4$	(or conformal) weight $w(h^{-1}) = 4$ (see Section II B), the
1510.06699_F00105	5	■ 1.000	w\left(L_{\mathrm{G}}\right)=-4	$w(L_G) = -4$	Lagrangian L_M must have a weight $w(L_G) = -4$.
1510.06699_F00106	5	■ 1.000	\left[\mathcal{D}_a{}^*,\mathcal{D}_b{}^*\right]\varphi	$[\mathcal{D}_a^*, \mathcal{D}_b^*] \varphi$	by considering $[\mathcal{D}_a^*, \mathcal{D}_b^*] \varphi$. One obtains $\mathcal{R}^{ab}{}_{cd}(h, A, \partial A)$
1510.06699_F00107	5	■ 1.000	\mathcal{T}^{*a}{}_{bc}(h,\partial h,A,B)	$\mathcal{T}^{*a}{}_{bc}(h, \partial h, A, B)$	and $\mathcal{T}^{*a}{}_{bc}(h, \partial h, A, B)$ as before (but where the form
1510.06699_F00108	5	■ 1.000	\mathcal{H}_{ab}(h,\partial B)	$\mathcal{H}_{ab}(h, \partial B)$	tional field strength $\mathcal{H}_{ab}(h, \partial B)$ corresponding to the new
1510.06699_F00109	5	■ 1.000	S_{\mathrm{G}}=	$S_G =$	free gravitational action then has the general form $S_G =$
1510.06699_F00110	5	■ 0.999	\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}^{ab}{}_{cd},\mathcal{T}^{*a}{}_{bc},\mathcal{H}_{ab}\right)d^4x	$\int h^{-1} L_G(\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^{*a}{}_{bc}, \mathcal{H}_{ab}) d^4x$	$\int h^{-1} L_G(\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^{*a}{}_{bc}, \mathcal{H}_{ab}) d^4x$. As usual, the total ac-
1510.06699_F00111	5	■ 1.000	\zeta^{\mu}\equiv\delta\mathcal{L}_{\mathrm{M}}/\delta B_{\mu}	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	$\sigma_{ab}{}^\mu \equiv \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ and dilation current $\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$
1510.06699_F00112	5	■ 1.000	w\left(\mathcal{R}^{ab}{}_{cd}\right)=w\left(\mathcal{H}_{ab}\right)=-2	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{ab}) = -2$	$w(\mathcal{R}^{ab}{}_{cd}) \equiv w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^{*a}{}_{bc}) = -1$, which
1510.06699_F00113	5	■ 1.000	w\left(\mathcal{T}^{*a}{}_{bc}\right)=-1	$w(\mathcal{T}^{*a}{}_{bc}) = -1$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^{*a}{}_{bc}) = -1$, which
1510.06699_F00114	5	■ 1.000	\mathcal{H}_{ab}	$\mathcal{H}_{ab}$	means that L_G can be quadratic in $\mathcal{R}^{ab}{}_{cd}$ and $\mathcal{H}_{ab}$, while
1510.06699_F00115	5	■ 1.000	\mathcal{T}^{*a}{}_{bc}	$\mathcal{T}^{*a}{}_{bc}$	terms linear in $\mathcal{R} \equiv \mathcal{R}^{ab}{}_{ab}$ or quadratic in $\mathcal{T}^{*a}{}_{bc}$ are
1510.06699_F00116	5	■ 1.000	\mathcal{R}^2+\mathcal{H}^2	$\mathcal{R}^2 + \mathcal{H}^2$	in the first derivatives of gauge fields, is of $\mathcal{R}^2 + \mathcal{H}^2$ type:
1510.06699_F00117	5	■ 1.000	\xi	ξ	where the α_i and ξ are dimensionless free parameters.
1510.06699_F00118	5	■ 1.000	\mathcal{T}^{*2}	$\mathcal{T}^{*2}$	in PGT; essentially the $\mathcal{R}$ and $\mathcal{T}^{*2}$ terms, which possess
1510.06699_F00119	5	■ 1.000	w=-4	$w = -4$	tal Lagrangian must have a Weyl weight $w \equiv -4$ clearly
1510.06699_F00120	5	■ 0.999	\phi	ϕ	ϕ with Weyl weight $w(\phi) = -1$, often termed the

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1510.06699_F00121	5	■ 0.999	$w(\phi)=-1$	$w(\phi) = -1$	ϕ with Weyl weight $w(\phi) = -1$, often termed the
1510.06699_F00122	5	■ 0.785	$(\mathrm{s})^{\underline{10}}$	$(s)^{10}$	compensator $(s)^{10}$. This opens up possibilities for the in-
1510.06699_F00123	5	■ 0.803	$\phi^2 \mathcal{R}$	$\phi^2 \mathcal{R}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00124	5	■ 0.803	$\phi^2 L_{\mathcal{T}^{*2}}$	$\phi^2 L_{\mathcal{T}^{*2}}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00125	5	■ 1.000	$\{ \}^{54,55}$	54,55	been widely investigated ^{54,55} , whereas the phenomeno-
1510.06699_F00126	5	■ 0.999	$\phi^2 L_{\mathcal{T}^{*2}}$	$\phi^2 L_{\mathcal{T}^{*2}}$	logical value of the $\phi^2 L_{\mathcal{T}^{*2}}$ terms has received relatively
1510.06699_F00127	5	■ 0.969	$\{ \}^{56,57}$	56,57	little attention ^{56,57} . In any case, the resulting total La-
1510.06699_F00128	5	■ 1.000	ψ	ψ	tion for a free Dirac field ψ , which has Weyl weight
1510.06699_F00129	5	■ 1.000	$w(\psi)=w(\bar{\psi})=-\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$. This action is not scale-invariant
1510.06699_F00130	5	■ 1.000	$m \bar{\psi} \psi$	$m \bar{\psi} \psi$	owing to the mass term $m \bar{\psi} \psi$, so it appears that one re-
1510.06699_F00131	5	■ 0.983	$\{ \}^{10} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	$^{10} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	vergence of the dilation current ¹⁰ $\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$. In
1510.06699_F00132	6	■ 1.000	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00133	6	■ 1.000	μ	μ	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00134	6	■ 1.000	$\mu \phi$	$\mu \phi$	is a dimensionless parameter but $\mu \phi$ has the dimensions
1510.06699_F00135	6	■ 1.000	G	G	‘constant’ G to vary with cosmic epoch. This idea can
1510.06699_F00136	6	■ 1.000	$\phi = \phi_0$	$\phi = \phi_0$	stein gauge. Setting $\phi = \phi_0$ in the equation of motion
1510.06699_F00137	6	■ 1.000	$m = \mu \phi_0$	$m = \mu \phi_0$	tation as a massive field with $m = \mu \phi_0$. It is usually
1510.06699_F00138	6	■ 1.000	w	w	$h_a{}^\mu$ and $A^a{}_\mu$ transform covariantly with weights of w ,
1510.06699_F00139	6	■ 1.000	$\mathcal{R}_{abcd}$	$\mathcal{R}_{abcd}$	strength (or ‘curvature’) $\mathcal{R}_{abcd}$ and translational gauge
1510.06699_F00140	6	■ 1.000	$\mathcal{T}_{abc}$	$\mathcal{T}_{abc}$	field strength (or ‘torsion’) $\mathcal{T}_{abc}$ are treated in a more
1510.06699_F00141	6	■ 0.617	$\{ \}^{59-62,89}$	$^{59-62,89}$	so is the total action) ^{59-62,89} .
1510.06699_F00142	6	■ 1.000	Y_4	Y_4	Weyl–Cartan Y_4 spacetime (although we do give a brief
1510.06699_F00143	7	■ 1.000	$\widehat{Y}_4$	$\widehat{Y}_4$	a new spacetime geometry (that we term $\widehat{Y}_4$), which is
1510.06699_F00144	7	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	We begin with a Minkowski spacetime $\mathcal{M}$, labelled
1510.06699_F00145	7	■ 1.000	x^μ	x^μ	using Cartesian inertial coordinates x^μ , on which the
1510.06699_F00146	7	■ 1.000	$\varphi(x)$	$\varphi(x)$	gravitational) fields $\varphi(x)$ is described by a matter action
1510.06699_F00147	7	■ 0.959	e_μ	e_μ	e_μ , which are tangents to the coordinate curves and for
1510.06699_F00148	7	■ 0.997	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	Cartesian inertial coordinates satisfy $e_\mu \cdot e_\nu = \eta_{\mu\nu}$, where
1510.06699_F00149	7	■ 1.000	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. At each point of $\mathcal{M}$, we can
1510.06699_F00150	7	■ 1.000	$\hat{e}_a(x)$	$\hat{e}_a(x)$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00151	7	■ 1.000	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00152	7	■ 1.000	$e_\mu(x)$	$e_\mu(x)$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00153	7	■ 1.000	$\hat{e}_a = \delta_a^\mu e_\mu$	$\hat{e}_a = \delta_a^\mu e_\mu$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00154	7	■ 1.000	$\text{SO}(3, 1)$	$\text{SO}(3, 1)$	$\text{SO}(3, 1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00155	7	■ 1.000	$\text{SL}(2, C)$	$\text{SL}(2, C)$	$\text{SO}(3, 1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00156	7	■ 1.000	$\{\varphi^a(x)\}$	$\{\varphi^a(x)\}$	ter field components $\{\varphi^a(x)\}$ simply carry a generic Ro-
1510.06699_F00157	8	■ 0.999	$\text{S}(\Lambda)$	$\text{S}(\Lambda)$	where $\text{S}(\Lambda)$ is the matrix corresponding to the element Λ
1510.06699_F00158	8	■ 1.000	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00159	8	■ 1.000	$a^\mu = 0$	$a^\mu = 0$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00160	8	■ 1.000	e^ρ	e^ρ	tor e^ρ . Adopting global inertial coordinates, such that
1510.06699_F00161	8	■ 0.999	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, the metric must therefore transform
1510.06699_F00162	8	■ 1.000	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	according to $\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$. Hence, under this
1510.06699_F00163	8	■ 1.000	$w=2$	$w = 2$	active dilation the metric has a Weyl weight of $w = 2$.
1510.06699_F00164	8	■ 1.000	$x'^\mu = e^\rho x^\mu$	$x'^\mu = e^\rho x^\mu$	ordinate transformation $x'^\mu = e^\rho x^\mu$, which ‘shrinks’ the

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1510.06699_F00165	8	■ 1.000	$\gamma_{\mu \nu}^{\prime} = e^{-2 \rho} \gamma_{\mu \nu} = \eta_{\mu \nu}$	$\gamma'_{\mu\nu} = e^{-2\rho}\gamma_{\mu\nu} = \eta_{\mu\nu}$	coordinate transformation, $\gamma'_{\mu\nu} = e^{-2\rho}\gamma_{\mu\nu} = \eta_{\mu\nu}$ and so,
1510.06699_F00166	8	■ 1.000	$e^{4 \rho}$	$e^{4\rho}$	transformation (6) (for which the Jacobian equals $e^{4\rho}$), a
1510.06699_F00167	8	■ 1.000	m	m	m , one has the Klein–Gordon Lagrangian
1510.06699_F00168	8	■ 1.000	$m=0$	$m=0$	dilation invariant only for $\overline{m}=0$ and provided the Weyl
1510.06699_F00169	8	■ 1.000	$\bar{\psi} \psi$	$\bar{\psi}\psi$	pretation of $\bar{\psi}\psi$ as a number density. As mentioned in the
1510.06699_F00170	8	■ 1.000	A_{μ}	A_{μ}	Finally, we consider the electromagnetic field A_{μ}
1510.06699_F00171	8	■ 1.000	$F_{\mu \nu} \equiv \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$	$F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$	where $\overline{F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}}$ and J^{μ} is some source cur-
1510.06699_F00172	8	■ 1.000	J^{μ}	J^{μ}	where $F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$ and J^{μ} is some source cur-
1510.06699_F00173	8	■ 1.000	w_J	w_J	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00174	8	■ 1.000	J_{μ}	J_{μ}	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00175	8	■ 1.000	$w(A_{\mu})=-1$	$w(A_{\mu})=-1$	(9), provided that $w(A_{\mu})=-1$ and $w(J_{\mu})=-3$. It is
1510.06699_F00176	8	■ 1.000	$w(J_{\mu})=-3$	$w(J_{\mu})=-3$	(9), provided that $w(A_{\mu})=-1$ and $w(J_{\mu})=-3$. It is
1510.06699_F00177	8	■ 1.000	$J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi$	$J^{\mu} \equiv \bar{\psi}\gamma^{\mu}\psi$	worth noting that the Dirac current $\overline{J^{\mu} \equiv \bar{\psi}\gamma^{\mu}\psi}$ satisfies
1510.06699_F00178	9	■ 1.000	a^{μ}	a^{μ}	the four translation parameters a^{μ} and the dilation pa-
1510.06699_F00179	9	■ 1.000	ρ	ρ	rameter ρ into eleven independent arbitrary functions of
1510.06699_F00180	9	■ 1.000	x	x	x . This leads to a complete decoupling of the translation
1510.06699_F00181	9	■ 1.000	$x^{\prime}$	x'	lation parts. Thus, it is customary to view x' as arising
1510.06699_F00182	9	■ 1.000	$\Lambda(x)$	$\Lambda(x)$	local Lorentz rotation $\Lambda(x)$ of the orthonormal tetrad
1510.06699_F00183	9	■ 0.997	$\rho(x)$	$\rho(x)$	$\hat{e}_a(x)$, or local (physical) dilation $\rho(x)$, at each point.
1510.06699_F00184	9	■ 1.000	$\Lambda^a{}_b(x)$	$\Lambda^a{}_b(x)$	Lorentz rotations $\Lambda^a{}_b(x)$. This means that any spinor,
1510.06699_F00185	9	■ 1.000	$\partial_{\mu} \varphi$	$\partial_{\mu}\varphi$	field. Consequently, the partial derivative, $\partial_{\mu}\varphi$, is co-
1510.06699_F00186	9	■ 1.000	$\eta_{a b}=\eta_{a b}=\operatorname{diag}(1,-1,-1,-1)$	$\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$	ric $\eta_{\overline{ab}} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$, whereas Greek in-
1510.06699_F00187	9	■ 1.000	$\gamma_{\mu \nu}=\boldsymbol{e}_{\mu} \cdot \boldsymbol{e}_{\nu}$	$\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00188	9	■ 1.000	$\gamma^{\mu \nu}$	$\gamma^{\mu\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00189	9	■ 1.000	$\hat{e}^a=\eta^{a b} \hat{e}_b$	$\hat{e}^a = \eta^{ab}\hat{e}_b$	by $\hat{\overline{e}}^a = \eta^{ab}\hat{\overline{e}}_b$ and $e^{\mu} = \gamma^{\mu\nu}e_{\nu}$, respectively. Moreover,
1510.06699_F00190	9	■ 1.000	$e^{\mu}=\gamma^{\mu \nu} e_{\nu}$	$e^{\mu} = \gamma^{\mu\nu}e_{\nu}$	by $\hat{e}^a = \eta^{ab}\hat{e}_b$ and $\overline{e^{\mu} = \gamma^{\mu\nu}e_{\nu}}$, respectively. Moreover,
1510.06699_F00191	9	■ 1.000	$\gamma_{\mu \nu}$	$\gamma_{\mu\nu}$	metric $\gamma_{\mu\nu}$ of the Minkowski spacetime obeys the trans-
1510.06699_F00192	9	■ 1.000	(Λ, ρ)	(Λ, ρ)	two steps. Firstly, one defines the (Λ, ρ) -covariant deriva-
1510.06699_F00193	9	■ 1.000	$D_{\mu} \varphi(x)$	$D_{\mu}\varphi(x)$	where $D_{\mu}\varphi(x)$ is the Λ -covariant derivative introduced
1510.06699_F00194	9	■ 1.000	$A^{a b}{}_{\mu}(x)$	$A^{ab}{}_{\mu}(x)$	of complex conjugation. Most importantly, $\overline{A^{ab}{}_{\mu}(x)}$ is
1510.06699_F00195	9	■ 1.000	$B_{\mu}(x)$	$B_{\mu}(x)$	tions and $B_{\mu}(x)$ is the gauge-field corresponding to local
1510.06699_F00196	9	■ 1.000	$\Sigma_{a b}=-\Sigma_{a b}$	$\Sigma_{ab} = -\Sigma_{ab}$	dilations. The matrices $\overline{\Sigma_{ab} = -\Sigma_{ab}}$ are the generators of
1510.06699_F00197	9	■ 0.483	$(2, C)$	$(2, C)$	the $\text{SL}(\overline{2, C})$ representation $\text{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00198	9	■ 0.483	(Λ)	(Λ)	the $\text{SL}(2, C)$ representation $\text{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00199	9	■ 1.000	$\Sigma_{a b}$	Σ_{ab}	Given the antisymmetry of the generator matrices Σ_{ab} ,
1510.06699_F00200	9	■ 1.000	$A^{a b}{}_{\mu}=$	$A^{ab}{}_{\mu} =$	to share this antisymmetry property, such that $A^{ab}{}_{\mu} =$
1510.06699_F00201	9	■ 0.890	$-A^{b a}{}_{\mu}$	$-A^{ba}{}_{\mu}$	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00202	9	■ 0.890	B	B	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00203	9	■ 1.000	$w=0$	$w=0$	local dilations (i.e. has Weyl weight $\overline{w}=0$).
1510.06699_F00204	10	■ 1.000	$\partial_{\mu}^* \varphi(x)$	$\partial_{\mu}^*\varphi(x)$	It is straightforward to show that $\partial_{\mu}^*\varphi(x)$ itself also trans-
1510.06699_F00205	10	■ 0.999	$D_{\mu}^* \varphi$	$D_{\mu}^*\varphi$	derivative, linearly related to $D_{\mu}^*\varphi$, by introducing a new,
1510.06699_F00206	10	■ 1.000	$h_a{}^{\mu}(x)$	$h_a{}^{\mu}(x)$	‘translational’ gauge field $h_a{}^{\mu}(x)$ that transforms via
1510.06699_F00207	10	■ 1.000	$w=-1$	$w=-1$	Lorentz four-vector, and has Weyl weight $\overline{w}=-1$ under
1510.06699_F00208	10	■ 1.000	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$ (which has Weyl weight $w=1$) and satisfies

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1510.06699_F00209	10	■ 1.000	w=1	$w = 1$	$b^a{}_\mu(x)$ (which has Weyl weight $\overline{w=1}$) and satisfies
1510.06699_F00210	10	■ 1.000	$\partial \varphi$	$\partial \varphi$	mations in the same way as $\partial \varphi$ does under global Weyl
1510.06699_F00211	10	■ 1.000	$x \rightarrow x'$	$x \rightarrow x'$	$w = -4$. However, since $x \rightarrow x'$ is a GCT, we need a
1510.06699_F00212	10	■ 0.997	$L_{\mathrm{M}}(\varphi; \mathcal{D}^* \varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^* \varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^* \varphi)$ by a factor which transforms by the appro-
1510.06699_F00213	10	■ 1.000	w=4	$w = 4$	weight $w = 4$. From the transformation properties of the
1510.06699_F00214	10	■ 1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$. We
1510.06699_F00215	10	■ 0.999	h, A	h, A	note that the gravitational gauge fields h, A and B con-
1510.06699_F00216	10	■ 0.613	$L_{\mathcal{T}^{*2}}$	$L_{\mathcal{T}^{*2}}$	weight $w = -4$ and so may be added to L_{M} (where $L_{\mathcal{T}^{*2}}$
1510.06699_F00217	10	■ 0.999	D_μ^*	D_μ^*	choice, one may extend the (Λ, ρ) -covariant derivative D_μ^*
1510.06699_F00218	10	■ 1.000	$(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	$(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	where $(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$ are the generators of the
1510.06699_F00219	10	■ 0.509	C	C	der GCT, in addition to $\mathrm{SL}(2, C)$ behaviour. In this case,
1510.06699_F00220	11	■ 1.000	$\{ \}^{\{0\}} \nabla_{\mu} \equiv \{ \}^{\{0\}} \nabla_{\mu} + w B_{\mu}, \{ \}^{\{0\}} \Gamma_{\mu\nu}^{\lambda} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma_{\mu\nu}^{\lambda} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	where ${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma_{\mu\nu}^{\lambda} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$
1510.06699_F00221	11	■ 0.709	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu}$) is the metric connection corresponding to the
1510.06699_F00222	11	■ 0.991	X^{ρ}_{σ}	X^{ρ}_{σ}	and X^{ρ}_{σ} are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00223	11	■ 0.991	$\mathrm{GL}(4, R)$	$\mathrm{GL}(4, R)$	and X^{ρ}_{σ} are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00224	11	■ 1.000	$\Gamma^{\sigma}_{\rho\mu}$	${}^0\Gamma^{\sigma}_{\rho\mu}$	spacetime, the connection coefficients ${}^0\Gamma^{\sigma}_{\rho\mu}$ are not dy-
1510.06699_F00225	11	■ 1.000	∇_{μ}	${}^0\nabla_{\mu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00226	11	■ 1.000	∇_{ν}	${}^0\nabla_{\nu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00227	11	■ 1.000	$\boldsymbol{h}$	$\boldsymbol{h}$	rank-2 tensor field, which we will denote by $\boldsymbol{h}$, defined on
1510.06699_F00228	11	■ 1.000	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	by $\overline{h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})}$. Similarly, the inverse of $\boldsymbol{h}$, which
1510.06699_F00229	11	■ 1.000	$\boldsymbol{b}$	$\boldsymbol{b}$	we denote by $\boldsymbol{b}$, is also a rank-2 tensor field defined on
1510.06699_F00230	11	■ 1.000	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	considered so far are given by $\overline{b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})}$. Other
1510.06699_F00231	11	■ 1.000	$\boldsymbol{J}$	$\boldsymbol{J}$	field $\boldsymbol{J}$ defined on the Minkowski spacetime, which may
1510.06699_F00232	11	■ 1.000	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	basis or its dual as $\overline{\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}}$. Replacing the
1510.06699_F00233	11	■ 1.000	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\boldsymbol{h}$, we may obtain the quantities $\overline{\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}}$ or
1510.06699_F00234	11	■ 1.000	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\overline{\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}}$. Adopting the viewpoint that $\boldsymbol{h}$ is
1510.06699_F00235	11	■ 0.967	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	basis of a <i>new</i> vector field $\overline{\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}}$. It is also worth
1510.06699_F00236	11	■ 0.820	$\boldsymbol{h}(\mathcal{J}, \cdot)$	$\boldsymbol{h}(\mathcal{J}, \cdot)$	the vector $\overline{\boldsymbol{h}(\mathcal{J}, \cdot)}$, which has contravariant components
1510.06699_F00237	11	■ 1.000	$h_a{}^{\mu} \mathcal{J}^a$	$h_a{}^{\mu} \mathcal{J}^a$	in the coordinate basis given by $h_a{}^{\mu} \mathcal{J}^a$; these are the
1510.06699_F00238	11	■ 1.000	$\mathcal{J}$	$\mathcal{J}$	example, if we place $\mathcal{J}$ into the first ‘slot’ of $\boldsymbol{b}$, we re-
1510.06699_F00239	11	■ 0.690	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	cover the <i>original</i> vector $\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$, since its covari-
1510.06699_F00240	11	■ 1.000	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	ant components in the coordinate basis are $\overline{b^a{}_{\mu} \mathcal{J}_a = J_{\mu}}$,
1510.06699_F00241	11	■ 1.000	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$, and their contravariant counterparts as
1510.06699_F00242	11	■ 1.000	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$\overline{J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a}$. In all these cases, the compo-
1510.06699_F00243	11	■ 1.000	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	tion: consider, for example, the vector $\boldsymbol{b}(\cdot, \boldsymbol{J})$; this has
1510.06699_F00244	11	■ 1.000	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$, but these are <i>not</i> , in general, equal to $\mathcal{J}^a$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00285	13	■ 0.990	$t^{\{d\}}$	t^d	$\epsilon_{abcd}t^d$ for some vector t^d , then the second term on the
1510.06699_F00286	13	■ 0.999	d	d	By contracting over the indices a and d in the ‘ $\mathcal{R}$ -
1510.06699_F00287	13	■ 1.000	b	b	Contracting again over b and e , one then finds the twice-
1510.06699_F00288	13	■ 1.000	e	e	Contracting again over b and e , one then finds the twice-
1510.06699_F00289	13	■ 0.770	$\mathrm{mathcal{T}}_{\{a\}}^{\{*\}}$	$\mathcal{T}_a^*$	torsion trace $\mathcal{T}_a^*$ vanishes, which clearly includes the spe-
1510.06699_F00290	13	■ 1.000	$\mathrm{mathcal{H}}$	$\mathcal{H}$	It is clear that the ‘ $\mathcal{H}$ -identity’ (53) has no non-trivial
1510.06699_F00291	13	■ 1.000	$\mathrm{mathcal{R}}_{\{a\ b\ c\ d\}}, \mathrm{mathcal{T}}_{\{a\ b\ c\}}^{\{*\}}$	$\mathcal{R}_{abcd}, \mathcal{T}_{abc}^*$	itational gauge fields. From $\mathcal{R}_{abcd}$, $\mathcal{T}_{abc}^*$ and $\mathcal{H}_{ab}$, one can
1510.06699_F00292	13	■ 1.000	$\mathrm{mathcal{L}}_{\{\mathrm{mathrm{G}}\}} \equiv h^{-1} L_{\mathrm{G}}$	$\mathcal{L}_G \equiv h^{-1} L_G$	sity $\mathcal{L}_G \equiv h^{-1} L_G$ has weight zero (i.e scale invari-
1510.06699_F00293	13	■ 1.000	$\mathrm{mathcal{R}}_{\{a\ b\ c\ d\}}, \mathrm{mathcal{H}}_{\{a\ b\}}$	$\mathcal{R}_{abcd}, \mathcal{H}_{ab}$	ant). It is trivial to show that $\mathcal{R}_{abcd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}_{abc}^*$
1510.06699_F00294	13	■ 1.000	$w(\mathrm{mathcal{R}}_{\{a\ b\ c\ d\}})=w(\mathrm{mathcal{H}}_{\{a\ b\}})=-2$	$w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00295	13	■ 1.000	$w(\mathrm{mathcal{T}}_{\{a\ b\ c\}}^{\{*\}})= -1$	$w(\mathcal{T}_{abc}^*) = -1$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00296	13	■ 0.988	$\{ \}^{\{\underline{70}\}}$	$\overline{70}$	respectively ⁷⁰ . Hence L_G can be quadratic in $\mathcal{R}_{abcd}$ and
1510.06699_F00297	13	■ 0.994	$L_{\{\mathrm{mathcal{H}}^{\{2\}}\}}$	$L_{\mathcal{H}^2}$	where the expressions for $L_{\mathcal{R}^2}$ and $L_{\mathcal{H}^2}$ are given in (4).
1510.06699_F00298	13	■ 1.000	$D \leq 4$	$D \leq 4$	contributes a total derivative to the action (in $D \leq 4$
1510.06699_F00299	13	■ 0.994	$\mathrm{mathcal{T}}_{\{a\ b\ c\}} \rightarrow \mathrm{mathcal{T}}_{\{a\ b\ c\}}^{\{*\}}$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$. However, since these terms depend upon the
1510.06699_F00300	13	■ 1.000	$\mathrm{mathcal{L}}_{\{\mathrm{mathrm{G}}\}}$	$\mathcal{L}_G$	sity $\mathcal{L}_G$. As an illustration, in Appendix D we give a
1510.06699_F00301	13	■ 1.000	$L_{\{\mathrm{mathrm{G}}\}}=L_{\{\mathrm{mathcal{H}}^{\{2\}}\}}$	$L_G = L_{\mathcal{H}^2}$	$L_G = L_{\mathcal{H}^2}$ and L_M contains a term proportional to $\phi^2 \mathcal{R}$.
1510.06699_F00302	13	■ 1.000	$S_{\{\mathrm{mathrm{T}}\}}$	S_T	The total action S_T is simply the sum of the matter and
1510.06699_F00303	13	■ 1.000	$h, A, \partial A$	$h, A, \partial A$	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00304	13	■ 1.000	∂B	∂B	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00305	13	■ 1.000	$\varphi, \partial \varphi, h, A$	$\varphi, \partial \varphi, h, A$	φ induce a dependence on $\varphi, \partial \varphi, h, A$ and B . We will
1510.06699_F00306	13	■ 1.000	∂A	∂A	$\phi^2 \mathcal{R}$, then this brings an additional dependence on ∂A .
1510.06699_F00307	13	■ 0.901	$\phi^2 L_{\{\mathrm{mathcal{T}}^{\{2\}}\}}$	$\phi^2 L_{\mathcal{T}^{\{2\}}}$	Finally, if one includes terms of the generic form $\phi^2 L_{\mathcal{T}^{\{2\}}}$,
1510.06699_F00308	13	■ 1.000	∂h	∂h	then it produces an additional dependence on ∂h . Con-
1510.06699_F00309	14	■ 1.000	$\mathcal{L}_T$	$\mathcal{L}_T$	Since $\mathcal{L}_T$ is at most quadratic in the field strength ten-
1510.06699_F00310	14	■ 1.000	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	sors $\mathcal{R}_{abcd}(h, A, \partial A)$, $\mathcal{T}_{abc}^*(h, \partial h, A, B)$ and $\mathcal{H}_{ab}(h, \partial B)$,
1510.06699_F00311	14	■ 1.000	$t^a_{\{\} \}_{\{\mu\}} \equiv$	$t^a_{\mu} \equiv$	In the free gravitational sector, we have defined $t^a_{\mu} \equiv$
1510.06699_F00312	14	■ 0.999	$\delta \mathcal{L}_G / \delta h_a^{\mu} = \partial \mathcal{L}_G / \partial h_a^{\mu}, s_{ab}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}_{\mu}$	$\delta \mathcal{L}_G / \delta h_a^{\mu} = \partial \mathcal{L}_G / \partial h_a^{\mu}, s_{ab}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}_{\mu}$	$\delta \mathcal{L}_G / \delta h_a^{\mu} = \partial \mathcal{L}_G / \partial h_a^{\mu}, s_{ab}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}_{\mu}$ and
1510.06699_F00313	14	■ 1.000	$j^{\mu} \equiv \delta \mathcal{L}_G / \delta B_{\mu}$	$j^{\mu} \equiv \delta \mathcal{L}_G / \delta B_{\mu}$	$j^{\mu} \equiv \delta \mathcal{L}_G / \delta B_{\mu}$, respectively. In the matter sector,
1510.06699_F00314	14	■ 1.000	$\tau^a_{\{\} \}_{\{\mu\}} \equiv \delta \mathcal{L}_M / \delta h_a^{\mu}$	$\tau^a_{\mu} \equiv \delta \mathcal{L}_M / \delta h_a^{\mu}$	the energy-momentum $\tau^a_{\mu} \equiv \delta \mathcal{L}_M / \delta h_a^{\mu}$, spin-angular-
1510.06699_F00315	14	■ 0.692	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta B_{\mu} = \partial \mathcal{L}_M / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta B_{\mu} = \partial \mathcal{L}_M / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta B_{\mu} = \partial \mathcal{L}_M / \partial B_{\mu}$ act as sources ⁷² . It is easy
1510.06699_F00316	14	■ 0.692	$\{ \}^{\{\underline{72}\}}$	$\overline{72}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta B_{\mu} = \partial \mathcal{L}_M / \partial B_{\mu}$ act as sources ⁷² . It is easy
1510.06699_F00317	14	■ 1.000	$w=1, w=0$	$w=1, w=0$	$w=1, w=0$ and $w=0$, respectively. It is, in fact, more
1510.06699_F00318	14	■ 1.000	$t^a_b \equiv t^a_{\mu} h_b^{\mu}, s_{ab}^c \equiv s_{ab}^{\mu} b^c_{\mu}$	$t^a_b \equiv t^a_{\mu} h_b^{\mu}, s_{ab}^c \equiv s_{ab}^{\mu} b^c_{\mu}$	where $t^a_b \equiv t^a_{\mu} h_b^{\mu}, s_{ab}^c \equiv s_{ab}^{\mu} b^c_{\mu}$ and $j^a \equiv j^{\mu} b^a_{\mu}$, and
1510.06699_F00319	14	■ 1.000	$j^a \equiv j^{\mu} b^a_{\mu}$	$j^a \equiv j^{\mu} b^a_{\mu}$	where $t^a_b \equiv t^a_{\mu} h_b^{\mu}, s_{ab}^c \equiv s_{ab}^{\mu} b^c_{\mu}$ and $j^a \equiv j^{\mu} b^a_{\mu}$, and
1510.06699_F00320	14	■ 0.992	$\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	$\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	where $\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$ so that φ
1510.06699_F00321	14	■ 1.000	$\mathcal{D}_{\mu}^* \varphi$	$\mathcal{D}_{\mu}^* \varphi$	and $\mathcal{D}_{\mu}^* \varphi$ are treated as independent variables, rather
1510.06699_F00322	14	■ 1.000	$\mathcal{L}_M = h^{-1} L_M$	$\mathcal{L}_M = h^{-1} L_M$	where we have defined $\mathcal{L}_M = h^{-1} L_M$ and $\bar{\partial} L_M / \partial \varphi \equiv$

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1510.06699_F00323	14	■ 1.000	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$	where we have defined $\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}}$ and $\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$
1510.06699_F00324	14	■ 0.974	$\left[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi \right]_{u=\varphi}$	$[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi]_{u=\varphi}$	$\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi _{u=\varphi}$, so that φ and $\mathcal{D}_a^* \varphi$ are treated
1510.06699_F00325	14	■ 1.000	S_{M}	S_{M}	the minimal-coupling procedure applied to the action S_{M} ,
1510.06699_F00326	14	■ 0.999	$\{ \}^{\{9,74\}}$	$9,74$	equations directly ^{9,74} , which would be given by (67a) and
1510.06699_F00327	14	■ 0.963	$2, \mathrm{C}$	$2, C$	S_{G} under (infinitesimal) local Lorentz (or $\mathrm{SL}(2, C)$) rota-
1510.06699_F00328	15	■ 0.718	$(69 \mathrm{~}\mathrm{c})$	$(69\mathrm{c})$	$(69\mathrm{c})$
1510.06699_F00329	15	■ 1.000	$w = -\frac{3}{2}$	$w = -\frac{3}{2}$	$w = -\frac{3}{2}$ and $w = -1$, respectively; we note that the
1510.06699_F00330	15	■ 1.000	$\psi \rightarrow \psi e^{i\alpha}$	$\psi \rightarrow \psi e^{i\alpha}$	mation $\psi \rightarrow \psi e^{i\alpha}$.
1510.06699_F00331	15	■ 1.000	γ	γ	where, on each γ -matrix, one has simply replaced the
1510.06699_F00332	15	■ 1.000	$\bar{\psi}$	$\bar{\psi}$	that, since both ψ and $\bar{\psi}$ have Weyl weight $w = -\frac{3}{2}$, the
1510.06699_F00333	15	■ 1.000	$\mathcal{D}_a$	$\mathcal{D}_a$	term in (71) can be replaced by its PGT counterpart $\mathcal{D}_a$,
1510.06699_F00334	15	■ 1.000	$\mathcal{D}_a^* \psi$	$\mathcal{D}_a^* \psi$	varying the action with respect to ψ . Since $\mathcal{D}_a^* \psi$ and $\mathcal{T}_a^*$
1510.06699_F00335	15	■ 1.000	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$ and $w = -1$ respectively, we see immediately
1510.06699_F00336	15	■ 1.000	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00337	15	■ 1.000	$\mathcal{T}_a =$	$\mathcal{T}_a =$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00338	15	■ 1.000	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$ to recover (73). We note that (73) and (74) are
1510.06699_F00339	16	■ 1.000	$\mathcal{T}_a$	$\mathcal{T}_a$	density $\mathcal{L}_{\mathrm{G}}$, but the expression for $\mathcal{T}_a$, or $\mathcal{T}_a^*$, will depend
1510.06699_F00340	16	■ 1.000	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	given by $\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$, after a short calculation one
1510.06699_F00341	16	■ 0.945	$\mathcal{T}_{[abc]}^* = 0$	$\mathcal{T}_{[abc]}^* = 0$	of the Dirac field ψ . In other words, if $\mathcal{T}_{[abc]}^* = 0$ then
1510.06699_F00342	16	■ 1.000	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	ric and has the trace $\tau = h^{-1} \mu \phi \bar{\psi} \psi$, whereas the spin-
1510.06699_F00343	16	■ 1.000	$\sigma_{abc} = \sigma_{[abc]}$	$\sigma_{abc} = \sigma_{[abc]}$	that $\sigma_{abc} = \sigma_{[abc]}$. Consequently, all contractions of the
1510.06699_F00344	16	■ 1.000	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	ply making the replacement $\mathcal{D}_a \rightarrow \mathcal{D}_a^*$ throughout; the
1510.06699_F00345	16	■ 1.000	μ, ν, λ	μ, ν, λ	where μ, ν, λ and a are dimensionless constants (usually
1510.06699_F00346	16	■ 0.669	β_1, β_2	β_1, β_2	stants β_1, β_2 and β_3 in $L_{\mathcal{T}^{*2}}$. Since none of the additional
1510.06699_F00347	16	■ 0.669	β_3	β_3	stants β_1, β_2 and β_3 in $L_{\mathcal{T}^{*2}}$. Since none of the additional
1510.06699_F00348	16	■ 1.000	$F_{\mu\nu} \equiv$	$F_{\mu\nu} \equiv$	where the electromagnetic field strength tensor $F_{\mu\nu} \equiv$
1510.06699_F00349	16	■ 1.000	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source current density. We
1510.06699_F00350	16	■ 1.000	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00351	16	■ 1.000	χ	χ	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00352	16	■ 1.000	$\partial_\mu J^\mu = 0$	$\partial_\mu J^\mu = 0$	ity equation $\partial_\mu J^\mu = 0$, which embodies the conservation
1510.06699_F00353	16	■ 1.000	$\mathcal{A}_a = h_a^\mu A_\mu$	$\mathcal{A}_a = h_a^\mu A_\mu$	field $\mathcal{A}_a = h_a^\mu A_\mu$. Similarly, we introduce the covari-
1510.06699_F00354	16	■ 1.000	$\mathcal{J}_a = h_a^\mu J_\mu$	$\mathcal{J}_a = h_a^\mu J_\mu$	ant current density $\mathcal{J}_a = h_a^\mu J_\mu$. This leads us, however,
1510.06699_F00355	16	■ 1.000	$w(\mathcal{A}_a) = -1$	$w(\mathcal{A}_a) = -1$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00356	16	■ 1.000	$w(\mathcal{J}_a) = -3$	$w(\mathcal{J}_a) = -3$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00357	16	■ 1.000	$w(A_\mu) = 0$	$w(A_\mu) = 0$	fields carrying a Greek index have weights $w(A_\mu) = 0$
1510.06699_F00358	16	■ 1.000	$w(J_\mu) = -2$	$w(J_\mu) = -2$	and $w(J_\mu) = -2$.
1510.06699_F00359	16	■ 1.000	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	where we have defined $\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$. Since
1510.06699_F00360	16	■ 1.000	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_{\mathrm{M}}$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_{\mathrm{M}}$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2$, $\mathcal{L}_{\mathrm{M}}$ has an overall Weyl weight of zero,
1510.06699_F00361	17	■ 1.000	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	where we have defined $\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$, in which
1510.06699_F00362	17	■ 1.000	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is

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1510.06699_F00363	17	■ 1.000	$\mathcal{T}^{*c}_{\{*\ c\}_{\{*\}_{\{a\ b\}}}}$	$\mathcal{T}^{*c}_{ab}$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is
1510.06699_F00364	17	■ 1.000	$\mathcal{F}_{\{a\ b\}}$	$\mathcal{F}_{ab}$	gauge field, as defined in (42). It is apparent that $\mathcal{F}_{ab}$
1510.06699_F00365	17	■ 1.000	$F_{\{\mu\ \nu\}}$	$F_{\mu\nu}$	ance of $F_{\mu\nu}$, whereas the second term on the RHS of (82)
1510.06699_F00366	17	■ 1.000	$\widehat{\mathcal{F}}_{\{a\ b\}}^{*}$	$\widehat{\mathcal{F}}_{ab}^*$	transformation, and so neither is $\widehat{\mathcal{F}}_{ab}^*$. Moreover, $\mathcal{F}_{ab}$ has
1510.06699_F00367	17	■ 1.000	$w\left(\mathcal{F}_{\{a\ b\}}\right)=-2$	$w(\mathcal{F}_{ab}) = -2$	a Weyl weight $w(\mathcal{F}_{ab}) = -2$. Thus, substituting (82)
1510.06699_F00368	17	■ 1.000	$\mathcal{L}_{\{L\}_{\{\mathrm{M}\}}}$	$\mathcal{L}_\mathrm{M}$	for $\mathcal{L}_\mathrm{M}$ is identical to that obtained in PGT, as should be
1510.06699_F00369	17	■ 0.999	$\delta\mathcal{L}_{\{L\}_{\{\mathrm{EM}\}}} / \delta A_{\{\mu\}}=0$	$\delta\mathcal{L}_\mathrm{EM}/\delta A_\mu = 0$	is given by $\delta\mathcal{L}_\mathrm{EM}/\delta A_\mu = \emptyset$, which is easily shown to be
1510.06699_F00370	17	■ 0.997	$\delta\mathcal{L}_{\{L\}_{\{\mathrm{EM}\}}} / \delta \mathcal{A}_{\{a\}}=0$	$\delta\mathcal{L}_\mathrm{EM}/\delta \mathcal{A}_a = 0$	equivalent to $\delta\mathcal{L}_\mathrm{EM}/\delta \mathcal{A}_a = \emptyset$, and in manifestly WGT-
1510.06699_F00371	17	■ 1.000	$\mathcal{D}_{\{a\}}\mathcal{F}^{ac}=\left(\mathcal{D}_{\{a\}}^{*}+\right.$	$\mathcal{D}_a\mathcal{F}^{ac} = (\mathcal{D}_a^* +$	is easily checked directly by noting that $\mathcal{D}_a\mathcal{F}^{ac} = (\mathcal{D}_a^* +$
1510.06699_F00372	17	■ 1.000	$\left.2\mathcal{B}_{\{a\}}\right)\mathcal{F}^{ac}$	$2\mathcal{B}_a)\mathcal{F}^{ac}$	$2\mathcal{B}_a)\mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3\mathcal{B}_a$, which on substitution
1510.06699_F00373	17	■ 1.000	$\mathcal{T}_{\{a\}}=\mathcal{T}^{*\{*\}_{\{*\}_{\{a\}}}-3\mathcal{B}_{\{a\}}}$	$\mathcal{T}_a = \mathcal{T}_a^* - 3\mathcal{B}_a$	$2\mathcal{B}_a)\mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3\mathcal{B}_a$, which on substitution
1510.06699_F00374	17	■ 1.000	$\mathcal{J}^{\{a\}}=0$	$\mathcal{J}^a = 0$	angular-momentum tensors. Setting $\mathcal{J}^a = \emptyset$ in (83), one
1510.06699_F00375	17	■ 1.000	$\sigma_{\{a\ b\}}^{\{*\}_{\{*\}_{\{c\}}}=0$	$\sigma_{ab}^{\ c} = 0$	whereas $\sigma_{ab}^{\ c} = 0$ immediately. We see that $\tau^a_{\ b}$ is the
1510.06699_F00376	17	■ 1.000	$\tau^{\{a\}_{\{*\}_{\{b\}}}$	$\tau^a_{\ b}$	whereas $\sigma_{ab}^{\ c} = 0$ immediately. We see that $\tau^a_{\ b}$ is the
1510.06699_F00377	17	■ 1.000	$\tau_{\{a\ b\}}$	τ_{ab}	gravity. Moreover, τ_{ab} is symmetric and traceless, as
1510.06699_F00378	17	■ 1.000	$\mathcal{J}^{\{a\}}=q\bar{\psi}\gamma^{\{a\}}\psi$	$\mathcal{J}^a = q\bar{\psi}\gamma^a\psi$	current density of the Dirac field is simply $\mathcal{J}^a = q\bar{\psi}\gamma^a\psi$,
1510.06699_F00379	17	■ 1.000	$w\left(\mathcal{J}^{\{a\}}\right)=-3$	$w(\mathcal{J}^a) = -3$	which has the required Weyl weight $w(\mathcal{J}^a) = -3$. Thus,
1510.06699_F00380	17	■ 1.000	$\phi\neq 0$	$\phi \neq 0$	(provided $\phi \neq 0$), which is known as the Einstein gauge.
1510.06699_F00381	17	■ 1.000	$\phi(x)=1$	$\phi(x) = 1$	Although it is most common simply to set $\phi(x) = \mathbb{1}$, this
1510.06699_F00382	17	■ 1.000	$\phi(x)=\phi_{\{0\}}$	$\phi(x) = \phi_0$	can be misleading and so we instead set $\phi(x) = \phi_\mathrm{f}$, where
1510.06699_F00383	17	■ 1.000	$\phi_{\{0\}}$	ϕ_0	ϕ_0 is a constant with the same units as ϕ . Adopting the
1510.06699_F00384	18	■ 0.638	(91 c)	(91c)	(91c)
1510.06699_F00385	18	■ 1.000	$\widehat{\mathcal{A}}_{\{a\ b\}}^{\{*\}_{\{*\}_{\{c\}}}\equiv\widehat{\mathcal{h}}_{\{a\}}^{\{*\}_{\{\mu\}}}\widehat{\mathcal{A}}_{\{a\ b\}}^{\{*\}_{\{\mu\}}}$	$\widehat{\mathcal{A}}^{ab}_{\ c} \equiv \widehat{h}_a^{\ \mu} \widehat{A}^{ab}_{\ \mu}$	ables $\widehat{\mathcal{A}}^{ab}_{\ c} \equiv \widehat{h}_a^{\ \mu} \widehat{A}^{ab}_{\ \mu}$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a^{\ \mu} \widehat{B}_\mu$.
1510.06699_F00386	18	■ 1.000	$\widehat{\mathcal{B}}_{\{a\}}\equiv\widehat{\mathcal{h}}_{\{a\}}^{\{*\}_{\{\mu\}}}\widehat{\mathcal{B}}_{\{a\}}^{\{*\}_{\{\mu\}}}$	$\widehat{\mathcal{B}}_a \equiv \widehat{h}_a^{\ \mu} \widehat{B}_\mu$	ables $\widehat{\mathcal{A}}^{ab}_{\ c} \equiv \widehat{h}_a^{\ \mu} \widehat{A}^{ab}_{\ \mu}$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a^{\ \mu} \widehat{B}_\mu$.
1510.06699_F00387	18	■ 1.000	$\mathcal{D}_{\{a\}}^{*}\phi=-\left(\phi^2/\phi_{\{0\}}\right)\widehat{\mathcal{B}}_{\{a\}}$	$\mathcal{D}_a^*\phi = -(\phi^2/\phi_0)\widehat{\mathcal{B}}_a$	the scalar field ϕ may be written as $\mathcal{D}_a^*\phi = -(\phi^2/\phi_0)\widehat{\mathcal{B}}_a$,
1510.06699_F00388	18	■ 1.000	$\widehat{\chi}=\left(\phi/\phi_{\{0\}}\right)^{-w}\chi$	$\widehat{\chi} = (\phi/\phi_0)^{-w}\chi$	where $\widehat{\chi} = (\phi/\phi_0)^{-w}\chi$ and we have defined the derivative
1510.06699_F00389	18	■ 0.996	$\widehat{\mathcal{D}}_{\{a\}}\equiv\widehat{\mathcal{h}}_{\{a\}}^{\{*\}_{\{\mu\}}}\mathcal{D}_{\{a\}}^{\{*\}_{\{\mu\}}}$	$\widehat{\mathcal{D}}_a \equiv \widehat{h}_a^{\ \mu} D_\mu$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a^{\ \mu} D_\mu$ and $\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w\widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00390	18	■ 0.996	$\widehat{\mathcal{D}}_{\{a\}}^{*}=\widehat{\mathcal{D}}_{\{a\}}+w\widehat{\mathcal{B}}_{\{a\}}$	$\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w\widehat{\mathcal{B}}_a$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a^{\ \mu} D_\mu$ and $\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w\widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00391	18	■ 1.000	$\left(\phi/\phi_{\{0\}}\right)^{1-w}$	$(\phi/\phi_0)^{1-w}$	all multiplicative factor $(\phi/\phi_0)^{1-w}$, the scale-invariant
1510.06699_F00392	18	■ 1.000	$\widehat{\mathcal{D}}_{\{a\}}^{*}\widehat{\chi}$	$\widehat{\mathcal{D}}_a^*\widehat{\chi}$	quantity $\widehat{\mathcal{D}}_a^*\widehat{\chi}$ has the <i>same</i> functional dependency on $\widehat{\chi}$,
1510.06699_F00393	18	■ 1.000	$\widehat{\chi}$	$\widehat{\chi}$	quantity $\widehat{\mathcal{D}}_a^*\widehat{\chi}$ has the <i>same</i> functional dependency on $\widehat{\chi}$,
1510.06699_F00394	18	■ 0.999	$\widehat{\mathcal{h}}_{\{a\}}^{\{*\}_{\{\mu\}}},\widehat{\mathcal{A}}_{\{a\ b\}}^{\{*\}_{\{\mu\}}}$	$\widehat{h}_a^{\ \mu}, \widehat{A}^{ab}_{\ \mu}$	$\widehat{h}_a^{\ \mu}$, $\widehat{A}^{ab}_{\ \mu}$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^*\chi$ does on χ , $h_a^{\ \mu}$,
1510.06699_F00395	18	■ 0.999	$\widehat{\mathcal{B}}_{\{a\}}^{\{*\}_{\{\mu\}}}$	$\widehat{B}_\mu$	$\widehat{h}_a^{\ \mu}$, $\widehat{A}^{ab}_{\ \mu}$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^*\chi$ does on χ , $h_a^{\ \mu}$,
1510.06699_F00396	18	■ 0.999	$\mathcal{D}_{\{a\}}^{*}\chi$	$\mathcal{D}_a^*\chi$	$\widehat{h}_a^{\ \mu}$, $\widehat{A}^{ab}_{\ \mu}$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^*\chi$ does on χ , $h_a^{\ \mu}$,
1510.06699_F00397	18	■ 0.999	$\chi,\mathcal{h}_{\{a\}}^{\{*\}_{\{\mu\}}}$	$\chi, h_a^{\ \mu}$	$\widehat{h}_a^{\ \mu}$, $\widehat{A}^{ab}_{\ \mu}$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^*\chi$ does on χ , $h_a^{\ \mu}$,
1510.06699_F00398	18	■ 1.000	$\mathcal{R}^{ab}_{\{a\ b\}}^{\{*\}_{\{c\ d\}}=\left(\phi/\phi_{\{0\}}\right)^2\widehat{\mathcal{R}}^{ab}_{\{a\ b\}}^{\{*\}_{\{c\ d\}}},\mathcal{T}^{*a}_{\{a\ b\}}^{\{*\}_{\{b\ c\}}=\left(\phi/\phi_{\{0\}}\right)\widehat{\mathcal{T}}^{*a}_{\{a\ b\}}^{\{*\}_{\{b\ c\}}}$	$\mathcal{R}^{ab}_{\ cd} = (\phi/\phi_0)^2 \widehat{\mathcal{R}}^{ab}_{\ cd}, \mathcal{T}^{*a}_{\ bc} = (\phi/\phi_0) \widehat{\mathcal{T}}^{*a}_{\ bc}$	ten as $\mathcal{R}^{ab}_{\ cd} = (\phi/\phi_0)^2 \widehat{\mathcal{R}}^{ab}_{\ cd}, \mathcal{T}^{*a}_{\ bc} = (\phi/\phi_0) \widehat{\mathcal{T}}^{*a}_{\ bc}$ and
1510.06699_F00399	18	■ 1.000	$\mathcal{H}_{\{a\ b\}}=\left(\phi/\phi_{\{0\}}\right)^2\widehat{\mathcal{H}}_{\{a\ b\}}$	$\mathcal{H}_{ab} = (\phi/\phi_0)^2 \widehat{\mathcal{H}}_{ab}$	$\mathcal{H}_{ab} = (\phi/\phi_0)^2 \widehat{\mathcal{H}}_{ab}$, where each quantity with a caret is

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1510.06699_F00400	18	■ 1.000	$\widehat{h}^{-1}=\left(\phi/\phi_0\right)^{-4}\widehat{h}^{-1}$	$h^{-1}=(\phi/\phi_0)^{-4}\widehat{h}^{-1}$	Noting that $\widehat{h}^{-1}=(\phi/\phi_0)^{-4}\widehat{h}^{-1}$, one is thus led to the
1510.06699_F00401	18	■ 1.000	$\varphi, h_a{}^\mu, A^{ab}{}_\mu$	$\varphi, h_a{}^\mu, A^{ab}{}_\mu$	Thus, if χ represents φ , $h_a{}^\mu$, $A^{ab}{}_\mu$ or B_μ , one may im-
1510.06699_F00402	18	■ 1.000	$\delta\mathcal{L}_\mathrm{T}/\delta\widehat{\chi}=0$	$\delta\mathcal{L}_\mathrm{T}/\delta\widehat{\chi}=0$	motion $\delta\mathcal{L}_\mathrm{T}/\delta\widehat{\chi}=0$ has the <i>same</i> functional form as
1510.06699_F00403	18	■ 1.000	$\delta\mathcal{L}_\mathrm{T}/\delta\chi _{\phi=\phi_0}=0$	$\delta\mathcal{L}_\mathrm{T}/\delta\chi _{\phi=\phi_0}=0$	(the corresponding term in) $\delta\mathcal{L}_\mathrm{T}/\delta\chi _{\phi=\phi_0}=0$, but with
1510.06699_F00404	18	■ 1.000	$\delta\mathcal{L}_\mathrm{T}/\delta\phi=\delta\mathcal{L}_\mathrm{M}/\delta\phi$	$\delta\mathcal{L}_\mathrm{T}/\delta\phi=\delta\mathcal{L}_\mathrm{M}/\delta\phi$	variables, one begins by noting that $\delta\mathcal{L}_\mathrm{T}/\delta\phi=\delta\mathcal{L}_\mathrm{M}/\delta\phi$,
1510.06699_F00405	18	■ 1.000	$\delta\mathcal{L}_\mathrm{T}/\delta\varphi=\delta\mathcal{L}_\mathrm{M}/\delta\varphi=0$	$\delta\mathcal{L}_\mathrm{T}/\delta\varphi=\delta\mathcal{L}_\mathrm{M}/\delta\varphi=0$	$\delta\mathcal{L}_\mathrm{T}/\delta\varphi=\delta\mathcal{L}_\mathrm{M}/\delta\varphi=0$ and set $\phi=\phi_0$ in the result-
1510.06699_F00406	18	■ 1.000	$\partial\mathcal{L}_\mathrm{M}/\partial\widehat{B}_\mu$	$\partial\mathcal{L}_\mathrm{M}/\partial\widehat{B}_\mu$	may replace $\partial\mathcal{L}_\mathrm{M}/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_\mathrm{M}/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00407	18	■ 1.000	$\delta\mathcal{L}_\mathrm{M}/\delta\widehat{B}_\mu$	$\delta\mathcal{L}_\mathrm{M}/\delta\widehat{B}_\mu$	may replace $\partial\mathcal{L}_\mathrm{M}/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_\mathrm{M}/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00408	19	■ 1.000	$m\rightarrow\mu\phi$	$m\rightarrow\mu\phi$	making the replacement $m\rightarrow\mu\phi$, as in (70). One begins
1510.06699_F00409	19	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	by constructing the appropriate action for a spin- $\frac{1}{2}$ point
1510.06699_F00410	19	■ 1.000	$p_\mu\rightarrow p_a\equiv h_a{}^\mu p_\mu$	$p_\mu\rightarrow p_a\equiv h_a{}^\mu p_\mu$	needs only to make the replacements $p_\mu\rightarrow p_a\equiv h_a{}^\mu p_\mu$,
1510.06699_F00411	19	■ 1.000	$\dot{x}^\mu\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$	$\dot{x}^\mu\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$	$\dot{x}^\mu\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$ and $m\rightarrow\mu\phi$, which gives ⁷⁵
1510.06699_F00412	19	■ 1.000	$p^a(\lambda)$	$p^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00413	19	■ 1.000	$v^a(\lambda)$	$v^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00414	19	■ 1.000	$e(\lambda)$	$e(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00415	19	■ 1.000	λ	λ	along the worldline, which is parameterised by λ . We
1510.06699_F00416	19	■ 1.000	$w(p^a)=-1, w(v^a)=0, w(e)=1, w(\lambda)=1$	$w(p^a)=-1, w(v^a)=0, w(e)=1, w(\lambda)=1$	(96) are $w(p^a)=-1, w(v^a)=0, w(e)=1, w(\lambda)=1$ and
1510.06699_F00417	19	■ 1.000	$p^2\equiv p_ap^a$	$p^2\equiv p_ap^a$	where $p^2\equiv p_ap^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00418	19	■ 1.000	$c^c{}_{ab}$	$c^c{}_{ab}$	where $p^2\equiv p_ap^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00419	19	■ 1.000	${}^0\mathcal{D}_c^*$	${}^0\mathcal{D}_c^*$	where ${}^0\mathcal{D}_c^*$ is the reduced WGT covariant derivative, de-
1510.06699_F00420	19	■ 1.000	$e=1/(\mu\phi)$	$e=1/(\mu\phi)$	one chooses $e=1/(\mu\phi)$, in which case $v^2=1$ and λ cor-
1510.06699_F00421	19	■ 1.000	$v^2=1$	$v^2=1$	one chooses $e=1/(\mu\phi)$, in which case $v^2=1$ and λ cor-
1510.06699_F00422	19	■ 1.000	τ	τ	responds to the proper time τ along the worldline; in this
1510.06699_F00423	19	■ 0.348	$\mathbf{76-78}$	$\mathbf{76-78}$	terms of the reduced PGT-covariant derivative as ⁷⁶⁻⁷⁸
1510.06699_F00424	19	■ 0.996	${}^0A^{ab}{}_\mu$	${}^0A^{ab}{}_\mu$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00425	19	■ 0.996	${}^0\mathcal{D}_b$	${}^0\mathcal{D}_b$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00426	19	■ 0.906	$v^b\mathcal{D}_bv_a=0$	$v^b\mathcal{D}_bv_a=0$	the equation of motion $v^b\mathcal{D}_bv_a=0$). We note that
1510.06699_F00427	19	■ 1.000	$\{ }^{79,80}$	79,80	this conclusion differs from some previous studies ^{79,80} ,
1510.06699_F00428	19	■ 1.000	$(q_\mathrm{W}/m)\mathcal{H}_{ab}v^b$	$(q_\mathrm{W}/m)\mathcal{H}_{ab}v^b$	$(q_\mathrm{W}/m)\mathcal{H}_{ab}v^b$, where q_W is the particle's Weyl charge
1510.06699_F00429	19	■ 1.000	q_W	q_W	$(q_\mathrm{W}/m)\mathcal{H}_{ab}v^b$, where q_W is the particle's Weyl charge
1510.06699_F00430	19	■ 1.000	$\mu=0$	$\mu=0$	By setting $\mu=0$ in the action (96) and choosing the
1510.06699_F00431	19	■ 1.000	$e=1$	$e=1$	einbein $e=1$, such that $v^2=0$, one finds that the
1510.06699_F00432	19	■ 1.000	$v^2=0$	$v^2=0$	einbein $e=1$, such that $v^2=0$, one finds that the
1510.06699_F00433	20	■ 1.000	$\widehat{p}_a, \widehat{v}^a$	$\widehat{p}_a, \widehat{v}^a$	where $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00434	20	■ 1.000	$\widehat{e}$	$\widehat{e}$	where $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00435	20	■ 0.990	$\widehat{\lambda}$	$\widehat{\lambda}$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00436	20	■ 0.990	$\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$	$\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00437	20	■ 1.000	$\widehat{\epsilon}=\left[1-\widehat{\lambda}\left(dx^\mu/d\widehat{\lambda}\right)\partial_\mu\ln(\phi/\phi_0)\right]\widehat{e}$	$\widehat{\epsilon}=\left[1-\widehat{\lambda}\left(dx^\mu/d\widehat{\lambda}\right)\partial_\mu\ln(\phi/\phi_0)\right]\widehat{e}$	bein $\widehat{\epsilon}=\left[1-\widehat{\lambda}(dx^\mu/d\widehat{\lambda})\partial_\mu\ln(\phi/\phi_0)\right]\widehat{e}$. Thus, in terms

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1510.06699_F00475	22	■ 1.000	$w\left(h_{\{a\}}^{\{\mu\}}\right)=-1$	$w\left(h_a^{\mu}\right)=-1$	have Weyl weights $w(g_{\mu\nu})=2$ and $w(h_a^{\mu})=-1$ respec-
1510.06699_F00476	22	■ 1.000	$w\left(\eta_{\{a\}b}\right)=0$	$w\left(\eta_{ab}\right)=0$	tively, and so (115) and (116) imply that $w(\eta_{ab})=0$, as
1510.06699_F00477	22	■ 1.000	$e_{\{a\}}^{\{\mu\}}$	e_a^{μ}	h_a^{μ} and b_{μ}^a as e_a^{μ} and e_{μ}^a , respectively.
1510.06699_F00478	22	■ 1.000	$e^{\{a\}}_{\{\mu\}}$	e_{μ}^a	h_a^{μ} and b_{μ}^a as e_a^{μ} and e_{μ}^a , respectively.
1510.06699_F00479	22	■ 1.000	$x+\delta x$	$x+\delta x$	and $x+\delta x$, which is accompanied by a local change in
1510.06699_F00480	22	■ 1.000	$J^{\{a\}}$	J^a	for some vector J^a of weight w is defined as
1510.06699_F00481	22	■ 1.000	$J^{\{a\}}(x)$	$J^a(x)$	which is required to compare vectors $J^a(x)$ and $J^a(x+\delta x)$
1510.06699_F00482	22	■ 1.000	$J^{\{a\}}(x+\delta x)$	$J^a(x+\delta x)$	which is required to compare vectors $J^a(x)$ and $J^a(x+\delta x)$
1510.06699_F00483	22	■ 1.000	$\hat{\boldsymbol{e}}_{\{a\}}(x+\delta x)$	$\hat{e}_a(x+\delta x)$	tetrad frames $\hat{e}_a(x)$ and $\hat{e}_a(x+\delta x)$ respectively. Hence,
1510.06699_F00484	22	■ 1.000	$\partial_{\{\mu\}}^{\{*\}}$	∂_{μ}^*	where we have used the partial derivative operator ∂_{μ}^*
1510.06699_F00485	22	■ 1.000	$\eta_{\{a\}b}$	η_{ab}	the Lorentz metric η_{ab} at each point. Then demanding
1510.06699_F00486	22	■ 1.000	$A^{\{a\}b}_{\{\mu\}}=-A^{ba}_{\{\mu\}}$	$A^{ab}_{\mu}=-A^{ba}_{\mu}$	antisymmetric, i.e. $A^{ab}_{\mu}=-A^{ba}_{\mu}$, as previously.
1510.06699_F00487	22	■ 0.528	$\nabla_{\{\mu\}}^{\{*\}}=\partial_{\{\mu\}}^{\{*\}}+\Gamma^{\{\sigma\}}_{\{\rho\mu\}}X^{\rho}_{\{\sigma\}}$	$\nabla_{\mu}^*=\partial_{\mu}^*+\Gamma^{\sigma}_{\rho\mu}X^{\rho}_{\sigma}$	where $\nabla_{\mu}^*=\partial_{\mu}^*+\Gamma^{\sigma}_{\rho\mu}X^{\rho}_{\sigma}$ and X^{ρ}_{σ} are the $\text{GL}(4,R)$ gen-
1510.06699_F00488	22	■ 1.000	$\Delta_{\{\mu\}}^{\{*\}}$	Δ_{μ}^*	of the field to which Δ_{μ}^* is applied. If a field ψ carries
1510.06699_F00489	22	■ 1.000	$\nabla_{\{\mu\}}^{\{*\}}\psi=\partial_{\{\mu\}}^{\{*\}}\psi$	$\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$	only Latin indices, then $\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=D_{\mu}^*\psi$;
1510.06699_F00490	22	■ 1.000	$\Delta_{\{\mu\}}^{\{*\}}\psi=D_{\{\mu\}}^{\{*\}}\psi$	$\Delta_{\mu}^*\psi=D_{\mu}^*\psi$	only Latin indices, then $\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=D_{\mu}^*\psi$;
1510.06699_F00491	22	■ 1.000	$D_{\{\mu\}}^{\{*\}}\psi=\partial_{\{\mu\}}^{\{*\}}\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$. When acting on an ob-
1510.06699_F00492	22	■ 1.000	$\Delta_{\{\mu\}}^{\{*\}}\psi=\nabla_{\{\mu\}}^{\{*\}}\psi$	$\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$. When acting on an ob-
1510.06699_F00493	22	■ 1.000	$\Gamma^{\{\sigma\}}_{\{\rho\mu\}}$	$\Gamma^{\sigma}_{\rho\mu}$	connection coefficients $\Gamma^{\sigma}_{\rho\mu}$ are themselves dynamical
1510.06699_F00494	22	■ 1.000	$J^{\{a\}}+\delta J^{\{a\}}$	$J^a+\delta J^a$	ported from x to $x+\delta x$, be equal to $J^a+\delta J^a$, i.e.
1510.06699_F00495	22	■ 1.000	Γ	Γ	which relates A and Γ (and B); in particular, we note that
1510.06699_F00496	22	■ 1.000	$w\left(\Gamma^{\sigma}_{\nu\mu}\right)=0$	$w\left(\Gamma^{\sigma}_{\nu\mu}\right)=0$	$w(\Gamma^{\sigma}_{\nu\mu})=0$. The relation (121) is sometimes known as
1510.06699_F00497	22	■ 1.000	$\nabla_{\{\sigma\}}^{\{*\}}g_{\{\mu\nu\}}=0$	$\nabla_{\sigma}^*g_{\mu\nu}=0$	Using (115) and (121), one finds that $\nabla_{\sigma}^*g_{\mu\nu}=0$, and
1510.06699_F00498	22	■ 1.000	$T^{*a}_{\{\mu\nu\}}$	$T^{*a}_{\mu\nu}$	the gauge field strengths $R^{ab}_{\mu\nu}$ and $T^{*a}_{\mu\nu}$, one finds that
1510.06699_F00499	22	■ 1.000	$R^{\{\rho\}}_{\{\sigma\mu\nu\}}=e_{\{a\}}^{\{\rho\}}e^{\{b\}}_{\{\sigma\}}R^a_{b\mu\nu}$	$R^{\rho}_{\sigma\mu\nu}=e_a^{\rho}e^b_{\sigma}R^a_{b\mu\nu}$	where $R^{\rho}_{\sigma\mu\nu}=e_a^{\rho}e^b_{\sigma}R^a_{b\mu\nu}$ and $T^{*\lambda}_{\mu\nu}=e_a^{\lambda}T^{*a}_{\mu\nu}$.
1510.06699_F00500	22	■ 1.000	$T^{*\lambda}_{\{\mu\nu\}}=e_{\{a\}}^{\{\lambda\}}T^{*a}_{\{\mu\nu\}}$	$T^{*\lambda}_{\mu\nu}=e_a^{\lambda}T^{*a}_{\mu\nu}$	where $R^{\rho}_{\sigma\mu\nu}=e_a^{\rho}e^b_{\sigma}R^a_{b\mu\nu}$ and $T^{*\lambda}_{\mu\nu}=e_a^{\lambda}T^{*a}_{\mu\nu}$.
1510.06699_F00501	22	■ 1.000	$T^{*\lambda}_{\{\mu\nu\}}$	$T^{*\lambda}_{\mu\nu}$	Thus, although we recognise $T^{*\lambda}_{\mu\nu}$ as (minus) the tor-
1510.06699_F00502	22	■ 1.000	$R^{\{\rho\}}_{\{\sigma\mu\nu\}}$	$R^{\rho}_{\sigma\mu\nu}$	sion tensor of the Y_4 spacetime, we see that $R^{\rho}_{\sigma\mu\nu}$ is not
1510.06699_F00503	22	■ 1.000	$\widetilde{R}_{\{\rho\sigma\mu\nu\}}$	$\widetilde{R}_{\rho\sigma\mu\nu}$	One should note that, although $\widetilde{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00504	22	■ 0.774	μ,ν	μ,ν	ric in (μ,ν) , it is no longer antisymmetric in (ρ,σ) (in-
1510.06699_F00505	22	■ 0.774	ρ,σ	ρ,σ	ric in (μ,ν) , it is no longer antisymmetric in (ρ,σ) (in-
1510.06699_F00506	22	■ 0.764	$\widetilde{R}_{\{(\rho\sigma)\mu\nu\}}=g_{\{\rho\sigma\}}H_{\mu\nu}$	$\widetilde{R}_{(\rho\sigma)\mu\nu}=g_{\rho\sigma}H_{\mu\nu}$	deed $\widetilde{R}_{(\rho\sigma)\mu\nu}=g_{\rho\sigma}H_{\mu\nu}$) and does not satisfy the fa-
1510.06699_F00507	22	■ 1.000	$V_{\{4\}}$	V_4	sor in a Riemannian V_4 spacetime. One may also show
1510.06699_F00508	22	■ 1.000	$\widetilde{R}^{\rho}_{\{\sigma\mu\nu\}}$	$\widetilde{R}^{\rho}_{\sigma\mu\nu}$	that, with the given arrangements of indices, both $\widetilde{R}^{\rho}_{\sigma\mu\nu}$
1510.06699_F00509	22	■ 1.000	$\widetilde{R}_{\{\mu\nu\}}\equiv\widetilde{R}_{\mu\lambda\nu}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$. As one
1510.06699_F00510	22	■ 1.000	$\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$	$\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$. As one
1510.06699_F00511	23	■ 1.000	$J^{\{\rho\}}$	J^{ρ}	operators acting on a vector J^{ρ} (say) of Weyl weight w ,
1510.06699_F00512	23	■ 1.000	$\{ \}^{\{0\}}\Gamma^{\{\lambda\}}_{\{\mu\nu\}}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_{\mu}g_{\nu\rho}+\partial_{\nu}g_{\mu\rho}-\partial_{\rho}g_{\mu\nu})$	${}^0\Gamma^{\lambda}_{\mu\nu}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_{\mu}g_{\nu\rho}+\partial_{\nu}g_{\mu\rho}-\partial_{\rho}g_{\mu\nu})$	in which ${}^0\Gamma^{\lambda}_{\mu\nu}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_{\mu}g_{\nu\rho}+\partial_{\nu}g_{\mu\rho}-\partial_{\rho}g_{\mu\nu})$ is the
1510.06699_F00513	23	■ 1.000	$K^{*\lambda}_{\{\mu\nu\}}$	$K^{*\lambda}_{\mu\nu}$	standard metric (Christoffel) connection and $K^{*\lambda}_{\mu\nu}$ is
1510.06699_F00514	23	■ 1.000	$K_{\{\lambda\mu\nu\}}^{\{*\}}=-K_{\{\mu\lambda\nu\}}^{\{*\}}$	$K_{\lambda\mu\nu}^*=-K_{\mu\lambda\nu}^*$	for which it is worth noting that $K_{\lambda\mu\nu}^*=-K_{\mu\lambda\nu}^*$.

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1510.06699_F00515	23	■ 1.000	$\{ \}^{\{0\}} \backslash \Gamma^{\{*\}}_{\backslash \sigma \{ \} }_{\backslash \rho \{ \mu \} }$	${}^0\Gamma^{*\sigma}{}_{\rho\mu}$	connection ${}^0\Gamma^{*\sigma}{}_{\rho\mu}$ represent the same geometrical object
1510.06699_F00516	23	■ 0.661	$\backslash \Gamma^{\{ \sigma \} }_{\{ \} _{\backslash \nu \{ \mu \} }} \backslash \rightarrow^{\{0\}} \backslash \Gamma^{\{ *\} }_{\backslash \sigma \{ \} }_{\{ \} _{\backslash \nu \{ \mu \} }}$	$\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}{}_{\nu\mu}$	again hold with the replacements $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}{}_{\nu\mu}$ and
1510.06699_F00517	23	■ 1.000	$A^{\{ a \} }_{\{ \} _{\backslash b \{ \mu \} }} \backslash \rightarrow^{\{ \} }_{\{0\}} A^{\{ *\} }_{a \{ \} _{\backslash b \{ \mu \} }}$	$A^a{}_{b\mu} \rightarrow {}^0A^{*a}{}_{b\mu}$	$A^a{}_{b\mu} \rightarrow {}^0A^{*a}{}_{b\mu}$, from which one can directly derive (131).
1510.06699_F00518	23	■ 1.000	$\{ \}^{\{0\}} \backslash \nabla_{\{ \sigma \} }^{\{ *\} } g_{\{ \mu \} \backslash \nu \{ \} }=0$	${}^0\nabla_{\sigma}^{*}g_{\mu\nu} = 0$	One also obtains the metricity condition ${}^0\nabla_{\sigma}^{*}g_{\mu\nu} = 0$. Fi-
1510.06699_F00519	23	■ 0.983	$R^{\{ \rho \} }_{\{ \} _{\backslash \sigma \{ \mu \} \backslash \nu \{ \} }} \backslash \rightarrow^{\{ \} }_{\{0\}} R^{\{ *\} }_{\backslash \rho \{ \} }_{\{ \} _{\backslash \sigma \{ \mu \} \backslash \nu \{ \} }}$	$R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}{}_{\sigma\mu\nu}$	but with $R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}{}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}{}_{\nu\mu}$, whereas
1510.06699_F00520	23	■ 0.983	$\backslash \Gamma^{\{ \sigma \} }_{\{ \} _{\backslash \nu \{ \mu \} }} \backslash \rightarrow^{\{ \} }_{\{0\}} \backslash \Gamma^{\{ *\} }_{\backslash \sigma \{ \} }_{\{ \} _{\backslash \nu \{ \mu \} }}$	$\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}{}_{\nu\mu}$	but with $R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}{}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}{}_{\nu\mu}$, whereas
1510.06699_F00521	23	■ 1.000	$\{ \}^{\{0\}} T^{\{ *\} }_{\backslash \lambda \{ \} }_{\{ \} _{\backslash \mu \{ \} \backslash \nu \{ \} }}=0$	${}^0T^{*\lambda}{}_{\mu\nu} = 0$	(126) becomes simply ${}^0T^{*\lambda}{}_{\mu\nu} = 0$, indicating the ab-
1510.06699_F00522	23	■ 0.941	$\backslash \nabla_{\{ \mu \} }^{\{ *\} } \backslash \rightarrow^{\{ \} }_{\{0\}} \backslash \nabla_{\{ \mu \} }^{\{ *\} }, R^{\{ \rho \} }_{\{ \} _{\backslash \sigma \{ \mu \} \backslash \nu \{ \} }} \backslash \rightarrow^{\{ \} }_{\{0\}} R^{\{ *\} }_{\backslash \rho \{ \} }_{\{ \} _{\backslash \sigma \{ \mu \} \backslash \nu \{ \} }}$	$\nabla_{\mu}^{*} \rightarrow {}^0\nabla_{\mu}^{*}, R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}{}_{\sigma\mu\nu}$	also valid, but with $\nabla_{\mu}^{*} \rightarrow {}^0\nabla_{\mu}^{*}, R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}{}_{\sigma\mu\nu}$ and
1510.06699_F00523	23	■ 0.976	$T^{\{ *\} }_{\backslash \sigma \{ \} }_{\{ \} _{\backslash \mu \{ \} \backslash \nu \{ \} }} \backslash \rightarrow^{\{ \} }_{\{0\}} T^{\{ *\} }_{\backslash \sigma \{ \} }_{\{ \} _{\backslash \mu \{ \} \backslash \nu \{ \} }}=0$	$T^{*\sigma}{}_{\mu\nu} \rightarrow {}^0T^{*\sigma}{}_{\mu\nu} = 0$	$T^{*\sigma}{}_{\mu\nu} \rightarrow {}^0T^{*\sigma}{}_{\mu\nu} = 0$.
1510.06699_F00524	23	■ 1.000	$P_{\{ \backslash \nu \} } \backslash \equiv \backslash \partial_{\{ \backslash \nu \} } \backslash \rho, \backslash \mathrm{cal} \{ P \} _{\{ a \} } \backslash \equiv \mathrm{h}_{\{ a \} }^{\{ \} _{\backslash \nu \} } P_{\{ \backslash \nu \} }$	$P_{\nu} \equiv \partial_{\nu}\rho, \mathcal{P}_a \equiv h_a{}^{\nu}P_{\nu}$	where $P_{\nu} \equiv \partial_{\nu}\rho, \mathcal{P}_a \equiv h_a{}^{\nu}P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00525	23	■ 1.000	$\backslash \theta$	θ	where $P_{\nu} \equiv \partial_{\nu}\rho, \mathcal{P}_a \equiv h_a{}^{\nu}P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00526	23	■ 1.000	$w=-2$	$w = -2$	forms covariantly with weight $w = -2$ and also acts as
1510.06699_F00527	24	■ 1.000	$\backslash \mathrm{cal} \{ T \}^{\{ a \} }_{\{ \} _{\backslash b \{ c \} }}=0$	$\mathcal{T}^a{}_{bc} = 0$	field originally has vanishing PGT torsion $\mathcal{T}^a{}_{bc} = 0$, the
1510.06699_F00528	24	■ 1.000	$\backslash \mathrm{cal} \{ T \}^{\{ \prime \} }_{\backslash \mathrm{prime} \{ a \} }_{\{ \} _{\backslash b \{ c \} }} \backslash \neq 0$	$\mathcal{T}^{\prime a}{}_{bc} \neq 0$	means that the local dilation leads to $\mathcal{T}^{\prime a}{}_{bc} \neq 0$.
1510.06699_F00529	24	■ 1.000	$\backslash \mathrm{cal} \{ T \}^{\{ \dagger \} }_{\backslash \mathrm{dagger} \{ a \} }_{\{ \} _{\backslash b \{ c \} }}$	$\mathcal{T}^{\dagger a}{}_{bc}$	Thus, for general values of θ , neither $\mathcal{R}^{ab}{}_{cd}$ nor $\mathcal{T}^{\dagger a}{}_{bc}$
1510.06699_F00530	24	■ 1.000	$\backslash \theta=0$	$\theta = 0$	transforms covariantly, but for $\theta = 0$ one recovers the
1510.06699_F00531	24	■ 1.000	$\backslash \theta=1$	$\theta = 1$	curvature under local dilations, whereas for $\theta = 1$ one
1510.06699_F00532	24	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{cal} \{ R \}^{\{ a \} }_{\{ \} _{\backslash b \{ \} \backslash c \{ d \} }}$	${}^0\mathcal{R}^{ab}{}_{cd}$	${}^0\mathcal{R}^{ab}{}_{cd}$ and its contractions (see Appendix B) are given
1510.06699_F00533	24	■ 0.971	$\{ \}^{\{0\}} \backslash \mathrm{cal} \{ T \}^{\{ a \} }_{\{ \} _{\backslash b \{ c \} }}$	${}^0\mathcal{T}^a{}_{bc}$	in Section V B; the ‘reduced’ PGT torsion ${}^0\mathcal{T}^a{}_{bc}$ vanishes
1510.06699_F00534	24	■ 1.000	$D_{\{ \backslash \mu \} }^{\{ \} _{\backslash \mathrm{dagger} \} } \backslash \varphi$	$D_{\mu}^{\dagger}\varphi$	tive $D_{\mu}^{\dagger}\varphi$ by introducing a dilation vector gauge field
1510.06699_F00535	24	■ 1.000	$V_{\{ \backslash \mu \} }$	V_{μ}	(called V_{μ} to distinguish it from the dilation gauge field
1510.06699_F00536	24	■ 0.999	$\backslash \mathrm{cal} \{ V \} _{\{ a \} }=\mathrm{h}_{\{ a \} }^{\{ \} _{\backslash \mu \} } V_{\{ \backslash \mu \} }$	$\mathcal{V}_a = h_a{}^{\mu}V_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu}V_{\mu}$. Clearly, by construction, $A^{\dagger ab}{}_{\mu}$ is
1510.06699_F00537	24	■ 0.999	$A^{\{ \mathrm{dagger} \{ a \} \} }_{\{ \} _{\backslash b \{ \} \backslash \mu \} }$	$A^{\dagger ab}{}_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu}V_{\mu}$. Clearly, by construction, $A^{\dagger ab}{}_{\mu}$ is
1510.06699_F00538	24	■ 1.000	$\backslash \partial_{\{ \backslash \mu \} } \backslash \varphi^{\{ \prime \} }$	$\partial_{\mu}\varphi'$	covariant derivative to cancel terms arising from $\partial_{\mu}\varphi'$
1510.06699_F00539	24	■ 1.000	$T_{\{ \backslash \mu \} }=$	$T_{\mu} =$	in which w is the Weyl weight of the field φ and $T_{\mu} =$
1510.06699_F00540	24	■ 0.999	$b^{\{ a \} }_{\{ \} _{\backslash \mu \} } \backslash \mathrm{cal} \{ T \} _{\{ a \} }$	$b^a{}_{\mu}\mathcal{T}_a$	$b^a{}_{\mu}\mathcal{T}_a$, where $\mathcal{T}_a$ is the trace of the PGT torsion (see
1510.06699_F00541	24	■ 1.000	$T_{\{ \backslash \mu \} }$	T_{μ}	equation (66). Here we consider T_{μ} merely as a short-
1510.06699_F00542	24	■ 1.000	$T_{\{ \backslash \mu \} }^{\{ \prime \} }=T_{\{ \backslash \mu \} }+3(1-\backslash \theta) P_{\{ \backslash \mu \} }$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$ and so (143) does indeed trans-
1510.06699_F00543	24	■ 0.898	$A^{\{ \mathrm{dagger} \{ a \} \} }_{\{ \} _{\backslash \mu \} }=A^{\{ \mathrm{dagger} \{ a \} \} }_{\{ \} _{\backslash \mu \} }$	$A^{\dagger ab}{}_{\mu} = A^{\dagger ab}{}_{\mu}$	tions (136)–(137), i.e. $A^{\dagger ab}{}_{\mu} = A^{\dagger ab}{}_{\mu}$ and so it has a
1510.06699_F00544	24	■ 1.000	V	V	covariant derivative (143) and the V -field transformation
1510.06699_F00545	25	■ 1.000	$V_{\{ \backslash \mu \} }=0$	$V_{\mu} = 0$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}{}_{\mu} = A^{ab}{}_{\mu}$
1510.06699_F00546	25	■ 1.000	$A^{\{ \mathrm{dagger} \{ a \} \} }_{\{ \} _{\backslash \mu \} }=A^{\{ a \} }_{\{ \} _{\backslash b \{ \} \backslash \mu \} }$	$A^{\dagger ab}{}_{\mu} = A^{ab}{}_{\mu}$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}{}_{\mu} = A^{ab}{}_{\mu}$
1510.06699_F00547	25	■ 1.000	$\backslash \left. D_{\{ \backslash \mu \} }^{\{ \mathrm{dagger} \} } \backslash \varphi \right _{\{ \backslash \theta =0 \} }=\backslash \left(D_{\{ \backslash \mu \} }-\frac{1}{3} w T_{\{ \backslash \mu \} } \right) \backslash \varphi$	$D_{\mu}^{\dagger}\varphi _{\theta=0} = \left(D_{\mu} - \frac{1}{3}wT_{\mu} \right)\varphi$	as $D_{\mu}^{\dagger}\varphi _{\theta=0} = (D_{\mu} - \frac{1}{3}wT_{\mu})\varphi$, in which the role played
1510.06699_F00548	25	■ 1.000	$T_{\{ \backslash \mu \} }=0$	$T_{\mu} = 0$	variantly set the PGT torsion to zero, and hence $T_{\mu} = 0$,
1510.06699_F00549	25	■ 1.000	$\backslash \left. D_{\{ \backslash \mu \} }^{\{ \mathrm{dagger} \} } \backslash \varphi \right _{\{ \backslash \theta =1 \} } \backslash \equiv \backslash \left(\backslash \partial_{\{ \backslash \mu \} }+\frac{1}{2} A^{\{ \mathrm{dagger} \{ a \} \} }_{\{ \} _{\backslash \mu \} } \backslash \Sigma_{\{ a \} \backslash b \} }-w V_{\{ \backslash \mu \} } \right) \backslash \varphi$	$D_{\mu}^{\dagger}\varphi _{\theta=1} \equiv \left(\partial_{\mu} + \frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab} - wV_{\mu} \right)\varphi$	$D_{\mu}^{\dagger}\varphi _{\theta=1} \equiv (\partial_{\mu} + \frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab} - wV_{\mu})\varphi$.
1510.06699_F00550	25	■ 1.000	$\backslash \partial_{\{ \backslash \mu \} }^{\{ \mathrm{dagger} \} } \backslash \varphi(x)$	$\partial_{\mu}^{\dagger}\varphi(x)$	It is straightforward to show that $\partial_{\mu}^{\dagger}\varphi(x)$ itself also trans-
1510.06699_F00551	25	■ 1.000	$w-1$	$w - 1$	which transforms covariantly with weight $w - 1$, as de-
1510.06699_F00552	25	■ 0.999	$\backslash \mathrm{cal} \{ L \} _{\{ \backslash \mathrm{M} \} } \backslash \equiv \mathrm{h}^{\{ -1 \} } L_{\{ \backslash \mathrm{M} \} } \backslash \left(\backslash \varphi, \backslash \mathrm{cal} \{ D \}^{\{ \mathrm{dagger} \} } \backslash \varphi \right)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1}L_{\mathrm{M}}(\varphi, \mathcal{D}^{\dagger}\varphi)$	action by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1}L_{\mathrm{M}}(\varphi, \mathcal{D}^{\dagger}\varphi)$.

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1510.06699_F00553	25	■ 1.000	$\phi^{\{2\}} \mathcal{R}^{\dagger}$	$\phi^2 \mathcal{R}^\dagger$	tional to $\phi^2 \mathcal{R}^\dagger$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00554	25	■ 1.000	$\phi^{\{2\}} L_{\mathcal{T}^{\dagger 2}}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	tional to $\phi^2 \mathcal{R}^\dagger$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00555	25	■ 0.968	$\operatorname{SU}(2,2)$	$\mathrm{SU}(2,2)$	of $\mathrm{SU}(2,2)$, to which $\mathrm{O}(4,2)$ is locally isomorphic. In
1510.06699_F00556	25	■ 0.968	$\mathrm{O}(4,2)$	$\mathrm{O}(4,2)$	of $\mathrm{SU}(2,2)$, to which $\mathrm{O}(4,2)$ is locally isomorphic. In
1510.06699_F00557	26	■ 1.000	$e_{\{b \ \mu\}} b_{\{a\}} - e_{\{a \ \mu\}} b_{\{b\}}$	$e_{b\mu} b_a - e_{a\mu} b_b$	in which a piece $e_{b\mu} b_a - e_{a\mu} b_b$ is added to the spin connec-
1510.06699_F00558	26	■ 1.000	$\omega_{\mu a b}$	$\omega_{\mu ab}$	tion $\omega_{\mu ab}$ (their b -field is the ‘dilatation field’, which in
1510.06699_F00559	26	■ 0.999	$\mathcal{V}$	$\mathcal{V}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00560	26	■ 0.999	$e_{\{a \ \mu\}}$	$e_{a\mu}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00561	26	■ 1.000	$c_{\{b \ \mu\}} x_{\{a\}} - c_{\{a \ \mu\}} x_{\{b\}}$	$c_{b\mu} x_a - c_{a\mu} x_b$	$c_{b\mu} x_a - c_{a\mu} x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00562	26	■ 1.000	$c_{\{a \ \mu\}}$	$c_{a\mu}$	$c_{b\mu} x_a - c_{a\mu} x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00563	26	■ 1.000	$x_{\{a\}}$	x_a	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00564	26	■ 1.000	$\mathcal{V}_{\{a\}}$	$\mathcal{V}_a$	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00565	26	■ 1.000	$x \cdot \nabla$	$x \cdot \nabla$	that the generator of global dilations is $x \cdot \nabla$), and so this
1510.06699_F00566	26	■ 1.000	$\{ \}^{\{0\}} \nabla_{\{ \mu\}}^{\dagger} \equiv \{ \}^{\{0\}} \nabla_{\{ \mu\}} - w \left(V_{\{ \mu\}} + \frac{1}{3} T_{\{ \mu\}} \right), \{ \}^{\{0\}} \Gamma^{\lambda}_{\{ \mu\} \{ \nu\}} \equiv \frac{1}{2} \gamma^{\lambda \rho} (\partial_{\{ \mu\}} \gamma_{\{ \nu\} \rho} +$	${}^0 \nabla_{\mu}^{\dagger} \equiv {}^0 \nabla_{\mu} - w \left(V_{\mu} + \frac{1}{3} T_{\mu} \right), {}^0 \Gamma^{\lambda}_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho} (\partial_{\mu} \gamma_{\nu \rho} +$	where ${}^0 \nabla_{\mu}^{\dagger} \equiv {}^0 \nabla_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\mu}), {}^0 \Gamma^{\lambda}_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho} (\partial_{\mu} \gamma_{\nu \rho} +$
1510.06699_F00567	26	■ 0.817	$\partial_{\{ \nu\}} \gamma_{\{ \mu\} \rho} - \partial_{\rho} \gamma_{\{ \mu\} \nu}$	$\partial_{\nu} \gamma_{\mu \rho} - \partial_{\rho} \gamma_{\mu \nu}$	$\partial_{\nu} \gamma_{\mu \rho} - \partial_{\rho} \gamma_{\mu \nu}$) is the metric connection corresponding to
1510.06699_F00568	26	■ 0.801	$D_{\{ \mu\}}$	D_{μ}	$D_{\mu}^{\dagger}$ is applied, and w is its Weyl weight.
1510.06699_F00569	26	■ 1.000	$T^a_{\{ \mu\} \{ \nu\}}$	$T^a_{\mu \nu}$	vature) in PGT (and WGT), $T^a_{\mu \nu}$ is the PGT transla-
1510.06699_F00570	26	■ 1.000	$D_{\{ \mu\}}$	D_{μ}	tional gauge field (torsion), and $\bar{D}_{\mu}$ is the PGT covariant
1510.06699_F00571	26	■ 1.000	$R^{\dagger ab}_{\{ \mu\} \{ \nu\}}$	$R^{\dagger ab}_{\mu \nu}$	As in WGT, both $R^{\dagger ab}_{\mu \nu}$ and $H^{\dagger}_{\mu \nu}$ transform covariantly
1510.06699_F00572	26	■ 1.000	$H_{\{ \mu\} \{ \nu\}}^{\dagger}$	$H^{\dagger}_{\mu \nu}$	As in WGT, both $R^{\dagger ab}_{\mu \nu}$ and $H^{\dagger}_{\mu \nu}$ transform covariantly
1510.06699_F00573	26	■ 1.000	$\mathcal{R}^{\dagger ab}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$	$\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$,
1510.06699_F00574	26	■ 1.000	$\mathcal{H}_{\{ \nu\} \{ \mu\}}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$	$\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$,
1510.06699_F00575	27	■ 1.000	$\mathcal{R}^{\dagger ab}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$	It is easy to show that, similarly to WGT, $\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$
1510.06699_F00576	27	■ 1.000	$\mathcal{T}^{\dagger a}_{\{ \nu\} \{ \mu\}}$	$\mathcal{T}^{\dagger a}_{cd}$	and $\mathcal{T}^{\dagger a}_{cd}$ transform covariantly under extended local
1510.06699_F00577	27	■ 1.000	$w \left(\mathcal{R}^{\dagger ab}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu} \right) = w \left(\mathcal{H}_{\{ \nu\} \{ \mu\}}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger} \right) = -2$	$w \left(\mathcal{R}^{\dagger ab}_{cd} \right) = w \left(\mathcal{H}_{cd}^{\dagger} \right) = -2$	dilations with weights $w(\mathcal{R}^{\dagger ab}_{cd}) = w(\mathcal{H}_{cd}^{\dagger}) = -2$ and
1510.06699_F00578	27	■ 0.999	$w \left(\mathcal{T}^{\dagger a}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} T^{\dagger a}_{\mu \nu} \right) = -1$	$w \left(\mathcal{T}^{\dagger a}_{cd} \right) = -1$	$w(\mathcal{T}^{\dagger a}_{cd}) = -1$ respectively.
1510.06699_F00579	27	■ 1.000	$\mathcal{R}^{\dagger ab}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$	$\mathcal{R}^{\dagger ab}_{cd}$	plicit forms of $\mathcal{R}^{\dagger ab}_{cd}$ and its contractions are given in
1510.06699_F00580	27	■ 1.000	$\mathcal{H}_{\{ \nu\} \{ \mu\}}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$	$\mathcal{H}_{ab}^{\dagger}$	strength $\mathcal{H}_{ab}^{\dagger}$ is antisymmetric in a and b and hence also
1510.06699_F00581	27	■ 0.725	$\mathcal{R}^{\dagger ab}_{\{ \nu\} \{ \mu\}} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu \nu}$	$\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$	tives are $\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$ and
1510.06699_F00582	27	■ 1.000	$\mathcal{H}_{\{ \nu\} \{ \mu\}}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu \nu}^{\dagger}$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$. It is important to note that
1510.06699_F00583	27	■ 1.000	$\partial h, \partial^2 h, A$	$\partial h, \partial^2 h, A$	strength $\mathcal{H}_{ab}^{\dagger}$, which depends on $\partial h, \partial^2 h, A$ and ∂A , in
1510.06699_F00584	27	■ 1.000	$\partial^2 h$	$\partial^2 h$	tive $\partial^2 h$ is a particularly unusual feature as compared
1510.06699_F00585	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}_{\{ \nu\} \{ \mu\}}^{\dagger} \equiv \{ \}^{\{0\}} \mathcal{A}_{\{ \nu\} \{ \mu\}} - \frac{1}{2} (c_{\{ \nu\} \{ \mu\}}^{\dagger} +$	${}^0 \mathcal{A}_{abc}^{\dagger} \equiv \frac{1}{2} (c_{abc}^{\dagger} +$	in which we have defined the quantities ${}^0 \mathcal{A}_{abc}^{\dagger} \equiv \frac{1}{2} (c_{abc}^{\dagger} +$
1510.06699_F00586	27	■ 0.973	$c_{\{ \nu\} \{ \mu\}}^{\dagger} - c_{\{ \mu\} \{ \nu\}}^{\dagger})$	$c_{bca}^{\dagger} - c_{cab}^{\dagger})$	$c_{bca}^{\dagger} - c_{cab}^{\dagger}$), where $c^{\dagger c}_{ab} \equiv h_a^{\mu} h_b^{\nu} (\partial_{\mu}^{\dagger} b_{\nu}^c - \partial_{\nu}^{\dagger} b_{\mu}^c)$, and

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1510.06699_F00587	27	■ 0.973	$c^{\dagger c}_{ab} \equiv h_a{}^\mu h_b{}^\nu (\partial_\mu^\dagger b^c{}_\nu - \partial_\nu^\dagger b^c{}_\mu)$	$c^{\dagger c}_{ab} \equiv h_a{}^\mu h_b{}^\nu (\partial_\mu^\dagger b^c{}_\nu - \partial_\nu^\dagger b^c{}_\mu)$	$c_{bca}^\dagger - c_{cab}^\dagger$), where $c_{ab}^{\dagger c} \equiv h_a{}^\mu h_b{}^\nu (\partial_\mu^\dagger b^c{}_\nu - \partial_\nu^\dagger b^c{}_\mu)$, and
1510.06699_F00588	27	■ 0.981	$\mathcal{K}_{abc} \equiv -\frac{1}{2}(\mathcal{T}_{abc} + \mathcal{T}_{bca} - \mathcal{T}_{cab})$	$\mathcal{K}_{abc}^\dagger \equiv -\frac{1}{2}(\mathcal{T}_{abc}^\dagger + \mathcal{T}_{bca}^\dagger - \mathcal{T}_{cab}^\dagger)$	$\mathcal{K}_{abc}^\dagger \equiv -\frac{1}{2}(\mathcal{T}_{abc}^\dagger + \mathcal{T}_{bca}^\dagger - \mathcal{T}_{cab}^\dagger)$. As before, both ${}^0\mathcal{A}_{abc}^\dagger$ and
1510.06699_F00589	27	■ 0.981	$\{ \}^{\{0\}} \mathcal{A}_{abc}^\dagger$	${}^0\mathcal{A}_{abc}^\dagger$	$\mathcal{K}_{abc}^\dagger \equiv -\frac{1}{2}(\mathcal{T}_{abc}^\dagger + \mathcal{T}_{bca}^\dagger - \mathcal{T}_{cab}^\dagger)$. As before, both ${}^0\mathcal{A}_{abc}^\dagger$ and
1510.06699_F00590	27	■ 0.840	$\mathcal{K}_{abc}^\dagger$	$\mathcal{K}_{abc}^\dagger$	$\mathcal{K}_{abc}^\dagger$ are antisymmetric in their first two indices. There
1510.06699_F00591	27	■ 0.568	$\{ \}^{\{0\}} \mathcal{A}_{abc}^\dagger$	${}^0\mathcal{A}_{abc}^\dagger$	${}^0\mathcal{A}_{abc}^\dagger$ itself depends on the A gauge field, whereas this
1510.06699_F00592	27	■ 0.551	$\mathcal{K}_{abc}^\dagger$	$\mathcal{K}_{abc}^\dagger$	was not the case previously. Less importantly, $\mathcal{K}_{abc}^\dagger$ is
1510.06699_F00593	27	■ 0.992	$\mathcal{A}_{abc}^\dagger$	$\mathcal{A}_{abc}^\dagger$	quantities ${}^0\mathcal{A}_{abc}^\dagger$ transform in the same way as $\mathcal{A}_{abc}^\dagger$,
1510.06699_F00594	27	■ 1.000	$\{ \}^{\{0\}} D_\mu^\dagger \varphi$	${}^0D_\mu^\dagger \varphi$	in eWGT, however, is that ${}^0D_\mu^\dagger \varphi$ still depends on the
1510.06699_F00595	27	■ 0.523	$\{ \}^{\{0\}} \mathcal{D}_a^\dagger \equiv h_a{}^\mu {}^0D_\mu^\dagger$	${}^0\mathcal{D}_a^\dagger \equiv h_a{}^\mu {}^0D_\mu^\dagger$	${}^0\mathcal{D}_a^\dagger \equiv h_a{}^\mu {}^0D_\mu^\dagger$, which consequently transforms in the
1510.06699_F00596	27	■ 1.000	$\mathcal{D}_a^\dagger$	$\mathcal{D}_a^\dagger$	same way as $\mathcal{D}_a^\dagger$, as required. The ‘full’ generalised co-
1510.06699_F00597	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{D}_a^\dagger$	${}^0\mathcal{D}_a^\dagger$	to ${}^0\mathcal{D}_a^\dagger$.
1510.06699_F00598	27	■ 1.000	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{T}^{\dagger a}_{bc}$	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{T}^{\dagger a}_{bc}$	the gravitational gauge field strengths $\mathcal{R}^{\dagger ab}_{cd}, \mathcal{T}^{\dagger a}_{bc}$ and
1510.06699_F00599	28	■ 0.999	$\mathcal{T}_{abc}^\dagger = \epsilon_{abcd} t^d$	$\mathcal{T}_{abc}^\dagger = \epsilon_{abcd} t^d$	sion’ is totally antisymmetric, such that $\mathcal{T}_{abc}^\dagger = \epsilon_{abcd} t^d$
1510.06699_F00600	28	■ 0.800	$\mathcal{R}_{abcd}^\dagger$	$\mathcal{R}_{abcd}^\dagger$	and discussed in Section II G. In principle, from $\mathcal{R}_{abcd}^\dagger$,
1510.06699_F00601	28	■ 0.899	$\mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc}^\dagger$ and $\mathcal{H}_{ab}^\dagger$, one can construct a free gravitational ac-
1510.06699_F00602	28	■ 1.000	$\mathcal{R}^\dagger \equiv$	$\mathcal{R}^\dagger \equiv$	be quadratic in $\mathcal{R}_{abcd}^\dagger$ and $\mathcal{H}_{ab}^\dagger$, but terms linear in $\mathcal{R}^\dagger \equiv$
1510.06699_F00603	28	■ 1.000	$\mathcal{R}^{\dagger ab}_{ab}$	$\mathcal{R}^{\dagger ab}_{ab}$	$\mathcal{R}^{\dagger ab}_{ab}$ or quadratic in $\mathcal{T}_{abc}^\dagger$ are not allowed. Similarly,
1510.06699_F00604	28	■ 1.000	$L_{\mathcal{R}^{\dagger 2}}$	$L_{\mathcal{R}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00605	28	■ 1.000	$L_{\mathcal{H}^{\dagger 2}}$	$L_{\mathcal{H}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00606	28	■ 1.000	$\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$	$\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00607	28	■ 1.000	$\mathcal{H}_{ab} \rightarrow$	$\mathcal{H}_{ab} \rightarrow$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00608	28	■ 0.829	$\{ \}^{\{\underline{71}\}}$	$\underline{71}$	$\mathcal{H}_{ab}^\dagger$. Performing a similar derivation to the WGT case ⁷¹ ,
1510.06699_F00609	28	■ 0.556	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2 \mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00610	28	■ 0.556	$L_{\mathcal{T}^{\dagger 2}}$	$L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2 \mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00611	28	■ 0.999	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^\dagger$ (but with no term proportional to β_3 ,
1510.06699_F00612	28	■ 1.000	$\mathcal{H}_{ab}^\dagger \mathcal{H}^{\dagger ab}$	$\mathcal{H}_{ab}^\dagger \mathcal{H}^{\dagger ab}$	tional to $\mathcal{H}_{ab}^\dagger \mathcal{H}^{\dagger ab}$ in (166). In the matter sector, covari-
1510.06699_F00613	28	■ 1.000	$\varphi, \partial\varphi, h, \partial h, A$	$\varphi, \partial\varphi, h, \partial h, A$	on $\varphi, \partial\varphi, h, \partial h, A$ and V ; the dependence on ∂h is new
1510.06699_F00614	28	■ 0.998	$\mathcal{D}_a^\dagger \varphi$	$\mathcal{D}_a^\dagger \varphi$	$\mathcal{T}_a$ in $\mathcal{D}_a^\dagger \varphi$. We will also consider here the inclusion of
1510.06699_F00615	28	■ 1.000	∂V	∂V	tional dependence on ∂A and ∂V ; the latter is new com-
1510.06699_F00616	29	■ 0.626	$\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^\dagger(h, \partial h, A)$	$\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^\dagger(h, \partial h, A)$	strength tensors $\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^\dagger(h, \partial h, A)$
1510.06699_F00617	29	■ 1.000	$\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^\dagger$	$\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^\dagger$	the term proportional to $\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^\dagger$ is included in $\mathcal{L}_G$, then
1510.06699_F00618	29	■ 1.000	$t^a{}_\mu \equiv \delta\mathcal{L}_G/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	$t^a{}_\mu \equiv \delta\mathcal{L}_G/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	sector, $t^a{}_\mu \equiv \delta\mathcal{L}_G/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ and $j^\mu \equiv$
1510.06699_F00619	29	■ 1.000	$j^\mu \equiv$	$j^\mu \equiv$	sector, $t^a{}_\mu \equiv \delta\mathcal{L}_G/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ and $j^\mu \equiv$
1510.06699_F00620	29	■ 1.000	$\delta\mathcal{L}_G/\delta V_\mu$	$\delta\mathcal{L}_G/\delta V_\mu$	$\delta\mathcal{L}_G/\delta V_\mu$ respectively. In the matter sector, the energy-

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1510.06699_F00621	29	■ 1.000	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta V_\mu$	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta V_\mu$	$\sigma_{ab}{}^\mu \equiv \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ and dilation current $\zeta^\mu \equiv \delta \mathcal{L}_M / \delta V_\mu$
1510.06699_F00622	29	■ 0.998	$t^a{}_b$	$t^a{}_b$	to WGT, however, the individual quantities $t^a{}_b$ and $\tau^a{}_b$
1510.06699_F00623	29	■ 1.000	$t^a{}_a$	$t^a{}_a$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00624	29	■ 1.000	$\tau^a{}_a$	$\tau^a{}_a$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00625	29	■ 1.000	t_{ab}	t_{ab}	momentum tensor. Nonetheless, although neither t_{ab} nor
1510.06699_F00626	29	■ 1.000	$\tau^{\dagger a}{}_a = \tau^a{}_a$	$\tau^{\dagger a}{}_a = \tau^a{}_a$	is invariant). It is also worth noting that $\tau^{\dagger a}{}_a = \tau^a{}_a$.
1510.06699_F00627	29	■ 0.999	$t_{ab}^{\dagger}$	$t_{ab}^{\dagger}$	One may construct a covariant $t_{ab}^{\dagger}$ in a similar way. In-
1510.06699_F00628	29	■ 1.000	$\mathcal{T}_a^{\dagger} \equiv 0$	$\mathcal{T}_a^{\dagger} \equiv 0$	ever, since $\mathcal{T}_a^{\dagger} \equiv 0$. Consequently, in eWGT, one finds
1510.06699_F00629	29	■ 1.000	$\bar{\partial} L_M / \partial \varphi \equiv [\partial L_M(\varphi, \mathcal{D}_a^{\dagger} u, \phi, \mathcal{D}_a^{\dagger} \phi, \dots) / \partial \varphi]_{u=\varphi}$	$\bar{\partial} L_M / \partial \varphi \equiv [\partial L_M(\varphi, \mathcal{D}_a^{\dagger} u, \phi, \mathcal{D}_a^{\dagger} \phi, \dots) / \partial \varphi]_{u=\varphi}$	where $\bar{\partial} L_M / \partial \varphi \equiv [\partial L_M(\varphi, \mathcal{D}_a^{\dagger} u, \phi, \mathcal{D}_a^{\dagger} \phi, \dots) / \partial \varphi]_{u=\varphi}$ so
1510.06699_F00630	30	■ 1.000	$H_{ab}^{\dagger}$	$H_{ab}^{\dagger}$	strength tensor $H_{ab}^{\dagger}$ of the vector gauge field (152), but
1510.06699_F00631	30	■ 1.000	$H_{ab} = h_a{}^\mu h_b{}^\nu (\partial_\mu V_\nu - \partial_\nu V_\mu)$	$H_{ab} = h_a{}^\mu h_b{}^\nu (\partial_\mu V_\nu - \partial_\nu V_\mu)$	instead the simpler object $H_{ab} = h_a{}^\mu h_b{}^\nu (\partial_\mu V_\nu - \partial_\nu V_\mu)$,
1510.06699_F00632	30	■ 1.000	$j^{\dagger a}$	$j^{\dagger a}$	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00633	30	■ 1.000	j^a	j^a	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00634	30	■ 1.000	$\varphi, \phi, h_a^\mu, A^{ab}{}_\mu, V_\mu$	$\varphi, \phi, h_a^\mu, A^{ab}{}_\mu, V_\mu$	from the set $\varphi, \phi, h_a^\mu, A^{ab}{}_\mu, V_\mu$ and their derivatives,
1510.06699_F00635	30	■ 1.000	φ, ϕ	φ, ϕ	with which we have worked so far, to the new set φ, ϕ ,
1510.06699_F00636	30	■ 1.000	$h_a^\mu, A^{\dagger ab}{}_\mu, V_\mu$	$h_a^\mu, A^{\dagger ab}{}_\mu, V_\mu$	$h_a^\mu, A^{\dagger ab}{}_\mu, V_\mu$ and their derivatives (i.e. we replace $A^{ab}{}_\mu$
1510.06699_F00637	30	■ 1.000	$h_a^\mu, A^{ab}{}_\mu, V_\mu$	$h_a^\mu, A^{ab}{}_\mu, V_\mu$	of $h_a^\mu, A^{ab}{}_\mu, V_\mu$ by its defining relationship (142), rather
1510.06699_F00638	30	■ 1.000	$\mathcal{T}_b^{\natural} \equiv \mathcal{T}^{\natural a}{}_{ba}$	$\mathcal{T}_b^{\natural} \equiv \mathcal{T}^{\natural a}{}_{ba}$	and the corresponding trace $\mathcal{T}_b^{\natural} \equiv \mathcal{T}^{\natural a}{}_{ba}$. It is straight-
1510.06699_F00639	30	■ 1.000	$\mathcal{T}_b^{\natural} = \mathcal{T}_b + 3\mathcal{V}_b$	$\mathcal{T}_b^{\natural} = \mathcal{T}_b + 3\mathcal{V}_b$	forward to show that $\mathcal{T}_b^{\natural} = \mathcal{T}_b + 3\mathcal{V}_b$, where $\mathcal{T}_b$ is the
1510.06699_F00640	31	■ 0.997	$h, \partial h$	$h, \partial h$	which is expressible wholly in terms of the fields $h, \partial h$
1510.06699_F00641	31	■ 1.000	$A^{\dagger}$	$A^{\dagger}$	and $A^{\dagger}$ (suppressing indices). Since (182) does not de-
1510.06699_F00642	31	■ 0.566	$\mathcal{R}_{abcd}^{\dagger}(h, A^{\dagger}, \partial A^{\dagger}), \mathcal{T}_{abc}^{\dagger}(h, \partial h, A^{\dagger})$	$\mathcal{R}_{abcd}^{\dagger}(h, A^{\dagger}, \partial A^{\dagger}), \mathcal{T}_{abc}^{\dagger}(h, \partial h, A^{\dagger})$	tensors $\mathcal{R}_{abcd}^{\dagger}(h, A^{\dagger}, \partial A^{\dagger}), \mathcal{T}_{abc}^{\dagger}(h, \partial h, A^{\dagger})$ and
1510.06699_F00643	31	■ 1.000	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A^{\dagger}, \partial A^{\dagger})$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A^{\dagger}, \partial A^{\dagger})$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A^{\dagger}, \partial A^{\dagger})$, where once again the last
1510.06699_F00644	31	■ 1.000	$\mathcal{L}$	$\mathcal{L}$	derivative of a Lagrangian $\mathcal{L}$ with respect to any one
1510.06699_F00645	31	■ 1.000	$(\delta \mathcal{L} / \delta \chi)_{\dagger}$	$(\delta \mathcal{L} / \delta \chi)_{\dagger}$	of the fields χ by $(\delta \mathcal{L} / \delta \chi)_{\dagger}$ to distinguish it from the
1510.06699_F00646	31	■ 1.000	$\delta \mathcal{L} / \delta \chi$	$\delta \mathcal{L} / \delta \chi$	variational derivative $\delta \mathcal{L} / \delta \chi$ obtained previously using
1510.06699_F00647	31	■ 1.000	$\sigma_{ab}{}^\mu$	$\sigma_{ab}{}^\mu$	ther that the spin-angular momentum tensor $\sigma_{ab}{}^\mu$ and
1510.06699_F00648	31	■ 1.000	$\zeta^{\dagger \mu} = 0$	$\zeta^{\dagger \mu} = 0$	gauge field V_μ or its derivative, then $\zeta^{\dagger \mu} = 0$ and so
1510.06699_F00649	31	■ 1.000	$\phi, h_a{}^\mu, A^{ab}{}_\mu, V_\mu$	$\phi, h_a{}^\mu, A^{ab}{}_\mu, V_\mu$	$\phi, h_a{}^\mu, A^{ab}{}_\mu, V_\mu$ and their derivatives, but we will use the
1510.06699_F00650	32	■ 1.000	$\mathcal{D}_a^* \rightarrow \mathcal{D}_a^{\dagger}$	$\mathcal{D}_a^* \rightarrow \mathcal{D}_a^{\dagger}$	with $\mathcal{D}_a^* \rightarrow \mathcal{D}_a^{\dagger}$. Similar to WGT, since both ψ and $\bar{\psi}$ have
1510.06699_F00651	32	■ 1.000	$\gamma^a \left(\mathcal{D}_a + \frac{1}{2} \mathcal{T}_a \right) \psi = \gamma^a \mathcal{D}_a^{\dagger} \psi$	$\gamma^a \left(\mathcal{D}_a + \frac{1}{2} \mathcal{T}_a \right) \psi = \gamma^a \mathcal{D}_a^{\dagger} \psi$	noting that $\gamma^a (\mathcal{D}_a + \frac{1}{2} \mathcal{T}_a) \psi = \gamma^a \mathcal{D}_a^{\dagger} \psi$, where we have
1510.06699_F00652	32	■ 1.000	$\gamma^a \Sigma_{ba} = \frac{1}{4} \gamma^a [\gamma_b, \gamma_a] = -\frac{3}{2} \gamma_b$	$\gamma^a \Sigma_{ba} = \frac{1}{4} \gamma^a [\gamma_b, \gamma_a] = -\frac{3}{2} \gamma_b$	$\gamma^a \Sigma_{ba} = \frac{1}{4} \gamma^a [\gamma_b, \gamma_a] = -\frac{3}{2} \gamma_b$. We also note that (188)
1510.06699_F00653	32	■ 0.997	$\mathcal{T}_{[abc]} = 0$	$\mathcal{T}_{[abc]} = 0$	presence of torsion (even in the case $\mathcal{T}_{[abc]} = 0$) will still

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1510.06699_F00654	32	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{a\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{a\}}^{\dagger}$	${}^0\mathcal{D}_a \rightarrow {}^0\mathcal{D}_a^\dagger$	again satisfied, but with the replacements ${}^0\mathcal{D}_a \rightarrow {}^0\mathcal{D}_a^\dagger$
1510.06699_F00655	32	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{[a\ b\ c]\}} \rightarrow \backslash \mathrm{mathcal{T}}_{\{[a\ b\ c]\}}^{\dagger}$	$\mathcal{T}_{[abc]} \rightarrow \mathcal{T}_{[abc]}^\dagger$	and $\overline{\mathcal{T}_{[abc]}} \rightarrow \overline{\mathcal{T}_{[abc]}^\dagger}$. Thus, only the total antisymmetric
1510.06699_F00656	32	■ 1.000	$\backslash \mathrm{sigma}_{\{a\ b\}}^{\{ \}^{\{c\}}}$	$\sigma_{ab}{}^c$	$\sigma_{ab}{}^c$ is already manifestly eWGT covariant. For the
1510.06699_F00657	32	■ 1.000	$\backslash \mathrm{sigma}_{\{a\ b\}}^{\{ \}^{\{b\}}} = 0$	$\sigma_{ab}{}^b = 0$	Recalling that $\sigma_{ab}{}^b = 0$ for Dirac matter, the transfor-
1510.06699_F00658	32	■ 0.444	$\backslash \mathrm{beta}_{\{1\}}$	β_1	stants β_1 and β_2 in $L_{\mathcal{T}^{12}}$. It is worth noting that $\mu\phi$ has
1510.06699_F00659	32	■ 0.444	$\backslash \mathrm{beta}_{\{2\}}$	β_2	stants β_1 and β_2 in $L_{\mathcal{T}^{12}}$. It is worth noting that $\mu\phi$ has
1510.06699_F00660	32	■ 0.444	$L_{\{ \backslash \mathrm{mathcal{D}}_{\{ \backslash \mathrm{mathcal{T}} \}^{\{ + \}^{\{ 2 \} } } \}}$	$L_{\mathcal{T}+\epsilon}$	stants β_1 and β_2 in $L_{\mathcal{T}^{12}}$. It is worth noting that $\mu\phi$ has
1510.06699_F00661	32	■ 0.998	$\widehat{\backslash \mathrm{mathcal{F}}_{\{a\ b\}}^{\{ \}^{\{ * \} } } \rightarrow \widehat{\backslash \mathrm{mathcal{F}}_{\{a\ b\}}^{\{ \}^{\{ \dagger \} } }}, \backslash \mathrm{mathcal{T}}_{\{ \}^{\{ * \}^{\{ c \} } }^{\{ \}^{\{ a \}^{\{ b \} } } } \rightarrow \backslash \mathrm{mathcal{T}}_{\{ \}^{\{ \dagger \}^{\{ c \} } }^{\{ \}^{\{ a \}^{\{ b \} } } } }$	$\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}^{*c}{}_{ab} \rightarrow \mathcal{T}^{\dagger c}{}_{ab}$	$\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}^{*c}{}_{ab} \rightarrow \mathcal{T}^{\dagger c}{}_{ab}$ and $\mathcal{D}_a^* \rightarrow \mathcal{D}_a^\dagger$, and recalling
1510.06699_F00662	33	■ 1.000	$\backslash \mathrm{tau}^{\{ \dagger \}^{\{ a \} } }_{\{ b \}} = \backslash \mathrm{tau}^{\{ a \}^{\{ \dagger \} } }_{\{ b \}}$	$\tau^{\dagger a}{}_b = \tau^a{}_b$	deed, from (170), one sees immediately that $\overline{\tau^{\dagger a}{}_b} = \overline{\tau^a{}_b}$,
1510.06699_F00663	33	■ 1.000	$\backslash \mathrm{varphi}, \backslash \mathrm{phi}, h_{\{a\}}^{\{ \}^{\{ \mu \} } }$	φ, ϕ, h_a^μ	set of field variables φ, ϕ, h_a^μ and $A^{\dagger ab}{}_\mu$ discussed in
1510.06699_F00664	33	■ 0.607	$\widehat{\backslash \mathrm{mathcal{A}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ c \} } } \equiv \widehat{\backslash \mathrm{mathcal{A}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ \mu \} } } \widehat{\backslash \mathrm{mathcal{A}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ \mu \} } } }$	$\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$	variables $\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$
1510.06699_F00665	33	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } \backslash \mathrm{chi} = \left(\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } - \frac{1}{3} w \backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } \right) \backslash \mathrm{chi}$	$\mathcal{D}_a^\dagger \chi = \left(\mathcal{D}_a^\dagger - \frac{1}{3} w \mathcal{T}_a^\dagger \right) \chi$	written in the form $\overline{\mathcal{D}_a^\dagger \chi} = (\overline{\mathcal{D}_a^\dagger} - \frac{1}{3} w \overline{\mathcal{T}_a^\dagger}) \overline{\chi}$, where $\mathcal{D}_a^\dagger$
1510.06699_F00666	33	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \natural \} } }$	$\mathcal{D}_a^\natural$	written in the form $\overline{\mathcal{D}_a^\dagger \chi} = (\overline{\mathcal{D}_a^\dagger} - \frac{1}{3} w \overline{\mathcal{T}_a^\dagger}) \overline{\chi}$, where $\mathcal{D}_a^\natural$
1510.06699_F00667	33	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \natural \} } }$	$\mathcal{T}_a^\natural$	and $\mathcal{T}_a^\natural$ are defined in (180) and (181), respectively. In
1510.06699_F00668	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \natural \} } } }$	$\widehat{\mathcal{D}}_a^\natural$	an analogous manner, we may define the quantities $\widehat{\mathcal{D}}_a^\natural$
1510.06699_F00669	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \natural \} } } }$	$\widehat{\mathcal{T}}_a^\natural$	and $\widehat{\mathcal{T}}_a^\natural$, for which each occurrence of $h_a{}^\mu$ and $A^{\dagger ab}{}_\mu$ in
1510.06699_F00670	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{A}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ \mu \} } }$	$\widehat{A}^{\dagger ab}{}_\mu$	terpart $\widehat{h}_a{}^\mu$ and $\widehat{A}^{\dagger ab}{}_\mu$, respectively. Thus, $\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$
1510.06699_F00671	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \natural \} } } } = \widehat{\backslash \mathrm{mathcal{H}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } }_{\{ \}^{\{ \mu \} } } \backslash \mathrm{D}_{\{ \mu \}}^{\{ \}^{\{ \natural \} } }$	$\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$	terpart $\widehat{h}_a{}^\mu$ and $\widehat{A}^{\dagger ab}{}_\mu$, respectively. Thus, $\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$
1510.06699_F00672	33	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } \backslash \mathrm{phi} = \frac{1}{3} (\phi^2 / \phi_0) \backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \natural \} } } / \backslash \mathrm{phi}_{\{0\}} \backslash \mathrm{right} \widehat{\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \natural \} } } }$	$\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$	be written as $\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$, whereas the covariant
1510.06699_F00673	33	■ 0.996	$\widehat{\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } } = \widehat{\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \natural \} } } } - \frac{1}{3} w \widehat{\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{ \}^{\{ \natural \} } } }$	$\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$	tive operator $\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$, which preserves the Weyl
1510.06699_F00674	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } } \widehat{\backslash \mathrm{mathcal{H}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } }$	$\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$	tor $(\phi / \phi_0)^{1-w}$, the scale-invariant quantity $\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$ has the
1510.06699_F00675	33	■ 1.000	$\widehat{\backslash \mathrm{mathcal{H}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } }, \widehat{\backslash \mathrm{mathcal{H}}_{\{a\}}^{\{ \}^{\{ \mu \} } } }$	$\widehat{\chi}, \widehat{h}_a{}^\mu$	same functional dependency on $\widehat{\chi}, \widehat{h}_a{}^\mu$, and $\widehat{A}^{\dagger ab}{}_\mu$, respec-
1510.06699_F00676	33	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{ \}^{\{ \dagger \} } } \backslash \mathrm{chi}$	$\mathcal{D}_a^\dagger \chi$	tively, as $\mathcal{D}_a^\dagger \chi$ does on $\chi, h_a{}^\mu$ and $A^{\dagger ab}{}_\mu$. Similarly, one
1510.06699_F00677	33	■ 0.999	$\backslash \mathrm{mathcal{R}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ c \}^{\{ d \} } } } = \left(\phi / \phi_0 \right)^2 \widehat{\backslash \mathrm{mathcal{R}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ c \}^{\{ d \} } } } }$	$\mathcal{R}^{\dagger ab}{}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{\dagger ab}{}_{cd}$	defined in (41) may be written as $\overline{\mathcal{R}^{\dagger ab}{}_{cd}} = (\phi / \phi_0)^2 \overline{\widehat{\mathcal{R}}^{\dagger ab}{}_{cd}}$,
1510.06699_F00678	33	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ c \}^{\{ d \} } } } = \left(\phi / \phi_0 \right) \backslash \mathrm{mathcal{T}}_{\{ \}^{\{ \dagger \}^{\{ a \}^{\{ b \} } } } }_{\{ \}^{\{ c \}^{\{ d \} } } } }$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$ and $\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$, where each
1510.06699_F00679	33	■ 1.000	$\backslash \mathrm{mathcal{H}}_{\{a\ b\}}^{\{ \}^{\{ \dagger \} } } = \left(\phi / \phi_0 \right)^2 \widehat{\backslash \mathrm{mathcal{H}}_{\{a\ b\}}^{\{ \}^{\{ \dagger \} } } }$	$\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$ and $\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$, where each
1510.06699_F00680	33	■ 1.000	$\backslash \mathrm{varphi}, h_{\{a\}}^{\{ \}^{\{ \mu \} } }$	$\varphi, h_a{}^\mu$	if χ represents $\varphi, h_a{}^\mu$, or $A^{\dagger ab}{}_\mu$, one may immediately
1510.06699_F00681	34	■ 1.000	$\backslash \mathrm{varphi}, h$	φ, h	for the φ, h and A (or $A^\dagger$) fields are the only three <i>in-</i>
1510.06699_F00682	34	■ 1.000	$\backslash \mathrm{varphi}, \backslash \mathrm{phi}, h_{\{a\}}^{\{ \}^{\{ \mu \} } }, A^{\{ \}^{\{ a \}^{\{ b \} } } }_{\{ \}^{\{ \mu \} } }$	$\varphi, \phi, h_a^\mu, A^{\dagger ab}{}_\mu$	ables $\varphi, \phi, h_a^\mu, A^{\dagger ab}{}_\mu$ and V_μ , rather than the alternative
1510.06699_F00683	34	■ 1.000	$\widehat{\backslash \mathrm{V}}_{\{ \}^{\{ \mu \} } } = \frac{1}{3} \widehat{\backslash \mathrm{T}}_{\{ \}^{\{ \mu \} } }^{\{ \}^{\{ \natural \} } }$	$\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$	where it is clear from the last definition that $\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$.

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1510.06699_F00684	34	■ 1.000	$\widehat{\mathcal{V}}_{\{a\}} \equiv \widehat{h}_{\{a\}}^{\mu} \widehat{V}_{\mu}$	$\widehat{V}_a \equiv \widehat{h}_a^{\mu} \widehat{V}_{\mu}$	variables $\widehat{\mathcal{A}}^{\dagger ab}_c \equiv \widehat{h}_a^{\mu} \widehat{A}^{\dagger ab}_{\mu}$ and $\widehat{\mathcal{V}}_a \equiv \widehat{h}_a^{\mu} \widehat{V}_{\mu}$. The situa-
1510.06699_F00685	34	■ 1.000	$\widehat{\varphi}, \widehat{h}_{\{a\}}^{\mu}$	φ, h_a^{μ}	Specifically, the equations of motion in terms of φ, h_a^{μ} ,
1510.06699_F00686	34	■ 1.000	$V_{\{\mu\}} + \frac{1}{3} T_{\mu}$	$V_{\mu} + \frac{1}{3} T_{\mu}$	tions $A^{\dagger ab}_{\mu}$ and $V_{\mu} + \frac{1}{3} T_{\mu}$ (which will always be possible).
1510.06699_F00687	34	■ 1.000	$\widehat{\varphi}, \widehat{h}_{\{a\}}^{\mu}, \widehat{A}^{\dagger ab}_{\mu}$	$\widehat{\varphi}, \widehat{h}_a^{\mu}, \widehat{A}^{\dagger ab}_{\mu}$	$\widehat{\varphi}, \widehat{h}_a^{\mu}, \widehat{A}^{\dagger ab}_{\mu}$ and $\widehat{V}_{\mu}$, respectively.
1510.06699_F00688	34	■ 1.000	$\widehat{V}_{\mu}$	$\widehat{V}_{\mu}$	$\widehat{\varphi}, \widehat{h}_a^{\mu}, \widehat{A}^{\dagger ab}_{\mu}$ and $\widehat{V}_{\mu}$, respectively.
1510.06699_F00689	34	■ 1.000	$v^c (\mathcal{D}_c^{\dagger} p_a -$	$v^c (\mathcal{D}_c^{\dagger} p_a -$	in a manifestly eWGT-covariant manner as $v^c (\mathcal{D}_c^{\dagger} p_a -$
1510.06699_F00690	34	■ 0.971	$\left. \mathcal{T}_{\text{cab}}^{\dagger} p^b \right) = e \mu^2 \phi \mathcal{D}_a^{\dagger} \phi$	$\mathcal{T}_{\text{cab}}^{\dagger} p^b) = e \mu^2 \phi \mathcal{D}_a^{\dagger} \phi$	$\mathcal{T}_{\text{cab}}^{\dagger} p^b) = e \mu^2 \phi \mathcal{D}_a^{\dagger} \phi$. By analogy with our treatment in
1510.06699_F00691	34	■ 0.997	$v^{c0} \mathcal{D}_c^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$	$v^{c0} \mathcal{D}_c^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$	$v^{c0} \mathcal{D}_c^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$, which is again manifestly eWGT-
1510.06699_F00692	34	■ 1.000	$\{ \}^{\dagger} D_{\mu}$	${}^0 D_{\mu}^{\dagger}$	covariant derivative operator ${}^0 D_{\mu}^{\dagger}$ in (159), to which the
1510.06699_F00693	34	■ 0.718	$\mathcal{T}_{abc}^{\dagger} \equiv 0$	$\mathcal{T}_{abc}^{\dagger} \equiv 0$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00694	34	■ 0.718	$\left. \mathcal{K}_{abc}^{\dagger} \equiv 0 \right)$	$\mathcal{K}_{abc}^{\dagger} \equiv 0$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00695	35	■ 1.000	$\{ \}^{\dagger} R^{\dagger ab}_{\mu\nu} (h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$	${}^0 R^{\dagger ab}_{\mu\nu} (h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$	${}^0 R^{\dagger ab}_{\mu\nu} (h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$, which is again given by
1510.06699_F00696	35	■ 1.000	$\{ \}^{\dagger} A^{\dagger ab}_{\mu}$	${}^0 A^{\dagger ab}_{\mu}$	the formula (39), but with $A^{\dagger ab}_{\mu}$ replaced by ${}^0 A^{\dagger ab}_{\mu}$.
1510.06699_F00697	35	■ 1.000	$\{ \}^{\dagger} R^{\dagger ab}_{\mu\nu}$	${}^0 R^{\dagger ab}_{\mu\nu}$	Unlike in WGT, however, ${}^0 R^{\dagger ab}_{\mu\nu}$ depends on all the
1510.06699_F00698	35	■ 1.000	$R^{\dagger ab}_{\mu\nu} (h, A, \partial A, V, \partial V)$	$R^{\dagger ab}_{\mu\nu} (h, A, \partial A, V, \partial V)$	$R^{\dagger ab}_{\mu\nu} (h, A, \partial A, V, \partial V)$. Considering instead the com-
1510.06699_F00699	35	■ 1.000	$\{ \}^{\dagger} \mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\mu} {}^0 R^{\dagger ab}_{\mu\nu}$	${}^0 \mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\mu} {}^0 R^{\dagger ab}_{\mu\nu}$	where ${}^0 \mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\mu} {}^0 R^{\dagger ab}_{\mu\nu}$, and the commuta-
1510.06699_F00700	35	■ 1.000	$\{ \}^{\dagger} \mathcal{T}^{\dagger a}_{bc} \equiv$	${}^0 \mathcal{T}^{\dagger a}_{bc} \equiv$	strength of the h gauge field (or ‘torsion’), since ${}^0 \mathcal{T}^{\dagger a}_{bc} \equiv$
1510.06699_F00701	35	■ 1.000	$h_b^{\mu} h_c^{\nu} ({}^0 D_{\mu}^{\dagger} b^a_{\nu} - {}^0 D_{\nu}^{\dagger} b^a_{\mu}) = 0$	$h_b^{\mu} h_c^{\nu} ({}^0 D_{\mu}^{\dagger} b^a_{\nu} - {}^0 D_{\nu}^{\dagger} b^a_{\mu}) = 0$	$h_b^{\mu} h_c^{\nu} ({}^0 D_{\mu}^{\dagger} b^a_{\nu} - {}^0 D_{\nu}^{\dagger} b^a_{\mu}) = 0$. As one might expect, the
1510.06699_F00702	35	■ 0.785	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}$	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}$	with the replacements $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}$ and
1510.06699_F00703	35	■ 0.831	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^0 \mathcal{T}_{abc}^{\dagger} = 0$	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^0 \mathcal{T}_{abc}^{\dagger} = 0$	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^0 \mathcal{T}_{abc}^{\dagger} = 0$.
1510.06699_F00704	35	■ 1.000	$\{ \}^{\dagger} \mathcal{R}^{ab}_{cd} \rightarrow$	${}^0 \mathcal{R}^{ab}_{cd} \rightarrow$	(110c), but with the replacements on the RHS ${}^0 \mathcal{R}^{ab}_{cd} \rightarrow$
1510.06699_F00705	35	■ 1.000	$\{ \}^{\dagger} \mathcal{R}^{\dagger ab}_{cd}, {}^0 \mathcal{D}_a \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$	${}^0 \mathcal{R}^{\dagger ab}_{cd}, {}^0 \mathcal{D}_a \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$	${}^0 \mathcal{R}^{\dagger ab}_{cd}, {}^0 \mathcal{D}_a \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$ and $\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^{\dagger}$, and
1510.06699_F00706	35	■ 1.000	$\mathcal{R}^{ab}_{cd} \rightarrow {}^0 \mathcal{R}^{\dagger ab}_{cd}$	$\mathcal{R}^{ab}_{cd} \rightarrow {}^0 \mathcal{R}^{\dagger ab}_{cd}$	which is based on (4) with $\mathcal{R}^{ab}_{cd} \rightarrow {}^0 \mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00707	35	■ 1.000	$\{ \}^{\dagger} \mathcal{R}^{\dagger ab}_{ab}$	${}^0 \mathcal{R}^{\dagger ab}_{ab}$	${}^0 \mathcal{R}^{\dagger ab}_{ab}$. As was the case for WGT, one can simplify (203)
1510.06699_F00708	35	■ 0.999	$\{ \}^{\dagger} \mathcal{R}^{\dagger abcd}$	${}^0 \mathcal{R}^{\dagger abcd}$	still further (in $D \leq 4$ dimensions), since ${}^0 \mathcal{R}^{\dagger abcd}$ and its
1510.06699_F00709	35	■ 1.000	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	form (59), but with $\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$. Thus, one can set to
1510.06699_F00710	35	■ 0.535	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}$	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}$	ment $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}$. As in WGT, however, the gravitational
1510.06699_F00711	35	■ 0.720	$\phi^{20} \mathcal{R}^{\dagger}$	$\phi^{20} \mathcal{R}^{\dagger}$	$\mathcal{L}_G$ and the term proportional to $\phi^2 {}^0 \mathcal{R}^{\dagger}$ in $\mathcal{L}_M$ depends

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1510.06699_F00712	35	0.806	$\mathcal{D}_{\{a\}^{\dagger}} \rightarrow \mathcal{D}_{\{a\}^{\dagger}}, \mathcal{R}_{\{a\}^{\dagger} \{b\}^{\dagger} \{c\}^{\dagger} \{d\}^{\dagger}} \rightarrow \mathcal{R}_{\{a\}^{\dagger} \{b\}^{\dagger} \{c\}^{\dagger} \{d\}^{\dagger}}$	$\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger, \mathcal{R}_{abcd}^\dagger \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$	replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger, \mathcal{R}_{abcd}^\dagger \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ and $\mathcal{T}_{abc}^\dagger \rightarrow$
1510.06699_F00713	35	0.806	$\mathcal{T}_{\{a\}^{\dagger} \{b\}^{\dagger} \{c\}^{\dagger}} \rightarrow$	$\mathcal{T}_{abc}^\dagger \rightarrow$	replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger, \mathcal{R}_{abcd}^\dagger \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ and $\mathcal{T}_{abc}^\dagger \rightarrow$
1510.06699_F00714	35	0.961	$\{ \}^{\dagger} \mathcal{T}_{\{a\}^{\dagger} \{b\}^{\dagger} \{c\}^{\dagger}} \equiv 0$	${}^0\mathcal{T}_{abc}^\dagger \equiv 0$	${}^0\mathcal{T}_{abc}^\dagger \equiv 0$. Furthermore, by performing analogous calcu-
1510.06699_F00715	35	0.891	$\mathcal{T}_{\{a\}^{\dagger} \{b\}^{\dagger} \{c\}^{\dagger}}=0$	$\mathcal{T}_{abc}^\dagger = 0$	erly eWGT-covariant) condition $\mathcal{T}_{abc}^\dagger = 0$ in the field
1510.06699_F00716	36	1.000	$b^{\dagger}_{\{b\}^{\dagger} \{ \}_{\nu}}$	b^b_{ν}	notate $h_a{}^\mu$ and b^b_{ν} by $e_a{}^\mu$ and e^b_{ν} , respectively.
1510.06699_F00717	36	1.000	$e^{\dagger}_{\{b\}^{\dagger} \{ \}_{\nu}}$	e^b_{ν}	notate $h_a{}^\mu$ and b^b_{ν} by $e_a{}^\mu$ and e^b_{ν} , respectively.
1510.06699_F00718	36	1.000	$e_{\{a\}^{\dagger} \{ \}_{\mu}}, A^{\dagger}_{\{a\}^{\dagger} \{b\}^{\dagger} \{ \}_{\mu}}$	$e_a{}^\mu, A^{ab}_{\mu}$	(142) of the three gauge fields $e_a{}^\mu$, A^{ab}_{μ} , and V_{μ} that
1510.06699_F00719	36	1.000	$\partial_{\{ \}_{\mu}^{\dagger}}$	$\partial_{\mu}^\dagger$	where we have used the partial derivative operator $\partial_{\mu}^\dagger$
1510.06699_F00720	36	1.000	$A^{\dagger}_{\{ \}_{\mu}^{\dagger} \{a\}^{\dagger} \{b\}^{\dagger}} = -A^{\dagger}_{\{ \}_{\mu}^{\dagger} \{b\}^{\dagger} \{a\}^{\dagger}}$	$A^{\dagger ab}_{\mu} = -A^{\dagger ba}_{\mu}$	$A^{\dagger ab}_{\mu} = -A^{\dagger ba}_{\mu}$, as previously.
1510.06699_F00721	36	0.545	$\nabla_{\{ \}_{\mu}^{\dagger}} = \partial_{\{ \}_{\mu}^{\dagger}} + \Gamma^{\dagger}_{\rho\mu} X^{\rho}_{\sigma}$	$\nabla_{\mu}^\dagger = \partial_{\mu}^\dagger + \Gamma^{\sigma}_{\rho\mu} X^{\rho}_{\sigma}$	where $\nabla_{\mu}^\dagger = \partial_{\mu}^\dagger + \Gamma^{\sigma}_{\rho\mu} X^{\rho}_{\sigma}$ and X^{ρ}_{σ} are the $\text{GL}(4, R)$ gen-
1510.06699_F00722	36	0.545	$4, R$	$4, R$	where $\nabla_{\mu}^\dagger = \partial_{\mu}^\dagger + \Gamma^{\sigma}_{\rho\mu} X^{\rho}_{\sigma}$ and X^{ρ}_{σ} are the $\text{GL}(4, R)$ gen-
1510.06699_F00723	36	1.000	$\Delta_{\{ \}_{\mu}^{\dagger}}$	$\Delta_{\mu}^\dagger$	of the field to which $\Delta_{\mu}^\dagger$ is applied. If a field ψ carries
1510.06699_F00724	36	1.000	$\nabla_{\{ \}_{\mu}^{\dagger}} \psi = \partial_{\{ \}_{\mu}^{\dagger}} \psi$	$\nabla_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$	only Latin indices, then $\nabla_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$ and so $\Delta_{\mu}^\dagger \psi = D_{\mu}^\dagger \psi$;
1510.06699_F00725	36	1.000	$\Delta_{\{ \}_{\mu}^{\dagger}} \psi = D_{\{ \}_{\mu}^{\dagger}} \psi$	$\Delta_{\mu}^\dagger \psi = D_{\mu}^\dagger \psi$	only Latin indices, then $\nabla_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$ and so $\Delta_{\mu}^\dagger \psi = D_{\mu}^\dagger \psi$;
1510.06699_F00726	36	1.000	$D_{\{ \}_{\mu}^{\dagger}} \psi = \partial_{\{ \}_{\mu}^{\dagger}} \psi$	$D_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$	$D_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$ and so $\Delta_{\mu}^\dagger \psi = \nabla_{\mu}^\dagger \psi$. When acting on an ob-
1510.06699_F00727	36	1.000	$\Delta_{\{ \}_{\mu}^{\dagger}} \psi = \nabla_{\{ \}_{\mu}^{\dagger}} \psi$	$\Delta_{\mu}^\dagger \psi = \nabla_{\mu}^\dagger \psi$	$D_{\mu}^\dagger \psi = \partial_{\mu}^\dagger \psi$ and so $\Delta_{\mu}^\dagger \psi = \nabla_{\mu}^\dagger \psi$. When acting on an ob-
1510.06699_F00728	36	1.000	T	T	which relates A and Γ (and V and T). Comparing this
1510.06699_F00729	36	1.000	$\partial_{\{ \}_{\mu}^{\dagger}}^* \rightarrow \partial_{\{ \}_{\mu}^{\dagger}}$	$\partial_{\mu}^* \rightarrow \partial_{\mu}^\dagger$	$\partial_{\mu}^* \rightarrow \partial_{\mu}^\dagger$ and $A_{\mu}^{ab} \rightarrow A^{\dagger ab}_{\mu}$. Consequently, the results
1510.06699_F00730	36	1.000	$A_{\{ \}_{\mu}^{\dagger} \{a\}^{\dagger} \{b\}^{\dagger}} \rightarrow A^{\dagger}_{\{ \}_{\mu}^{\dagger} \{a\}^{\dagger} \{b\}^{\dagger}}$	$A_{\mu}^{ab} \rightarrow A^{\dagger ab}_{\mu}$	$\partial_{\mu}^* \rightarrow \partial_{\mu}^\dagger$ and $A_{\mu}^{ab} \rightarrow A^{\dagger ab}_{\mu}$. Consequently, the results
1510.06699_F00731	36	1.000	$\nabla_{\{ \}_{\sigma}^{\dagger}} g_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}} = 0$	$\nabla_{\sigma}^\dagger g_{\mu\nu} = 0$	$\nabla_{\sigma}^\dagger g_{\mu\nu} = 0$, and so this derivative operator commutes
1510.06699_F00732	36	1.000	$R^{\dagger}_{\{ \}_{\sigma}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}} = e_{\{ \}_{\sigma}^{\dagger}}^{\rho} e_{\{ \}_{\mu}^{\dagger}}^b R^{\dagger a}_{\{ \}_{\nu}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\sigma}^{\dagger}}$	$R^{\dagger\rho}_{\sigma\mu\nu} = e_a{}^\rho e^b_{\sigma} R^{\dagger a}_{b\mu\nu}$	where $R^{\dagger\rho}_{\sigma\mu\nu} = e_a{}^\rho e^b_{\sigma} R^{\dagger a}_{b\mu\nu}$ and $T^{\dagger\lambda}_{\mu\nu} = e_a{}^\lambda T^{\dagger a}_{\mu\nu}$.
1510.06699_F00733	36	1.000	$T^{\dagger}_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}} = e_{\{ \}_{\mu}^{\dagger}}^{\lambda} T^{\dagger a}_{\{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}}$	$T^{\dagger\lambda}_{\mu\nu} = e_a{}^\lambda T^{\dagger a}_{\mu\nu}$	where $R^{\dagger\rho}_{\sigma\mu\nu} = e_a{}^\rho e^b_{\sigma} R^{\dagger a}_{b\mu\nu}$ and $T^{\dagger\lambda}_{\mu\nu} = e_a{}^\lambda T^{\dagger a}_{\mu\nu}$.
1510.06699_F00734	37	1.000	$T^{\dagger}_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}} = 0$	$T^{\dagger\lambda}_{\mu\lambda} = 0$	torsion vanishes, $T^{\dagger\lambda}_{\mu\lambda} = 0$, so that the affine connection
1510.06699_F00735	37	1.000	$R^{\dagger}_{\{ \}_{\sigma}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}}$	$R^{\dagger\rho}_{\sigma\mu\nu}$	From (212), we see that $R^{\dagger\rho}_{\sigma\mu\nu}$ is not simply its Riemann
1510.06699_F00736	37	0.999	$\{ \}^{\dagger} \Gamma^{\dagger}_{\rho\mu}$	${}^0\Gamma^{\dagger\sigma}_{\rho\mu}$	nection ${}^0\Gamma^{\dagger\sigma}_{\rho\mu}$ represent the same geometrical object in
1510.06699_F00737	37	1.000	$\widehat{R}_{\{ \}_{\rho}^{\dagger} \{ \}_{\sigma}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}}$	$\widehat{R}_{\rho\sigma\mu\nu}$	One should note that, although $\widehat{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00738	37	1.000	$\widehat{R}_{\{ \}_{(\rho}^{\dagger} \{ \}_{\sigma)}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}} = -g_{\{ \}_{\rho}^{\dagger} \{ \}_{\sigma)}^{\dagger} H_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}}$	$\widehat{R}_{(\rho\sigma)\mu\nu} = -g_{\rho\sigma} H_{\mu\nu}^\dagger$	$\widehat{R}_{(\rho\sigma)\mu\nu} = -g_{\rho\sigma} H_{\mu\nu}^\dagger$. Indeed, it does not satisfy the
1510.06699_F00739	37	1.000	$\widehat{R}^{\rho}_{\{ \}_{\sigma}^{\dagger} \{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger}}$	$\widehat{R}^{\rho}_{\sigma\mu\nu}$	that, with the given arrangements of indices, both $\widehat{R}^{\rho}_{\sigma\mu\nu}$
1510.06699_F00740	37	1.000	$T^{\dagger}_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}}$	$T^{\dagger\lambda}_{\mu\nu}$	(or $R^{\dagger\rho}_{\sigma\mu\nu}$) and $T^{\dagger\lambda}_{\mu\nu}$ transform covariantly with weight
1510.06699_F00741	37	1.000	$\widehat{R}_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}} \equiv \widehat{R}_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}} = R_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}}^\dagger + H_{\{ \}_{\mu}^{\dagger} \{ \}_{\nu}^{\dagger} \{ \}_{\lambda}^{\dagger}}^\dagger$	$\widehat{R}_{\mu\nu} \equiv \widehat{R}_{\mu\lambda\nu}{}^\lambda = R_{\mu\nu}^\dagger + H_{\mu\nu}^\dagger$	$\widehat{R}_{\mu\nu} \equiv \widehat{R}_{\mu\lambda\nu}{}^\lambda = R_{\mu\nu}^\dagger + H_{\mu\nu}^\dagger$ and $\widehat{R} \equiv \widehat{R}^\mu_{\mu} = R^\dagger$. In a
1510.06699_F00742	37	1.000	$\widehat{R} \equiv \widehat{R}^\mu_{\mu} = R^\dagger$	$\widehat{R} \equiv \widehat{R}^\mu_{\mu} = R^\dagger$	$\widehat{R}_{\mu\nu} \equiv \widehat{R}_{\mu\lambda\nu}{}^\lambda = R_{\mu\nu}^\dagger + H_{\mu\nu}^\dagger$ and $\widehat{R} \equiv \widehat{R}^\mu_{\mu} = R^\dagger$. In a
1510.06699_F00743	37	1.000	$\{ \}^{\dagger} \Delta_{\{ \}_{\mu}^{\dagger}} e^a_{\{ \}_{\nu}^{\dagger}} \equiv \partial_{\{ \}_{\mu}^{\dagger}} e^a_{\{ \}_{\nu}^{\dagger}} - {}^0\Gamma^{\dagger\sigma}_{\nu\mu} e^a_{\sigma} + {}^0A^{\dagger a}_{b\mu} e^b_{\nu} = 0$	${}^0\Delta_{\mu}^\dagger e^a_{\nu} \equiv \partial_{\mu}^\dagger e^a_{\nu} - {}^0\Gamma^{\dagger\sigma}_{\nu\mu} e^a_{\sigma} + {}^0A^{\dagger a}_{b\mu} e^b_{\nu} = 0$	${}^0\Delta_{\mu}^\dagger e^a_{\nu} \equiv \partial_{\mu}^\dagger e^a_{\nu} - {}^0\Gamma^{\dagger\sigma}_{\nu\mu} e^a_{\sigma} + {}^0A^{\dagger a}_{b\mu} e^b_{\nu} = 0$.
1510.06699_F00744	37	0.992	$\partial_{\{ \}_{\mu}^{\dagger}}^* \rightarrow \partial_{\{ \}_{\mu}^{\dagger}}, \Gamma^{\sigma}_{\nu\mu} \rightarrow$	$\partial_{\mu}^* \rightarrow \partial_{\mu}^\dagger, \Gamma^{\sigma}_{\nu\mu} \rightarrow$	again hold with the replacements $\partial_{\mu}^* \rightarrow \partial_{\mu}^\dagger, \Gamma^{\sigma}_{\nu\mu} \rightarrow$

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1510.06699_F00745	37	■ 1.000	$\{ \}^{\{0\}} \backslash \Gamma^{\{\dagger\}}_{\sigma} \{ \}_{\nu \mu}$	${}^0\Gamma^{\dagger\sigma}_{\nu\mu}$	${}^0\Gamma^{\dagger\sigma}_{\nu\mu}$ and $A^a_{b\mu} \rightarrow {}^0A^{\dagger a}_{b\mu}$, from which one can di-
1510.06699_F00746	37	■ 1.000	$A^{\{a\}}\{ \}_{b \mu} \backslash \rightarrow \{ \}^{\{0\}} A^{\{\dagger\}}_{\{a\}} \{ \}_{b \mu}$	$A^a_{b\mu} \rightarrow {}^0A^{\dagger a}_{b\mu}$	${}^0\Gamma^{\dagger\sigma}_{\nu\mu}$ and $A^a_{b\mu} \rightarrow {}^0A^{\dagger a}_{b\mu}$, from which one can di-
1510.06699_F00747	37	■ 1.000	$\{ \}^{\{0\}} \backslash \nabla_{\sigma}^{\{\dagger\}} g_{\mu\nu} = 0$	${}^0\nabla^{\dagger}_{\sigma} g_{\mu\nu} = 0$	dition ${}^0\nabla^{\dagger}_{\sigma} g_{\mu\nu} = 0$. Finally, the expression (212) for
1510.06699_F00748	37	■ 1.000	$R^{\{\dagger\}}_{\rho} \{ \}_{\sigma \mu \nu} \backslash \rightarrow \{ \}^{\{0\}} R^{\{\dagger\}}_{\rho} \{ \}_{\sigma \mu \nu}$	$R^{\dagger\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}_{\sigma\mu\nu}$	the curvature also holds, but with $R^{\dagger\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}_{\sigma\mu\nu}$
1510.06699_F00749	37	■ 0.999	$\backslash \Gamma^{\{\sigma\}}_{\nu \mu} \{ \}_{\nu \mu} \backslash \rightarrow \{ \}^{\{0\}} \backslash \Gamma^{\{\dagger\}}_{\sigma} \{ \}_{\nu \mu}$	$\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{\dagger\sigma}_{\nu\mu}$	and $\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{\dagger\sigma}_{\nu\mu}$, whereas (213) becomes simply
1510.06699_F00750	37	■ 1.000	$\{ \}^{\{0\}} T^{\{\dagger\}}_{\mu\nu} \{ \}_{\mu \nu} = 0$	${}^0T^{\dagger\lambda}_{\mu\nu} = 0$	${}^0T^{\dagger\lambda}_{\mu\nu} = 0$, indicating the absence of torsion, as ex-
1510.06699_F00751	37	■ 0.986	$\backslash \nabla_{\mu}^{\{\dagger\}} \{ \}_{\mu} \backslash \rightarrow \{ \}^{\{0\}} \backslash \nabla_{\mu}^{\{\dagger\}} \{ \}_{\mu}$, $R^{\{\rho\}}_{\sigma \mu \nu} \{ \}_{\sigma \mu \nu} \backslash \rightarrow \{ \}^{\{0\}} R^{\{\dagger\}}_{\sigma \mu \nu}$	$\nabla^{\dagger}_{\mu} \rightarrow {}^0\nabla^{\dagger}_{\mu}, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}_{\sigma\mu\nu}$	$\nabla^{\dagger}_{\mu} \rightarrow {}^0\nabla^{\dagger}_{\mu}, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}_{\mu\nu} \rightarrow {}^0T^{\dagger\sigma}_{\mu\nu} = 0$.
1510.06699_F00752	37	■ 0.986	$T^{\{\dagger\}}_{\sigma} \{ \}_{\mu \nu} \backslash \rightarrow \{ \}^{\{0\}} T^{\{\dagger\}}_{\sigma} \{ \}_{\mu \nu} = 0$	$T^{\dagger\sigma}_{\mu\nu} \rightarrow {}^0T^{\dagger\sigma}_{\mu\nu} = 0$	$\nabla^{\dagger}_{\mu} \rightarrow {}^0\nabla^{\dagger}_{\mu}, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}_{\mu\nu} \rightarrow {}^0T^{\dagger\sigma}_{\mu\nu} = 0$.
1510.06699_F00753	37	■ 1.000	$L_{\mathrm{T}} =$	$L_{\mathrm{T}} =$	we now consider in detail has a total Lagrangian $L_{\mathrm{T}} =$
1510.06699_F00754	37	■ 1.000	$L_{\mathrm{G}} + L_{\mathrm{M}}$	$L_{\mathrm{G}} + L_{\mathrm{M}}$	$L_{\mathrm{G}} + L_{\mathrm{M}}$ corresponding to the general eWGT form (167).
1510.06699_F00755	37	■ 1.000	$\{ \}^{\{0\}} \backslash \Gamma^{\lambda}_{\mu\nu} \{ \}_{\mu \nu}$	${}^0\Gamma^{\lambda}_{\mu\nu}$	in which ${}^0\Gamma^{\lambda}_{\mu\nu}$ is the standard metric (Christoffel) con-
1510.06699_F00756	37	■ 1.000	$K^{\{\dagger\}}_{\mu\nu} \{ \}_{\mu \nu}$	$K^{\dagger\lambda}_{\mu\nu}$	nection and $K^{\dagger\lambda}_{\mu\nu}$ is the $\hat{Y}_4$ contortion tensor
1510.06699_F00757	37	■ 1.000	$L_{\mathrm{G}} = L_{\mathrm{R}^{\dagger 2}} + L_{\mathrm{H}^{\dagger 2}}$	$L_{\mathrm{G}} = L_{\mathcal{R}^{\dagger 2}} + L_{\mathcal{H}^{\dagger 2}}$	namely $L_{\mathrm{G}} = L_{\mathcal{R}^{\dagger 2}} + L_{\mathcal{H}^{\dagger 2}}$, where
1510.06699_F00758	37	■ 1.000	$L_{\mathrm{M}} =$	$L_{\mathrm{M}} =$	of gravity. In particular, we adopt the form $L_{\mathrm{M}} =$
1510.06699_F00759	38	■ 1.000	ν, λ, a	ν, λ, a	in which ν, λ, a and β_i are again dimensionless parame-
1510.06699_F00760	38	■ 1.000	ν, λ	ν, λ	ters (with ν, λ and a usually positive) and some factors
1510.06699_F00761	38	■ 1.000	L_{φ}	L_{φ}	the moment we allow the form of L_{φ} for the matter field
1510.06699_F00762	38	■ 1.000	$\left(\tau^{\dagger}_{\varphi} \right)^a_b = h_b{}^{\mu} (\delta L_{\varphi} / \delta h_a{}^{\mu})_{\dagger}$	$(\tau^{\dagger}_{\varphi})^a_b = h_b{}^{\mu} (\delta L_{\varphi} / \delta h_a{}^{\mu})_{\dagger}$	and in the matter sector, in addition to $(\tau^{\dagger}_{\varphi})^a_b = h_b{}^{\mu} (\delta L_{\varphi} / \delta h_a{}^{\mu})_{\dagger}$, one has
1510.06699_F00763	38	■ 1.000	$\left(t^{\dagger}_{\mathcal{H}^2} \right)^a_b$	$(t^{\dagger}_{\mathcal{H}^2})^a_b$	In particular, we note that $(t^{\dagger}_{\mathcal{H}^2})^a_b$ in (230) is linear
1510.06699_F00764	39	■ 1.000	$\left(\sigma^{\dagger}_{\varphi} \right)^c_{ab} =$	$(\sigma^{\dagger}_{\varphi})^c_{ab} =$	and in the matter sector, in addition to $(\sigma^{\dagger}_{\varphi})^c_{ab} =$
1510.06699_F00765	39	■ 1.000	$b^c_{\mu} \delta L_{\varphi} / \delta A^{ab}_{\mu}$	$b^c_{\mu} \delta L_{\varphi} / \delta A^{ab}_{\mu}$	
1510.06699_F00766	39	■ 0.860	$(s_{\mathcal{H}^2})^c_{ab}$	$(s_{\mathcal{H}^2})^c_{ab}$	We note that $(s_{\mathcal{H}^2})^c_{ab}$ in (236) is linear in third-order
1510.06699_F00767	39	■ 0.554	$\mathcal{T}^{\dagger}_{cab} + 2\mathcal{T}^{\dagger}_{[ab]c} - 3\mathcal{T}^{\dagger}_{[cab]} = 0$	$\mathcal{T}^{\dagger}_{cab} + 2\mathcal{T}^{\dagger}_{[ab]c} - 3\mathcal{T}^{\dagger}_{[cab]} = 0$	strength satisfies the identity $\mathcal{T}^{\dagger}_{cab} + 2\mathcal{T}^{\dagger}_{[ab]c} - 3\mathcal{T}^{\dagger}_{[cab]} = 0$.
1510.06699_F00768	39	■ 1.000	α_4	α_4	in the action (namely those in (235) proportional to α_4 ,
1510.06699_F00769	39	■ 1.000	α_5	α_5	α_5 and α_6 , respectively) have a form similar to the LHS
1510.06699_F00770	39	■ 0.606	$\overline{\mathcal{R}}^{\dagger}_{abcd} \equiv \mathcal{R}^{\dagger}_{cdab}$	$\overline{\mathcal{R}}^{\dagger}_{abcd} \equiv \mathcal{R}^{\dagger}_{cdab}$	$\overline{\mathcal{R}}^{\dagger}_{abcd} \equiv \mathcal{R}^{\dagger}_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}_b = \mathcal{R}^{\dagger a}_b$, one also sees that
1510.06699_F00771	39	■ 0.606	$\overline{\mathcal{R}}^{\dagger a}_b = \mathcal{R}^{\dagger a}_b$	$\overline{\mathcal{R}}^{\dagger a}_b = \mathcal{R}^{\dagger a}_b$	$\overline{\mathcal{R}}^{\dagger}_{abcd} \equiv \mathcal{R}^{\dagger}_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}_b = \mathcal{R}^{\dagger a}_b$, one also sees that
1510.06699_F00772	39	■ 1.000	α_2	α_2	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00773	39	■ 1.000	α_3	α_3	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00774	39	■ 1.000	α_1	α_1	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger} = \mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00775	39	■ 1.000	$\overline{\mathcal{R}}^{\dagger} = \mathcal{R}^{\dagger}$	$\overline{\mathcal{R}}^{\dagger} = \mathcal{R}^{\dagger}$	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger} = \mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00776	39	■ 1.000	$(\zeta_{\varphi})_a =$	$(\zeta_{\varphi})_a =$	
1510.06699_F00777	39	■ 1.000	$b_{a\mu} \delta L_{\varphi} / \delta V_{\mu}$	$b_{a\mu} \delta L_{\varphi} / \delta V_{\mu}$	$b_{a\mu} \delta L_{\varphi} / \delta V_{\mu}$,
1510.06699_F00778	39	■ 1.000	c	c	equation (234) contracted on the indices b and c , in ac-
1510.06699_F00779	39	■ 1.000	$j^{\dagger}_a$	$j^{\dagger}_a$	so $j^{\dagger}_a$ and $\zeta^{\dagger}_a$ vanish identically for each part of the La-

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1510.06699_F00780	39	■ 1.000	$\backslash\mathrm{zeta}_{\mathrm{a}}^{\mathrm{\dag}}$	$\zeta_a^\dagger$	so $j_a^\dagger$ and $\zeta_a^\dagger$ vanish identically for each part of the La-
1510.06699_F00781	39	■ 1.000	$\backslash\mathrm{partial}_{\mathrm{\phi}}\equiv\backslash\mathrm{partial}/\backslash\mathrm{partial}\mathrm{\phi}$	$\partial_\phi \equiv \partial/\partial\phi$	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00782	39	■ 1.000	$\backslash\mathrm{delta}_{\mathrm{\phi}}\equiv\backslash\mathrm{delta}/\backslash\mathrm{delta}\mathrm{\phi}$	$\delta_\phi \equiv \delta/\delta\phi$	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00783	39	■ 1.000	$\mathrm{L}_{\mathrm{\phi}}$	L_ϕ	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00784	40	■ 1.000	$-\mathrm{\phi}$	$-\phi$	by $-\phi$, in accordance with the general results (177) and
1510.06699_F00785	40	■ 0.999	$\backslash\mathrm{delta}_{\mathrm{\varphi}}\mathrm{L}_{\mathrm{\varphi}}=0$	$\delta_\varphi L_\varphi = 0$	(178d), assuming the φ -equation $\delta_\varphi L_\varphi = 0$ is satisfied.
1510.06699_F00786	40	■ 1.000	$\backslash\mathrm{x}=0$	$\xi = 0$	tional Lagrangian (by setting $\xi = 0$). Indeed, one can
1510.06699_F00787	40	■ 1.000	$\backslash\mathrm{left}.\backslash\mathrm{mathcal{D}}_{\mathrm{a}}^{\mathrm{\dag}}\backslash\mathrm{phi}\backslash\mathrm{right} _{\mathrm{\phi}=\mathrm{\phi}_0}=\backslash\mathrm{phi}_{\mathrm{0}}\backslash\mathrm{left}(\backslash\mathrm{mathcal{V}}_{\mathrm{a}}+\backslash\mathrm{frac{1}{3}}\backslash\mathrm{mathcal{T}}_{\mathrm{a}}\backslash\mathrm{right})$	$\mathcal{D}_a^\dagger\phi _{\phi=\phi_0}=\phi_0\left(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a\right)$	the replacement $\mathcal{D}_a^\dagger\phi _{\phi=\phi_0}=\phi_0(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a)$ throughout,
1510.06699_F00788	40	■ 1.000	$\backslash\mathrm{widehat{\varphi}},\backslash\mathrm{widehat{h}}_{\mathrm{a}}^{\mathrm{\mu}}$	$\widehat{\varphi},\widehat{h}_a^\mu$	expressed in terms of the scale-invariant variables $\widehat{\varphi},\widehat{h}_a^\mu$,
1510.06699_F00789	41	■ 1.000	$\mathrm{h}^{\mathrm{\prime}}_{\mathrm{a}}^{\mathrm{\mu}}=\mathrm{e}^{-\mathrm{\rho}}\mathrm{h}_{\mathrm{a}}^{\mathrm{\mu}}$	$h_a^{\prime\mu}=e^{-\rho}h_a^\mu$	fields are given by $\overline{h}_a^{\prime\mu}=e^{-\rho}\overline{h}_a^\mu$ and $A^{\prime ab}{}_\mu=A^{ab}{}_\mu+$
1510.06699_F00790	41	■ 1.000	$\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}}=\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}}+$	$A^{\prime ab}{}_\mu=A^{ab}{}_\mu+$	fields are given by $h_a^{\prime\mu}=e^{-\rho}h_a^\mu$ and $\overline{A}^{\prime ab}{}_\mu=\overline{A}^{ab}{}_\mu+$
1510.06699_F00791	41	■ 1.000	$\backslash\mathrm{theta}\backslash\mathrm{left}(\mathrm{b}^{\mathrm{a}}_{\mathrm{\mu}}\backslash\mathrm{mathcal{P}}^{\mathrm{b}}-\mathrm{b}^{\mathrm{b}}_{\mathrm{\mu}}\backslash\mathrm{mathcal{P}}^{\mathrm{a}}\backslash\mathrm{right})$	$\theta(b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a)$	$\theta(b^a{}_\mu\mathcal{P}^b-\overline{b}^b{}_\mu\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a^\mu\partial_\mu\rho$. Here
1510.06699_F00792	41	■ 1.000	$\backslash\mathrm{mathcal{P}}_{\mathrm{a}}=\mathrm{h}_{\mathrm{a}}^{\mathrm{\mu}}\backslash\mathrm{partial}_{\mathrm{\mu}}\mathrm{\rho}$	$\mathcal{P}_a=h_a^\mu\partial_\mu\rho$	$\theta(b^a{}_\mu\mathcal{P}^b-\overline{b}^b{}_\mu\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a^\mu\partial_\mu\rho$. Here
1510.06699_F00793	41	■ 1.000	$\backslash\mathrm{mathcal{P}}^{\mathrm{2}}\equiv\backslash\mathrm{mathcal{P}}_{\mathrm{a}}\backslash\mathrm{mathcal{P}}^{\mathrm{a}}$	$\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$	where $\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$, and the translational gauge field
1510.06699_F00794	41	■ 1.000	$(\backslash\mathrm{theta}=0)$	$(\theta=0)$	is covariant with weight $w=-2$ under ‘normal’ ($\theta=0$)
1510.06699_F00795	41	■ 1.000	$(\backslash\mathrm{theta}=1)$	$(\theta=1)$	$w=-1$ under ‘special’ transformations ($\theta=1$).
1510.06699_F00796	41	■ 1.000	$\backslash\mathrm{theta}\neq 0$	$\theta\neq 0$	extended transformation law for the A -field, with $\theta\neq 0$,
1510.06699_F00797	41	■ 1.000	$\backslash\mathrm{alpha}_{\mathrm{i}}=0$	$\alpha_i=0$	forms covariantly (aside from the trivial case $\overline{\alpha}_i=0$ for
1510.06699_F00798	41	■ 0.996	i	i	all i) and so L_G must vanish. Nonetheless, by contrast,
1510.06699_F00799	41	■ 0.998	$\mathrm{L}_{\mathrm{\mathcal{T}}^2}$	$L_{\mathcal{T}^2}$	parameters β_1 , β_2 and β_3 in $L_{\mathcal{T}^2}$.
1510.06699_F00800	42	■ 0.989	$\backslash\mathrm{mathcal{L}}_{\mathrm{\mathcal{M}}}\equiv\backslash\mathrm{mathcal{L}}_{\mathrm{\psi}}+\backslash\mathrm{mathcal{L}}_{\mathrm{\phi}}+\phi^2\backslash\mathrm{mathcal{L}}_{\mathrm{\mathcal{R}}}+\phi^2\backslash\mathrm{mathcal{L}}_{\mathrm{\mathcal{T}}^2}$	$\mathcal{L}_\mathrm{M}\equiv\mathcal{L}_\psi+\mathcal{L}_\phi+\phi^2\mathcal{L}_\mathcal{R}+\phi^2\mathcal{L}_{\mathcal{T}^2}$	Let us denote (256) by $\overline{\mathcal{L}}_\mathrm{M}\equiv\mathcal{L}_\psi+\mathcal{L}_\phi+\phi^2\mathcal{L}_\mathcal{R}+\phi^2\mathcal{L}_{\mathcal{T}^2}$.
1510.06699_F00801	42	■ 0.990	$\backslash\mathrm{mathcal{L}}_{\mathrm{\psi}}$	$\mathcal{L}_\psi$	As discussed in Section III K, $\mathcal{L}_\psi$ is invariant under ex-
1510.06699_F00802	42	■ 1.000	$\backslash\mathrm{nu},\mathrm{a}$	ν,a	rameters ν , a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00803	42	■ 1.000	$\backslash\mathrm{beta}_{\mathrm{i}}(\mathrm{i}=1,2,3)$	$\beta_i(i=1,2,3)$	rameters ν , a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00804	42	■ 1.000	$\backslash\mathrm{nu}$	ν	the parameters ν and a must be of opposite sign, which
1510.06699_F00805	42	■ 1.000	$\backslash\mathrm{nu}=0=2\backslash\mathrm{beta}_{\mathrm{1}}+\backslash\mathrm{beta}_{\mathrm{2}}+3\backslash\mathrm{beta}_{\mathrm{3}}$	$\nu=0=2\beta_1+\beta_2+3\beta_3$	$\nu=0=2\beta_1+\beta_2+3\beta_3$, and a ‘special’ transforma-
1510.06699_F00806	42	■ 0.997	$\backslash\mathrm{nu}=0=\mathrm{a}$	$\nu=0=a$	tion ($\theta=1$), for which $\nu=0=a$. Thus, in both special
1510.06699_F00807	42	■ 1.000	$\backslash\mathrm{nu}=\backslash\mathrm{beta}_{\mathrm{1}}=\backslash\mathrm{beta}_{\mathrm{2}}=\backslash\mathrm{beta}_{\mathrm{3}}=0$	$\nu=\beta_1=\beta_2=\beta_3=0$	Obukhov ⁸⁹ , who set $\overline{\nu}=\beta_1=\beta_2=\beta_3=0$. Adopting
1510.06699_F00808	42	■ 1.000	$2\backslash\mathrm{beta}_{\mathrm{1}}+\backslash\mathrm{beta}_{\mathrm{2}}+3\backslash\mathrm{beta}_{\mathrm{3}}=0$	$2\beta_1+\beta_2+3\beta_3=0$	with $\overline{2\beta_1+\beta_2+3\beta_3}=0$ in $L_{\mathcal{T}^{+2}}$, which is merely a special
1510.06699_F00809	42	■ 0.706	$\backslash\mathrm{mathcal{D}}_{\mathrm{a}}\backslash\mathrm{phi}\backslash\mathrm{rightarrow}\backslash\mathrm{left}(\backslash\mathrm{mathcal{D}}_{\mathrm{a}}+\backslash\mathrm{frac{1}{3}}\backslash\mathrm{mathcal{T}}_{\mathrm{a}}\backslash\mathrm{right})\backslash\mathrm{phi}$	$\mathcal{D}_a\phi\rightarrow\left(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a\right)\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\overline{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00810	42	■ 0.706	$\backslash\mathrm{mathcal{D}}^{\mathrm{a}}\backslash\mathrm{phi}$	$\mathcal{D}^a\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\overline{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00811	42	■ 1.000	$\backslash\mathrm{frac{1}{2}}\backslash\mathrm{nu}+6\mathrm{a}=0$	$\frac{1}{2}\nu+6a=0$	vided only that $\frac{1}{2}\nu+6a=0$ and $2\beta_1+\beta_2+3\beta_3=0$
1510.06699_F00812	42	■ 0.994	$2\backslash\mathrm{beta}_{\mathrm{1}}+\backslash\mathrm{beta}_{\mathrm{2}}=0$	$2\beta_1+\beta_2=0$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00813	42	■ 0.994	$\mathrm{L}_{\mathrm{\mathcal{T}^{+2}}}$	$L_{\mathcal{T}^{+2}}$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00814	43	■ 1.000	$\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\backslash\mathrm{mathcal{D}}_{\mathrm{a}}$	${}^0\mathcal{D}_a$	derivative ${}^0\mathcal{D}_a$, to which the full PGT covariant deriva-
1510.06699_F00815	43	■ 1.000	$\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\backslash\mathrm{mathcal{T}}^{\mathrm{a}}_{\mathrm{bc}}\equiv\mathrm{h}_{\mathrm{b}}^{\mathrm{\mu}}\mathrm{h}_{\mathrm{c}}^{\mathrm{\nu}}({}^0\mathrm{D}_{\mathrm{\mu}}\mathrm{b}^{\mathrm{a}}_{\mathrm{\nu}}-{}^0\mathrm{D}_{\mathrm{\nu}}\mathrm{b}^{\mathrm{a}}_{\mathrm{\mu}})=0$	${}^0\mathcal{T}^a{}_{bc}\equiv h_b^\mu h_c^\nu({}^0D_\mu b^a{}_\nu-{}^0D_\nu b^a{}_\mu)=0$	${}^0\mathcal{T}^a{}_{bc}\equiv h_b^\mu h_c^\nu({}^0D_\mu b^a{}_\nu-{}^0D_\nu b^a{}_\mu)=0$. Thus, reduced
1510.06699_F00816	43	■ 1.000	$\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}}=\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}}+\mathrm{b}^{\mathrm{a}}_{\mathrm{\mu}}\backslash\mathrm{mathcal{P}}^{\mathrm{b}}-\mathrm{b}^{\mathrm{b}}_{\mathrm{\mu}}\backslash\mathrm{mathcal{P}}^{\mathrm{a}}$	${}^0A^{\prime ab}{}_\mu={}^0A^{ab}{}_\mu+b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a$	${}^0A^{\prime ab}{}_\mu={}^0A^{ab}{}_\mu+b^a{}_\mu\mathcal{P}^b-b^b{}_\mu\mathcal{P}^a$, which is independent
1510.06699_F00817	43	■ 1.000	$\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}},\{ \}^{\mathrm{\dag}}_{\mathrm{\phi}}\backslash\mathrm{mathcal{D}}_{\mathrm{c}}\mathrm{\varphi}$	${}^0A^{ab}{}_\mu,{}^0\mathcal{D}_c\varphi$	${}^0A^{ab}{}_\mu,{}^0\mathcal{D}_c\varphi$ and ${}^0\mathcal{R}^{ab}{}_{cd}$ all transform independently of
1510.06699_F00818	43	■ 0.999	$\mathrm{A}^{\mathrm{\prime}}_{\mathrm{a}}\mathrm{b}_{\mathrm{\mu}},\backslash\mathrm{mathcal{D}}_{\mathrm{c}}\mathrm{\varphi}$	$A^{ab}{}_\mu,\mathcal{D}_c\varphi$	$A^{ab}{}_\mu,\mathcal{D}_c\varphi$ and $\mathcal{R}^{ab}{}_{cd}$ do assuming the ‘special’ transfor-

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1510.06699_F00853	45	■ 0.971	$\tau_{a\ b}^{\dagger}$	$\tau_{ab}^{\dagger}$	versions $t_{ab}^{\dagger}$ and $\tau_{ab}^{\dagger}$ of free-gravitational and mat-
1510.06699_F00854	45	■ 1.000	$A^{\dagger\dagger\dagger a\ b\{\}\ }_{\{\}\ \mu}\equiv$	$A^{\dagger ab}_{\ \mu}\equiv$	variables such that $A^{ab}_{\ \mu}$ is replaced by $A^{\dagger ab}_{\ \mu}\equiv$
1510.06699_F00855	45	■ 0.932	$A^{\{a\ b\}\ \}_{\{\mu\}}+\left(\mathrm{mathcal{V}}^{\{a\}\ b^{\{b\}\ \}_{\{\mu\}}}\cdot\right.$ $\left.\mathrm{mathcal{V}}^{\{V\}^{\{b\}\ }b^{\{a\}\ \}_{\{\mu\}}}\right)$	$A^{ab}_{\ \mu}+(\mathcal{V}^ab^b_{\ \mu}-\mathcal{V}^bb^a_{\ \mu})$	$A^{ab}_{\ \mu}+(\mathcal{V}^ab^b_{\ \mu}-\mathcal{V}^bb^a_{\ \mu})$.
1510.06699_F00856	46	■ 1.000	$t^{\dagger\dagger\dagger a\ }_{\{\}\ a}$	$t^{\dagger a}_{\ a}$	trace $t^{\dagger a}_{\ a}$ of the free-gravitational energy momen-
1510.06699_F00857	46	■ 1.000	$\tau^{\dagger\dagger\dagger a\ }_{\{\}\ a}$	$\tau^{\dagger a}_{\ a}$	$\tau^{\dagger a}_{\ a}$ of the total matter energy-momentum tensor
1510.06699_F00858	46	■ 0.948	$\{\ \}^{\{0\}\ }\mathrm{mathcal{R}}_{a\ b\ c\ d}^{\dagger\dagger\dagger}\left(h,\ \partial\mathrm{partial}\right.$ $\left.h,\ \partial^2\mathrm{partial}^{\{2\}\ }h,\ A,\ \partial A,\ V,\ \partial V\right)$	${}^0\mathcal{R}_{abcd}^{\dagger}(h,\partial h,\partial^2h,A,\partial A,V,\partial V)$	${}^0\mathcal{R}_{abcd}^{\dagger}(h,\partial h,\partial^2h,A,\partial A,V,\partial V)$, depends on all the
1510.06699_F00859	47	■ 1.000	$\not\ \partial\equiv\gamma^{\mu}\partial_{\mu}$	$\not{\partial}\equiv\gamma^{\mu}\partial_{\mu}$	positive energy on quantisation) and $\not{\partial}\equiv\gamma^{\mu}\partial_{\mu}$. Our
1510.06699_F00860	47	■ 0.938	$\bar{\psi}/t$	$\bar{\psi}/t$	We begin by considering the quantity $\bar{\psi}\not{J}$, where $J^{\mu}=$
1510.06699_F00861	47	■ 0.938	$J^{\{\mu\}}=$	$J^{\mu}=$	We begin by considering the quantity $\bar{\psi}\not{J}$, where $J^{\mu}=$
1510.06699_F00862	47	■ 1.000	$\bar{\psi}\gamma^{\mu}\psi$	$\bar{\psi}\gamma^{\mu}\psi$	$\bar{\psi}\gamma^{\mu}\psi$ is the Dirac current, which we may write as
1510.06699_F00863	47	■ 1.000	$\psi\bar{\psi}$	$\psi\bar{\psi}$	The spinor outer product $\psi\bar{\psi}$ may be rewritten using the
1510.06699_F00864	47	■ 1.000	$\gamma^{\{5\}}=\gamma_{\{5\}}=i\ \gamma^{\{0\}\ }\gamma^{\{1\}\ }\gamma^{\{2\}\ }\gamma^{\{3\}\ }$	$\gamma^5=\gamma_5=i\gamma^0\gamma^1\gamma^2\gamma^3$	where $\gamma^5=\gamma_5=i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho}=\frac{i}{2}[\gamma^{\nu},\gamma^{\rho}]$ (and
1510.06699_F00865	47	■ 1.000	$\sigma^{\{\nu\}\ \rho\}=\frac{i}{2}\left[\gamma^{\nu},\gamma^{\rho}\right]$	$\sigma^{\nu\rho}=\frac{i}{2}[\gamma^{\nu},\gamma^{\rho}]$	where $\gamma^5=\gamma_5=i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho}=\frac{i}{2}[\gamma^{\nu},\gamma^{\rho}]$ (and
1510.06699_F00866	47	■ 1.000	$\rho\equiv\bar{\psi}\psi$	$\rho\equiv\bar{\psi}\psi$	It is convenient to define the real scalar $\rho\equiv\bar{\psi}\psi$ and
1510.06699_F00867	47	■ 0.671	$\beta\equiv\bar{\psi}i\gamma^5\psi$	$\beta\equiv\bar{\psi}i\gamma^5\psi$	real pseudoscalar $\beta\equiv\bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J}=$
1510.06699_F00868	47	■ 0.671	$\bar{\psi}/\hbar=$	$\bar{\psi}/\hbar=$	real pseudoscalar $\beta\equiv\bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J}=$
1510.06699_F00869	47	■ 1.000	$\bar{\psi}\left(\rho+\beta i\gamma^5\right)$	$\bar{\psi}\left(\rho+\beta i\gamma^5\right)$	$\bar{\psi}(\rho+\beta i\gamma^5)$. Postmultiplying this expression by $(\rho-\beta i\gamma^5)$
1510.06699_F00870	47	■ 1.000	$\left(\rho-\beta i\gamma^5\right)$	$(\rho-\beta i\gamma^5)$	$\bar{\psi}(\rho+\beta i\gamma^5)$. Postmultiplying this expression by $(\rho-\beta i\gamma^5)$
1510.06699_F00871	47	■ 0.982	$\not{J}\not{\partial}\psi$	$\not{J}\not{\partial}\psi$	In particular, we note that the quantity $\not{J}\not{\partial}\psi$ appearing
1510.06699_F00872	47	■ 1.000	$J^{\mu}\partial_{\mu}\psi$	$J^{\mu}\partial_{\mu}\psi$	The quantity $J^{\mu}\partial_{\mu}\psi$ is clearly interpreted as a derivative
1510.06699_F00873	47	■ 1.000	$n^{\nu}\equiv-i\sigma^{\mu\nu}J_{\mu}$	$n^{\nu}\equiv-i\sigma^{\mu\nu}J_{\mu}$	the direction $\not{n}^{\nu}\equiv-i\sigma^{\mu\nu}J_{\mu}$, which is orthogonal to J^{μ} ,
1510.06699_F00874	47	■ 1.000	$J_{\nu}n^{\nu}=-i\sigma^{\mu\nu}J_{\mu}J_{\nu}=0$	$J_{\nu}n^{\nu}=-i\sigma^{\mu\nu}J_{\mu}J_{\nu}=0$	since $\not{J}_{\nu}\not{n}^{\nu}=-i\sigma^{\mu\nu}J_{\mu}\not{J}_{\nu}=0$. Thus, $-i\sigma^{\mu\nu}J_{\mu}\partial_{\nu}\psi$ is in-
1510.06699_F00875	47	■ 1.000	$-i\sigma^{\mu\nu}J_{\mu}\partial_{\nu}\psi$	$-i\sigma^{\mu\nu}J_{\mu}\partial_{\nu}\psi$	since $J_{\nu}n^{\nu}=-i\sigma^{\mu\nu}J_{\mu}J_{\nu}=0$. Thus, $-i\sigma^{\mu\nu}J_{\mu}\partial_{\nu}\psi$ is in-
1510.06699_F00876	47	■ 1.000	$x^{\mu}(\lambda)$	$x^{\mu}(\lambda)$	the particle is described by a worldline $\not{x}^{\mu}(\lambda)$, together
1510.06699_F00877	47	■ 1.000	$R(\lambda)$	$R(\lambda)$	with a ‘rotor’ $R(\lambda)$, which is a Dirac spinor defined along
1510.06699_F00878	47	■ 1.000	$\bar{R}R=1$	$\bar{R}R=1$	$\bar{R}R=1$ and $\bar{R}i\gamma^5R=0$, and contains information about
1510.06699_F00879	47	■ 1.000	$\bar{R}i\gamma^5R=0$	$\bar{R}i\gamma^5R=0$	$\bar{R}R=1$ and $\not{R}i\gamma^5R=0$, and contains information about
1510.06699_F00880	47	■ 1.000	R	R	one replaces ψ by R in (A6), restricts derivatives to lie
1510.06699_F00881	47	■ 1.000	$\dot{x}^{\mu}\equiv dx^{\mu}/d\lambda$	$\dot{x}^{\mu}\equiv dx^{\mu}/d\lambda$	locity $\dot{x}^{\mu}\equiv dx^{\mu}/d\lambda$ with $\bar{R}\gamma^{\mu}R$. The last identification
1510.06699_F00882	47	■ 1.000	$\bar{R}\gamma^{\mu}R$	$\bar{R}\gamma^{\mu}R$	locity $\dot{x}^{\mu}\equiv dx^{\mu}/d\lambda$ with $\bar{R}\gamma^{\mu}R$. The last identification
1510.06699_F00883	47	■ 1.000	$p_{\mu}(\lambda)$	$p_{\mu}(\lambda)$	multiplier $p_{\mu}(\lambda)$, which is identified with the momentum
1510.06699_F00884	47	■ 1.000	p_{μ}	p_{μ}	of the particle; note that, in general, p_{μ} and $\dot{x}^{\mu}$ are not
1510.06699_F00885	47	■ 1.000	$\dot{x}^{\mu}$	$\dot{x}^{\mu}$	of the particle; note that, in general, p_{μ} and $\dot{x}^{\mu}$ are not
1510.06699_F00886	47	■ 0.999	$me(\lambda)$	$me(\lambda)$	ten as $me(\lambda)$ for later convenience) to ensure reparam-
1510.06699_F00887	47	■ 1.000	$x^{\mu}(\lambda),R(\lambda),p^{\mu}(\lambda)$	$x^{\mu}(\lambda),R(\lambda),p^{\mu}(\lambda)$	to the dynamical variables $x^{\mu}(\lambda)$, $R(\lambda)$, $p^{\mu}(\lambda)$ and $e(\lambda)$
1510.06699_F00888	47	■ 1.000	p^{μ}/m	p^{μ}/m	replacing $\bar{R}\gamma^{\mu}R$ by p^{μ}/m and then neglecting the particle
1510.06699_F00889	47	■ 1.000	$e(\lambda)\rightarrow\frac{1}{2}e(\lambda)$	$e(\lambda)\rightarrow\frac{1}{2}e(\lambda)$	einbein $e(\lambda)\rightarrow\frac{1}{2}e(\lambda)$ for later convenience, this process
1510.06699_F00890	47	■ 1.000	$x^{\mu}(\lambda),\ p^{\mu}(\lambda)$	$x^{\mu}(\lambda),p^{\mu}(\lambda)$	variables $x^{\mu}(\lambda)$, $p^{\mu}(\lambda)$ and $e(\lambda)$ leads to the classical
1510.06699_F00891	47	■ 1.000	$\rho=0$	$\rho=0$	ticular, by setting $\rho=0$ in the global Weyl transforma-
1510.06699_F00892	47	■ 1.000	$\rho(x)=0$	$\rho(x)=0$	reduce to local Poincaré transformations if $\rho(x)=0$. In
1510.06699_F00893	48	■ 1.000	$D_{\mu}\varphi$	$D_{\mu}\varphi$	derivative $D_{\mu}\varphi$. Thus, all the results given in Sec-
1510.06699_F00894	48	■ 0.981	$\rho(x)=0=$	$\rho(x)=0=$	tions II B and II C also hold in PGT if one sets $\rho(x)=0=$
1510.06699_F00895	48	■ 1.000	$B_{\mu}=0$	$B_{\mu}=0$	valid in PGT if one sets $\not{B}_{\mu}=0$ and removes all asterisks.
1510.06699_F00896	48	■ 1.000	$H_{\mu\nu}=0$	$H_{\mu\nu}=0$	prescription of setting $B_{\mu}=0$ (so that $H_{\mu\nu}=0$) and re-

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1510.06699_F00897	48	■ 0.999	$\backslash\mathrm{mathcal}\{R\} \equiv \backslash\mathrm{mathcal}\{R\}^{\{a\}b\}_{\{a\}b\}, \backslash\Lambda$	$\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}, \Lambda$	where $\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}$, Λ is a cosmological constant, a is
1510.06699_F00898	48	■ 1.000	$a, \backslash\Lambda, \backslash\mathrm{left}\{\backslash\alpha_i\}\backslash\mathrm{right}\}$	$a, \Lambda, \{\alpha_i\}$	depends on the choice of the parameters $a, \Lambda, \{\alpha_i\}$ and
1510.06699_F00899	48	■ 1.000	$\backslash\mathrm{left}\{\backslash\beta_i\}\backslash\mathrm{right}\}$	$\{\beta_i\}$	$\{\beta_i\}$ in L_G . As an illustration, in Appendix C we give
1510.06699_F00900	48	■ 1.000	$\backslash\mathrm{varphi}, \backslash\mathrm{partial} \backslash\mathrm{varphi}, h$	$\varphi, \partial\varphi, h$	induce a dependence on $\varphi, \partial\varphi, h$ and A , but one typically
1510.06699_F00901	48	■ 0.546	$\backslash\mathrm{mathcal}\{H\}_{a\,b}, j^a, \backslash\mathrm{zeta}^a$	$\mathcal{H}_{ab}, j^a, \zeta^a$	tion III, by removing all asterisks and setting $\mathcal{H}_{ab}, j^a, \zeta^a$
1510.06699_F00902	48	■ 1.000	$\backslash\mathrm{mu} \backslash\mathrm{phi} \backslash\mathrm{bar}\{\backslash\mathrm{psi}\} \backslash\mathrm{psi} \backslash\mathrm{rightarrow} m \backslash\mathrm{bar}\{\backslash\mathrm{psi}\} \backslash\mathrm{psi}$	$\mu\phi\bar{\psi}\psi \rightarrow m\bar{\psi}\psi$	$\mu\phi\bar{\psi}\psi \rightarrow m\bar{\psi}\psi$ for the Dirac field. Thus, after this change
1510.06699_F00903	48	■ 1.000	$\backslash\mathrm{phi}=1$	$\phi = 1$	isks and setting $B_\mu = 0$ and $\phi = 1$, all the results in these
1510.06699_F00904	48	■ 0.992	$\{ \}^{\{0\}} \backslash\mathrm{mathcal}\{T\}^{\{a\}}\{ \}_{b\,c} \equiv 0$	${}^0\mathcal{T}^a_{bc} \equiv 0$	tically, ${}^0\mathcal{T}^a_{bc} \equiv 0$, only the reduced ‘rotational’ field
1510.06699_F00905	48	■ 1.000	$\{ \}^{\{0\}} \backslash\mathrm{mathcal}\{R\}^{\{a\}b\}\{ \}_{c\,d} \backslash\mathrm{left}(h, \backslash\mathrm{partial} h, \backslash\mathrm{partial}^{\{2\}} h \backslash\mathrm{right})$	${}^0\mathcal{R}^{ab}_{cd}(h, \partial h, \partial^2 h)$	strength ${}^0\mathcal{R}^{ab}_{cd}(h, \partial h, \partial^2 h)$ remains, which depends on
1510.06699_F00906	49	■ 1.000	$\{ \}^{\{0\}} \backslash\mathrm{mathcal}\{D\}_{a\,} \backslash\mathrm{mathcal}\{T\}^{\{a\}}=h \backslash\mathrm{partial}_{\backslash\mathrm{mu}} \backslash\mathrm{left}(h^{-1} h_{a\,}\{ \}^{\backslash\mathrm{mu}} \backslash\mathrm{mathcal}\{T\}^{\{a\}}\backslash\mathrm{right})$	${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$	terms of a total derivative as ${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$,
1510.06699_F00907	49	■ 0.980	$\{ \}^{\{\backslash\mathrm{underline}\{105\}\}}$	$\overline{105}$	whereas this is not possible in WGT ¹⁰⁵ .
1510.06699_F00908	49	■ 0.935	$L_{\{0\}} \backslash\mathrm{mathcal}\{R\}^{\{2\}}$	$L_0\mathcal{R}^2$	where $L_{\mathcal{R}^2}$ is based on (2) with $\mathcal{R}^{ab}_{cd}$ replaced by
1510.06699_F00909	49	■ 0.606	$L_{\{0^{\backslash\mathrm{mathcal}\{R\}^{\{2\}}\}}}$	$L_{0\mathcal{R}^2}$	tinct terms in $L_{\mathcal{R}^2}$ is reduced to
1510.06699_F00910	49	■ 0.999	$\backslash\mathrm{mathcal}\{R\}_{a\,b\,c\,d} \backslash\mathrm{rightarrow} \{ \}^{\{0\}} \backslash\mathrm{mathcal}\{R\}_{a\,b\,c\,d}$	$\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}$	$\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}$. Thus, one can set to zero any one of the
1510.06699_F00911	49	■ 1.000	$\backslash\alpha_3=0$	$\alpha_3 = 0$	
1510.06699_F00912	49	■ 0.999	$\backslash\alpha_1=\backslash\alpha_2=0$	$\alpha_1 = \alpha_2 = 0$	h , then one must set $\alpha_1 = \alpha_2 = 0$ (at least in $D \leq 4$
1510.06699_F00913	49	■ 1.000	$a=\backslash\mathrm{frac}\{1\}\{2\}$	$a = \frac{1}{2}$	spacetime dimensions), and on choosing $a = \frac{1}{2}$ the re-
1510.06699_F00914	49	■ 0.999	$a=0=\backslash\Lambda, \backslash\alpha_1=-\backslash\mathrm{frac}\{1\}\{3\} \backslash\alpha_2$	$a = 0 = \Lambda, \alpha_1 = -\frac{1}{3}\alpha_2$	is $a = 0 = \Lambda, \alpha_1 = -\frac{1}{3}\alpha_2$, which corresponds to the free
1510.06699_F00915	49	■ 1.000	$U_{\{4\}}$	U_4	that the spacetime has, in general, a Riemann–Cartan U_4
1510.06699_F00916	49	■ 1.000	$\{ \}^{\{0\}} R^{\{\backslash\mathrm{rho}\}}\{ \}_{\backslash\mathrm{sigma} \backslash\mathrm{mu} \backslash\mathrm{nu}}$	${}^0R^\rho{}_{\sigma\mu\nu}$	leads one to recognise ${}^0R^\rho{}_{\sigma\mu\nu}$ as the standard curvature
1510.06699_F00917	49	■ 1.000	$R \equiv R^{\backslash\mathrm{mu} \backslash\mathrm{nu}}\{ \}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}$	$R \equiv R^{\mu\nu}{}_{\mu\nu}$	which relates the Ricci–Cartan scalar $R \equiv R^{\mu\nu}{}_{\mu\nu}$ in
1510.06699_F00918	49	■ 0.999	$\{ \}^{\{0\}} R \equiv \{ \}^{\{0\}} R^{\backslash\mathrm{mu} \backslash\mathrm{nu}}\{ \}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}$	${}^0R \equiv {}^0R^{\mu\nu}{}_{\mu\nu}$	U_4 spacetime to the Ricci scalar ${}^0R \equiv {}^0R^{\mu\nu}{}_{\mu\nu}$ in V_4
1510.06699_F00919	49	■ 1.000	$L_{\{\mathrm{mathrm}\{G\}}=-R/(2\,\backslash\mathrm{kappa})$	$L_G = -R/(2\kappa)$	taken to be $L_G = -R/(2\kappa)$ and the theory is interpreted
1510.06699_F00920	49	■ 1.000	$T_{\backslash\mathrm{mu} \backslash\mathrm{nu} \backslash\mathrm{sigma}}$	$T_{\mu\nu\sigma}$	that includes a tensor field $T_{\mu\nu\sigma}$ (obeying the appropri-
1510.06699_F00921	49	■ 1.000	$T_{\backslash\mathrm{mu} \backslash\mathrm{nu} \backslash\mathrm{lambda}}$	$T_{\mu\nu\lambda}$	Lagrangian is given by the terms containing $T_{\mu\nu\lambda}$ in (B6)
1510.06699_F00922	49	■ 1.000	$\backslash\mathrm{mathcal}\{R\}=\backslash\mathrm{mathcal}\{R\}^{\{a\}b\}\{ \}_{a\,b}$	$\mathcal{R} = \mathcal{R}^{ab}_{ab}$	$\mathcal{R} = \mathcal{R}^{ab}_{ab}$. Thus, in Kibble’s original theory, one has
1510.06699_F00923	49	■ 1.000	$\backslash\Lambda=\backslash\alpha_i=\backslash\beta_i=0$	$\Lambda = \alpha_i = \beta_i = 0$	which corresponds to setting $a = \frac{1}{2}$ and $\Lambda = \alpha_i = \beta_i = 0$
1510.06699_F00924	49	■ 0.983	$\{ \}^{\{\backslash\mathrm{underline}\{106\}\}}$	$\overline{106}$	theory ¹⁰⁶ , which, when re-interpreted geometrically, is a
1510.06699_F00925	50	■ 1.000	$\backslash\mathrm{mathcal}\{T\}_{a\,} \equiv \backslash\mathrm{mathcal}\{T\}^{\{c\}}\{ \}_{a\,c}$	$\mathcal{T}_a \equiv \mathcal{T}^c_{ac}$	where $\mathcal{T}_a \equiv \mathcal{T}^c_{ac}$, as defined in (66). In particular, by
1510.06699_F00926	50	■ 0.999	$\backslash\mathrm{mathcal}\{A\}^{\{a\}b\}\{ \}_{c\,}=h_{c\,}\{ \}^{\backslash\mathrm{mu}} A^{\{a\}b\}\{ \}_{\backslash\mathrm{mu}}$	$\mathcal{A}^{ab}_c = h_c{}^\mu A^{ab}_\mu$	its fully Lorentz-index counterpart $\mathcal{A}^{ab}_c = h_c{}^\mu A^{ab}_\mu$. In
1510.06699_F00927	50	■ 1.000	$\backslash\mathrm{sigma}_{a\,b}\{ \}^{\{c\}} \backslash\mathrm{neq} 0$	$\sigma_{ab}{}^c \neq 0$	isks). In the general case, in which $\sigma_{ab}{}^c \neq 0$, this is not,
1510.06699_F00928	50	■ 1.000	$\backslash\mathrm{sigma}_{a\,b\,c}$	σ_{abc}	derivatives of the matter field(s), then σ_{abc} is indepen-
1510.06699_F00929	50	■ 1.000	$\backslash\mathrm{mathcal}\{T\}^{\{c\}}\{ \}_{a\,b}$	$\mathcal{T}^c_{ab}$	field strength $\mathcal{T}^c_{ab}$, and so the h -field itself must satisfy
1510.06699_F00930	50	■ 0.997	$\backslash\mathrm{mu} \backslash\mathrm{phi} \equiv m$	$\mu\phi \equiv m$	Lagrangian density is given by (70) with $\mu\phi \equiv m$, sev-
1510.06699_F00931	50	■ 1.000	$\backslash\mathrm{mathcal}\{T\}^{\{c\}}\{ \}_{a\,b}=2\,\backslash\mathrm{kappa} h \backslash\mathrm{sigma}_{a\,b}\{ \}^{\{c\}}$	$\mathcal{T}^c_{ab} = 2\kappa h\sigma_{ab}{}^c$	$\mathcal{T}^c_{ab} = 2\kappa h\sigma_{ab}{}^c$. More interestingly, the condition (C9)
1510.06699_F00932	50	■ 1.000	$R_{\backslash\mathrm{mu} \backslash\mathrm{nu}} \equiv R^{\{\backslash\mathrm{lambda}\}}\{ \}_{\backslash\mathrm{mu} \backslash\mathrm{lambda} \backslash\mathrm{nu}}$	$R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00933	50	■ 1.000	$R \equiv R^{\backslash\mathrm{mu}}\{ \}_{\backslash\mathrm{mu}}$	$R \equiv R^\mu{}_\mu$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00934	50	■ 1.000	$\backslash\mathrm{Gamma}^{\{\backslash\mathrm{lambda}\}}\{ \}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}$	$\Gamma^\lambda{}_{\mu\nu}$	but one must remember that the affine connection $\Gamma^\lambda{}_{\mu\nu}$
1510.06699_F00935	51	■ 1.000	$\backslash\mathrm{tau}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}=$	$\tau_{\mu\nu} =$	is not, in general, symmetric in μ and ν . Also, $\tau_{\mu\nu} =$
1510.06699_F00936	51	■ 1.000	$e^{\{a\}}\{ \}_{\backslash\mathrm{mu}} e^{\{b\}}\{ \}_{\backslash\mathrm{nu}} \backslash\mathrm{tau}_{a\,b}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00937	51	■ 1.000	$\backslash\mathrm{sigma}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}\{ \}^{\{\backslash\mathrm{lambda}\}}=e^{\{a\}}\{ \}_{\backslash\mathrm{mu}} e^{\{b\}}\{ \}_{\backslash\mathrm{nu}} e_{c\,}\{ \}^{\{\backslash\mathrm{lambda}\}} \backslash\mathrm{sigma}_{a\,b}\{ \}^{\{c\}}$	$\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00938	51	■ 1.000	$\backslash\mathrm{sigma}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}\{ \}^{\{\backslash\mathrm{lambda}\}}$	$\sigma_{\mu\nu}{}^\lambda$	angular-momentum $\sigma_{\mu\nu}{}^\lambda$ of the matter field vanishes,
1510.06699_F00939	51	■ 1.000	$T^{\{\backslash\mathrm{lambda}\}}\{ \}_{\backslash\mathrm{mu} \backslash\mathrm{nu}}$	$T^\lambda{}_{\mu\nu}$	then so too does the torsion $T^\lambda{}_{\mu\nu}$ and hence the affine
1510.06699_F00940	51	■ 1.000	$R_{\backslash\mathrm{mu} \backslash\mathrm{nu}}$	$R_{\mu\nu}$	case, $R_{\mu\nu}$ reduces to the standard symmetric Ricci ten-

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1510.06699_F00941	51	■ 1.000	$\{ \}^{\{0\}} R_{\{\mu \nu\}} \equiv R_{\{\mu \nu\}} \backslash \left(V_{\{4\}} \right)$	${}^0 R_{\mu \nu} \equiv R_{\mu \nu}(V_4)$	so ${}^0 R_{\mu \nu} \equiv R_{\mu \nu}(V_4)$ in a Riemann spacetime, expressed
1510.06699_F00942	51	■ 0.979	$\{ \}^{\{ \backslash \underline{108} \}}$	$\frac{108}{}$	grangian suggested by Dirac has the simple form ¹⁰⁸
1510.06699_F00943	51	■ 1.000	$-\frac{1}{4}\{ \}$	$-\frac{1}{4}$	sionless constant (usually positive); the factor of $-\frac{1}{4}$ is
1510.06699_F00944	51	■ 1.000	$\phi^{\{4\}}$	ϕ^4	ϕ^4 , and a term proportional to $\phi^2 \mathcal{R}$ that non-minimally
1510.06699_F00945	51	■ 1.000	$L_{\{\backslash \varphi \}}(\backslash \varphi, \backslash \partial \varphi, h, A, B, \backslash \phi)$	$L_{\varphi}(\varphi, \partial \varphi, h, A, B, \phi)$	for the matter field φ are $\mathcal{L}_{\varphi}(\varphi, \partial \varphi, h, A, B, \phi)$; note that
1510.06699_F00946	51	■ 1.000	$\backslash \mathrm{mathcal{L}}_{\{T\}} = h^{\{-1\}} \mathcal{L}_{\{T\}}$	$\mathcal{L}_T = h^{-1} L_T$	The total Lagrangian density is then $\mathcal{L}_T = h^{-1} L_T$,
1510.06699_F00947	51	■ 1.000	$\mathcal{L}_{\{T\}}$	L_T	where L_T has the form
1510.06699_F00948	51	■ 0.993	$(a, \backslash \xi, \backslash \nu, \backslash \lambda)$	(a, ξ, ν, λ)	$\langle a, \xi, \nu, \lambda \rangle$ are dimensionless (and usually positive). In
1510.06699_F00949	51	■ 1.000	$\phi^{\{-2\}}$	ϕ^{-2}	plied ϕ^{-2} should replace Newton’s gravitational constant
1510.06699_F00950	51	■ 0.967	$h_{\{a\}\{ \}^{\{\mu\}}, A^{\{a\} b\}\{ \}_{\{\mu\}}, B_{\{\mu\}}$	$h_a{}^{\mu}, A^{ab}{}_{\mu}, B_{\mu}$	independent fields $h_a{}^{\mu}$, $A^{ab}{}_{\mu}$, B_{μ} and ϕ respectively ⁵⁰ ,
1510.06699_F00951	51	■ 1.000	$B_{\{\mu\}}, \backslash \phi$	B_{μ}, ϕ	with respect to B_{μ}, ϕ and the spacetime metric $g_{\mu \nu}$. Our
1510.06699_F00952	52	■ 1.000	$\backslash \left(\backslash \tau_{\{\backslash \varphi \}} \backslash \right)^{\{a\}\{ \}_{\{b\}}}$	$(\tau_{\varphi})^a{}_b$	where $(\tau_{\varphi})^a{}_b$ is the energy-momentum tensor of the mat-
1510.06699_F00953	52	■ 1.000	$\backslash \left(\backslash \sigma_{\{\backslash \varphi \}} \backslash \right)^{\{a\} b\}\{ \}^{\{c\}}$	$(\sigma_{\varphi})_{ab}{}^c$	where $(\sigma_{\varphi})_{ab}{}^c$ is the spin-angular-momentum tensor of
1510.06699_F00954	52	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{*\}} \rightarrow \backslash \mathrm{mathcal{D}}_{\{a\}}, \backslash \mathrm{mathcal{T}}^{\{*\} c\}\{ \}_{\{a\} b\} \rightarrow \backslash \mathrm{mathcal{T}}^{\{c\}\{ \}_{\{a\} b\}}$	$\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}{}_{ab} \rightarrow \mathcal{T}^c{}_{ab}$	if one makes the replacements $\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}{}_{ab} \rightarrow \mathcal{T}^c{}_{ab}$
1510.06699_F00955	52	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{*\}} \rightarrow \backslash \mathrm{mathcal{T}}_{\{a\}}$	$\mathcal{T}_a^* \rightarrow \mathcal{T}_a$	and $\mathcal{T}_a^* \rightarrow \mathcal{T}_a$ to PGT quantities, although the resulting
1510.06699_F00956	52	■ 1.000	$\backslash \left(\backslash \zeta_{\{\backslash \varphi \}} \backslash \right)^{\{c\}}$	$(\zeta_{\varphi})^c$	where $(\zeta_{\varphi})^c$ is the dilation current of the φ matter field.
1510.06699_F00957	52	■ 1.000	$\backslash \left. \backslash \mathrm{mathcal{D}}_{\{a\}}^{\{*\}} \backslash \phi \right _{\{\phi = \phi_0\}} = -\phi_0 \backslash \mathrm{mathcal{B}}_{\{a\}}$	$\mathcal{D}_a^* \phi _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$	replacement $\mathcal{D}_a^* \phi _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$ throughout. It is also
1510.06699_F00958	52	■ 1.000	$a=1 / \backslash \left(\backslash \kappa \phi_0^{\{2\}} \backslash \right)$	$a = 1/(\kappa \phi_0^2)$	convenient to set $a = 1/(\kappa \phi_0^2)$, where κ is Einstein’s grav-
1510.06699_F00959	52	■ 1.000	$\backslash \Lambda = \backslash \lambda \phi_0^{\{2\}} \backslash \phi / a = \backslash \kappa \backslash \lambda$	$\Lambda = \lambda \phi_0^2/a = \kappa \lambda$	where $\Lambda = \lambda \phi_0^2/a = \kappa \lambda$ has the correct dimensions of
1510.06699_F00960	52	■ 1.000	$m^{\{2\}} = \backslash \nu \phi_0^{\{2\}} / \backslash \xi$	$m^2 = \nu \phi_0^2/\xi$	vector field (or Proca field) of mass-squared $m^2 = \nu \phi_0^2/\xi$.
1510.06699_F00961	53	■ 1.000	$\backslash \mathrm{mathcal{R}}^{\{a\} b\}\{ \}_{\{c\} d\}, \backslash \mathrm{mathcal{T}}^{\{a\}\{ \}_{\{b\} c\}} = \backslash \mathrm{mathcal{T}}^{\{*\} a\}\{ \}_{\{b\} c\} - \backslash \delta_{\{c\}}^{\{a\}} \backslash \mathrm{mathcal{B}}_{\{b\}} + \backslash \delta_{\{b\}}^{\{a\}} \backslash \mathrm{mathcal{B}}_{\{c\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^a{}_{bc} = \mathcal{T}^{*a}{}_{bc} - \delta_c^a \mathcal{B}_b + \delta_b^a \mathcal{B}_c$	tensors $\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^a{}_{bc} = \mathcal{T}^{*a}{}_{bc} - \delta_c^a \mathcal{B}_b + \delta_b^a \mathcal{B}_c$ and $\mathcal{H}_{ab}$
1510.06699_F00962	53	■ 1.000	$\backslash \left(\backslash \zeta_{\{\backslash \psi \}} \backslash \right)^{\{\mu\}} = 0$	$(\zeta_{\psi})^{\mu} = 0$	does not appear in (D12), the dilation current $(\zeta_{\psi})^{\mu} = 0$
1510.06699_F00963	53	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{a\}} = 0$	$\mathcal{T}_a = 0$	theory, from (C10), we again have $\mathcal{T}_a = 0$. Thus, the
1510.06699_F00964	53	■ 1.000	$\backslash \tau_{\{\backslash \psi \}} = h^{\{-1\}} \backslash \mu \phi \backslash \psi \backslash \psi$	$\tau_{\psi} = h^{-1} \mu \phi \bar{\psi} \psi$	of the Dirac field is easily found to be $\tau_{\psi} = h^{-1} \mu \phi \bar{\psi} \psi$.
1510.06699_F00965	53	■ 1.000	$m^{\{2\}} \equiv \backslash \nu \phi_0^{\{2\}} / \backslash \xi$	$m^2 \equiv \nu \phi_0^2/\xi$	where we have defined $m^2 \equiv \nu \phi_0^2/\xi$ and $\sigma_{ab}{}^c$ is the spin-
1510.06699_F00966	53	■ 1.000	$\backslash \mathrm{mathcal{B}}_{\{a\}}$	$\mathcal{B}_a$	field $\mathcal{B}_a$ has a non-minimal coupling to the gravitational
1510.06699_F00967	53	■ 1.000	$\mathcal{D}_{\{\backslash \mu\}}^{\{*\}} = \mathcal{D}_{\{\backslash \mu\}} + w \mathcal{B}_{\{\backslash \mu\}}$	$D_{\mu}^* = D_{\mu} + w B_{\mu}$	variant derivative (20), namely $\mathcal{D}_{\mu}^* = D_{\mu} + w B_{\mu}$, the
1510.06699_F00968	54	■ 1.000	$\mathcal{P}_{\{\backslash \mu\}} \equiv \partial_{\{\backslash \mu\}} \rho$	$P_{\mu} \equiv \partial_{\mu} \rho$	where $P_{\mu} \equiv \partial_{\mu} \rho$. This transformation for V_{μ} implies
1510.06699_F00969	54	■ 1.000	$(w=0)$	$(w = 0)$	variant ($w = 0$) under extended local dilations. One
1510.06699_F00970	54	■ 1.000	$\backslash \mathrm{mathcal{R}}^{\{\dagger a\} b\}\{ \}_{\{c\} d\} \equiv h_c{}^{\mu} h_d{}^{\nu} \backslash R^{\{\dagger a\} b}{}_{\mu \nu}$	$\mathcal{R}^{\dagger ab}{}_{cd} \equiv h_c{}^{\mu} h_d{}^{\nu} R^{\dagger ab}{}_{\mu \nu}$	can then define $\mathcal{R}^{\dagger ab}{}_{cd} \equiv h_c{}^{\mu} h_d{}^{\nu} R^{\dagger ab}{}_{\mu \nu}$ as before, which
1510.06699_F00971	54	■ 0.997	$\mathcal{D}_{\{\backslash \mu\}}^{\{\natural\}} \backslash \varphi$	$D_{\mu}^{\natural} \varphi$	ever, one quickly finds that $D_{\mu}^{\natural} \varphi$ does <i>not</i> transform co-
1510.06699_F00972	54	■ 1.000	$\mathcal{D}_{\{\backslash \mu\}}^{\{\natural\}}$	$D_{\mu}^{\natural}$	ing our reluctance thus far to call $D_{\mu}^{\natural}$ (or $\mathcal{D}_a^{\natural}$) a covariant
1510.06699_F00973	54	■ 1.000	$\backslash \nu = \backslash \beta_1 = \backslash \beta_2 = 0$	$\nu = \beta_1 = \beta_2 = 0$	gravitational Lagrangian L_G and $\nu = \beta_1 = \beta_2 = 0$ in the
1510.06699_F00974	54	■ 0.994	$\backslash \xi = \backslash \nu = \backslash \beta_1 = \backslash \beta_2 = 0$	$\xi = \nu = \beta_1 = \beta_2 = 0$	in Section IV A, but with $\xi = \nu = \beta_1 = \beta_2 = 0$. These
1510.06699_F00975	54	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{\natural\}} \backslash \chi$	$\mathcal{D}_a^{\natural} \chi$	to $\mathcal{D}_a^{\natural} \chi$ to assemble the appropriate <i>covariant</i> derivative
1510.06699_F00976	54	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{\dagger\}} \backslash \chi = h_a{}^{\mu} \backslash \mathcal{D}_{\{\backslash \mu\}}^{\dagger} \chi$	$\mathcal{D}_a^{\dagger} \chi = h_a{}^{\mu} D_{\mu}^{\dagger} \chi$	$\mathcal{D}_a^{\dagger} \chi$. In each case, one finds that $\mathcal{D}_a^{\dagger} \chi = h_a{}^{\mu} D_{\mu}^{\dagger} \chi$, where
1510.06699_F00977	54	■ 0.991	$\mathcal{D}_{\{\backslash \mu\}}^{\{\dagger\}}$	$D_{\mu}^{\dagger}$	$D_{\mu}^{\dagger}$ has the form (143), but with the appropriate value of
1510.06699_F00978	54	■ 0.997	$\backslash \left(U_{\{4\}} \backslash \right)$	(U_4)	respectively, of a Riemann–Cartan (U_4) spacetime.
1510.06699_F00979	54	■ 1.000	$\backslash \delta \backslash \varphi$	$\delta \varphi$	given action, for example, then the ‘total’ variation $\delta \varphi$ and the

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1510.06699_F00980	54	■ 1.000	$\left[\mathrm{D}_{\mathrm{c}}, \mathrm{D}_{\mathrm{d}}\right] \varphi \equiv \frac{1}{2} \mathcal{R}^{a b}{}_{c d} \Sigma_{c d} \varphi - \mathcal{T}^a{}_{c d} \mathcal{D}_a \varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{a c}{}_{b c}$	$[\mathcal{D}_c, \mathcal{D}_d] \varphi \equiv \frac{1}{2} \mathcal{R}^{ab}{}_{cd} \Sigma_{cd} \varphi - \mathcal{T}^a{}_{cd} \mathcal{D}_a \varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$	signature. $[\mathcal{D}_c, \mathcal{D}_d] \varphi \equiv \frac{1}{2} \mathcal{R}^{ab}{}_{cd} \Sigma_{cd} \varphi - \mathcal{T}^a{}_{cd} \mathcal{D}_a \varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$
1510.06699_F00981	54	■ 1.000	$\mathrm{T}_{\mathrm{a}} \equiv \mathrm{T}^b{}_{a b}$	$\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$	and $\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$.
1510.06699_F00982	55	■ 1.000	$\widehat{\mathrm{D}}_{\mathrm{a}}^*$	$\widehat{\mathcal{D}}_a^*$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^\mu$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see
1510.06699_F00983	55	■ 1.000	$\widehat{\mathrm{R}}^{a b}{}_{c d}$	$\widehat{\mathcal{R}}^{ab}{}_{cd}$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^\mu$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see
1510.06699_F00991	55	■ 0.926	a^{a}	a	PGT (generalised) covariant derivative ^a
1510.06699_F00993	55	■ 0.820	$\mathrm{a}^{\mathrm{a}} \mathrm{b}^{\mathrm{b}}$	$\mathrm{a} \mathrm{b}$	eWGT rotational gauge field strength ^{a, b}
1510.06699_F00994	55	■ 0.666	b^{b}	b	PGT translational gauge field strength ^b
1510.06699_F00995	55	■ 0.957	b^{b}	b	WGT translational gauge field strength ^b
1510.06699_F01002	55	■ 0.957	$\mathrm{T}_{b c} \equiv 0$	$\mathcal{T}^a{}_{bc} \equiv 0$	PGT reduced A -field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
1510.06699_F01003	55	■ 0.998	$\mathrm{T}^*{}_{b c} \equiv 0$	$\mathcal{T}^{*a}{}_{bc} \equiv 0$	WGT reduced A -field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
1510.06699_F01004	55	■ 1.000	$\mathrm{T}^{\dagger}{}_{b c} \equiv 0$	$\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$	eWGT reduced $A^\dagger$ -field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
1510.06699_F01017	55	■ 0.931	c^{c}	c	gravitational sector energy-momentum tensor ^c
1510.06699_F01018	55	■ 0.562	c^{c}	c	gravitational sector eWGT covariant dilation current ^c
1510.06699_F01019	55	■ 0.999	$\mathrm{R}^a{}_b \equiv \mathcal{R}^{a c}{}_{b c}, \mathcal{R} \equiv \mathcal{R}^a{}_a$	$\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$	^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01020	55	■ 0.999	$\mathrm{T}_{\mathrm{a}}=\mathrm{T}^b{}_{a b}$	$\mathcal{T}_a = \mathcal{T}^b{}_{ab}$	^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01021	56	■ 1.000	$S L(2, C)$	$S L(2, C)$	Lorentz transformations by Λ , rather than $S L(2, C)$ elements.
1510.06699_F01022	56	■ 1.000	$h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$	$h^\mu{}_a J_\mu \neq \mathcal{J}_a$	$\mathcal{J}$, so that, for example, $h^\mu{}_a J_\mu \neq \mathcal{J}_a$. We will not, however,
1510.06699_F01023	56	■ 1.000	$\mathrm{A}_{a b c}$	${}^0 \mathcal{A}_{abc}$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01024	56	■ 1.000	$\mathcal{K}_{a b c}$	$\mathcal{K}_{abc}$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01025	56	■ 1.000	$\mathrm{A}^{* a b}{}_c =$	${}^0 \mathcal{A}^{*ab}{}_c =$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01026	56	■ 1.000	$\mathrm{A}^{a b}{}_c+\delta_c^a \mathcal{B}^b-\delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01027	56	■ 1.000	$\mathcal{K}^{* a b}{}_c=\mathcal{K}^{a b}{}_c-\delta_c^a \mathcal{B}^b+\delta_c^b \mathcal{B}^a$	$\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01028	56	■ 1.000	$t^a{}_{\mu} \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{a b}{}^\mu \equiv$	$t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$	⁷² One could alternatively define $t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$
1510.06699_F01029	56	■ 1.000	$h \delta \mathcal{L}_G / \delta A^{a b}{}_{\mu}$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01030	56	■ 1.000	$j^{\mu} \equiv h \delta \mathcal{L}_G / \delta B_{\mu}$	$j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01031	56	■ 1.000	$\dot{R} \equiv$	$\dot{R} \equiv$	point particle one must, in addition, make the replacement $\dot{R} \equiv$
1510.06699_F01032	56	■ 1.000	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$.
1510.06699_F01033	56	■ 0.994	$\{77,78\}$	77,78	theory ^{77,78} . In the standard interpretation, the scalar field in
1510.06699_F01034	56	■ 0.999	$\bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a^* \psi=\bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a \psi$	⁸¹ It is easily shown that $\bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a 0} \overleftrightarrow{\mathcal{D}}_a \psi$ (which does not
1510.06699_F01035	56	■ 1.000	$\theta(x)$	$\theta(x)$	a field $\theta(x)$. Provided it has Weyl weight $w = 0$, one would ob-
1510.06699_F01036	57	■ 1.000	$D_{\mu}^{\dagger} \varphi =$	$D_\mu^\dagger \varphi =$	ticular, we note that the eWGT covariant derivative $D_\mu^\dagger \varphi =$
1510.06699_F01037	57	■ 1.000	$\left[D_{\mu}+\frac{1}{2}\left(\mathcal{V}^a b_{\mu}^b-\mathcal{V}^b b_{\mu}^a\right) \Sigma_{a b}-w V_{\mu}-\frac{1}{3} w T_{\mu}\right] \varphi$	$\left[D_\mu + \frac{1}{2} \left(\mathcal{V}^a b_{\mu}^b - \mathcal{V}^b b_{\mu}^a\right) \Sigma_{ab} - w V_\mu - \frac{1}{3} w T_\mu\right] \varphi$	$[D_\mu + \frac{1}{2} (\mathcal{V}^a b_{\mu}^b - \mathcal{V}^b b_{\mu}^a) \Sigma_{ab} - w V_\mu - \frac{1}{3} w T_\mu] \varphi$ represents a far
1510.06699_F01038	57	■ 1.000	$D_{\mu}^* \varphi=\left(D_{\mu}+w B_{\mu}\right) \varphi$	$D_\mu^* \varphi = (D_\mu + w B_\mu) \varphi$	derivative $D_\mu^* \varphi = (D_\mu + w B_\mu) \varphi$
1510.06699_F01039	57	■ 0.995	$\mathrm{A}^{\dagger a b}{}_c={ }^0 \mathcal{A}^{a b}{}_c-\delta_c^a\left(\mathcal{V}^b+\frac{1}{3} \mathcal{T}^b\right)+\delta_c^b\left(\mathcal{V}^a+\right.$	${}^0 \mathcal{A}^{\dagger ab}{}_c = {}^0 \mathcal{A}^{ab}{}_c - \delta_c^a \left(\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b\right) + \delta_c^b \left(\mathcal{V}^a +$	pendix B), one has ${}^0 \mathcal{A}^{\dagger ab}{}_c = {}^0 \mathcal{A}^{ab}{}_c - \delta_c^a (\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b) + \delta_c^b (\mathcal{V}^a +$

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1510.06699_F01040	57	0.600	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$) and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$, so confirming that
1510.06699_F01041	57	0.600	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$) and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$ so confirming that
1510.06699_F01042	57	0.965	$\bar{\psi} \gamma^{a0} \mathcal{D}_a^\dagger \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^{a0} \mathcal{D}_a^\dagger \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	⁹⁴ It is easily shown that $\bar{\psi} \gamma^{a0} \mathcal{D}_a^\dagger \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$ (which does not
1510.06699_F01043	57	1.000	$\nabla_\sigma g_{\mu\nu} =$	$\nabla_\sigma g_{\mu\nu} =$	that given in (124) for WGT can both be written as $\nabla_\sigma g_{\mu\nu} =$
1510.06699_F01044	57	1.000	$-\frac{1}{2} (\Gamma^\lambda{}_{\lambda\sigma} - {}^0\Gamma^\lambda{}_{\lambda\sigma}) g_{\mu\nu}$	$-\frac{1}{2} (\Gamma^\lambda{}_{\lambda\sigma} - {}^0\Gamma^\lambda{}_{\lambda\sigma}) g_{\mu\nu}$	$-\frac{1}{2} (\Gamma^\lambda{}_{\lambda\sigma} - {}^0\Gamma^\lambda{}_{\lambda\sigma}) g_{\mu\nu}$, where ${}^0\Gamma^\lambda{}_{\mu\nu}$ is the metric connection;
1510.06699_F01045	57	1.000	$L_{\mathcal{R}^2} =$	$L_{\mathcal{R}^2} =$	which its symmetries are more manifest, namely $L_{\mathcal{R}^2} =$
1510.06699_F01046	57	1.000	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^\dagger \mathcal{R}^{\dagger(ab)} + \alpha_3' \mathcal{R}_{[ab]}^\dagger \mathcal{R}^{\dagger[ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^\dagger \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^\dagger \mathcal{R}^{\dagger(ab)} + \alpha_3' \mathcal{R}_{[ab]}^\dagger \mathcal{R}^{\dagger[ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^\dagger \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^\dagger \mathcal{R}^{\dagger(ab)} + \alpha_3' \mathcal{R}_{[ab]}^\dagger \mathcal{R}^{\dagger[ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^\dagger \mathcal{R}^{\dagger a(bc)d} +$
1510.06699_F01047	57	1.000	$\alpha_5' \mathcal{R}_{a[bc]d}^\dagger \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^\dagger \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^\dagger \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^\dagger \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^\dagger \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^\dagger \mathcal{R}^{\dagger cdab}$, where the new coefficients
1510.06699_F01048	57	0.994	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01049	57	0.994	$\alpha_5' = \alpha_4 - \alpha_5$	$\alpha_5' = \alpha_4 - \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01050	57	1.000	$\alpha_2' = \alpha_3'$	$\alpha_2' = \alpha_3'$	allows one to set α_1 or α_6 to zero as before, or set $\alpha_2' = \alpha_3'$.
1510.06699_F01051	57	0.999	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^\dagger \mathcal{T}^{\dagger(ab)c} + \beta_2' \mathcal{T}_{[ab]c}^\dagger \mathcal{T}^{\dagger[ab]c}$	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^\dagger \mathcal{T}^{\dagger(ab)c} + \beta_2' \mathcal{T}_{[ab]c}^\dagger \mathcal{T}^{\dagger[ab]c}$	symmetric form $L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^\dagger \mathcal{T}^{\dagger(ab)c} + \beta_2' \mathcal{T}_{[ab]c}^\dagger \mathcal{T}^{\dagger[ab]c}$ where
1510.06699_F01052	57	0.998	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01053	57	0.998	$\beta_2' = \beta_1 - \beta_2$	$\beta_2' = \beta_1 - \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01054	57	1.000	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	lations, but the total action $S_{\mathrm{T}} \equiv S_{\mathrm{G}} + S_{\mathrm{M}}$ is so. We will not
1510.06699_F01055	57	0.999	${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	quantity ${}^0\mathcal{R}^{*ab}{}_{cd}$ by ${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$
1510.06699_F01056	57	1.000	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$, from which one finds that their
1510.06699_F01057	57	1.000	${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	contractions are related by ${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$
1510.06699_F01058	57	1.000	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 ({}^0\mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$. Similarly,
1510.06699_F01059	57	1.000	${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 ({}^0\mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$	${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 ({}^0\mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 ({}^0\mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$. Similarly,
1510.06699_F01060	57	0.996	${}^0\mathcal{R}^{*ab}{}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}{}_{cd}$	${}^0\mathcal{R}^{*ab}{}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}{}_{cd}$	placements ${}^0\mathcal{R}^{*ab}{}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}{}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a)$.
1510.06699_F01061	57	0.996	$\mathcal{B}_a \rightarrow -\left(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a\right)$	$\mathcal{B}_a \rightarrow -\left(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a\right)$	placements ${}^0\mathcal{R}^{*ab}{}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}{}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a)$.
1510.06699_F01062	57	1.000	${}^0\mathcal{R}^*$	${}^0\mathcal{R}^*$	‘reduced’ versions ${}^0\mathcal{D}_a^*$ and ${}^0\mathcal{R}^*$, respectively, as described in